Advanced Statistical Methods II

Lecture I: Why Incomplete Data Changes Inference

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Outline of Today's Lecture

Course orientation

Motivating examples

Complete data versus observed data

Why the missingness mechanism matters

First look at imputation

Where EM will enter

Key Takeaway Messages

Where does this course fit?

Advanced Statistical Methods II is about statistical methods that go beyond the clean textbook regression/inference setup.

In many real problems, one or more of the following happens:

- the data are incomplete, censored, or partly latent;
- the usual squared-error or likelihood criterion is not appropriate;
- the estimator is defined by an objective function that is hard to optimize directly.

This course studies methods designed for exactly these situations.

The unifying message

Many modern statistical methods are built by changing one of three things.

1. Data structure

Complete observations are replaced by missing, latent, censored, or high-dimensional observations.

2. Loss/objective

Squared error may be replaced by absolute loss, quantile loss, penalized loss, or a censored likelihood.

3. Optimization

Direct maximization/minimization may be replaced by EM, MM, Newton, or gradient-based iterations.

A large part of the course is learning how to recognize which part of the usual statistical setup has changed.

Situation ⟶ methodological response

Situation	Methodological response
Missing or latent data	EM algorithm
Hard objective function	MM algorithm / iterative optimization
Censored response	Survival likelihood, Kaplan–Meier, Cox model
Multicollinearity or many predictors	Ridge, PCR, PLS
Need sparsity	LASSO, LARS
Outliers or non-mean effects	LAD, robust regression, quantile regression
Nonlinear mean structure	Nonlinear least squares, Newton/Gauss–Newton/gradient descent

A semester in two connected arcs

Arc 1: before mid-sem

Incomplete-data thinking: missing data → imputation → EM → MM → censored survival data → Kaplan–Meier.

Arc 2: after mid-sem

Advanced regression methodology: Cox proportional hazards → regularized regression → robust/quantile regression → nonlinear regression.

	Lectures	**Tutorials**
Before mid-sem	12 sessions = 18 hours	6 sessions = 6 hours
After mid-sem	12 sessions = 18 hours	6 sessions = 6 hours

The tutorial in Week j will be linked to the lectures in Week j.

Suggested books and notes (till midsem)

1. Little, R. and Rubin, D. (2019). *Statistical Analysis with Missing Data*, (Third ed.). Wiley.
2. van Buuren, S. (2018). *Flexible Imputation of Missing Data* (Second ed.). Chapman and Hall/CRC. https://stefvanbuuren.name/fimd/.
3. Julie Josse's notes on handling missing values (Josse, 2018).

Example 1: a survey with missing income

Suppose we want to estimate average household income from a sample survey.

Person	Age	Education	Income
1	23	Graduate	42
2	45	School	?
3	36	Graduate	78
4	58	School	?
5	30	Postgraduate	95

What should we do with the two missing incomes?

Possible quick answers:

- delete rows 2 and 4;
- replace ? by the sample mean;
- predict missing income using age and education;
- write a likelihood for the observed data.

Why deletion may be misleading

Complete-case analysis keeps only observations with no missing entries.

If people with very high or very low income are less likely to report income, then complete cases are not a representative subsample.

- Missingness can change the distribution of the observed sample.
- The estimator may be biased even when the original sample was well-designed.
- The loss of observations also reduces efficiency.

The key question is not only *how much* is missing, but also *why* it is missing.

Real-world consequences

Public health: delayed COVID-19 case reporting

Public Health England reported that a technical issue meant **15,841 positive COVID-19 cases** from 25 September–2 October 2020 were not included in daily reported figures and contact-tracing data flows at the time (Public Health England, 2020).

Finance: JPMorgan's "London Whale"

The JPMorgan task-force report on the 2012 CIO losses described risk-model implementation and control failures; later summaries highlight manual Excel copy-paste workflows and a formula error affecting a Value-at-Risk calculation (JPMorgan Chase & Co. Management Task Force, 2013; Milmo, 2024).

In both cases, the statistical issue is not only a missing cell. It is the chain from data capture to analysis to consequential decisions.

Example 2: hidden groups in a population

Suppose the data come from two subpopulations, but group labels are unobserved.

Observation	Measurement X	Group label Z
1	1.2	?
2	1.8	?
3	5.4	?
4	6.1	?
5	5.9	?

Statistical model

$$X_i \sim \pi N(\mu_1, \sigma_1^2) + (1-\pi) N(\mu_2, \sigma_2^2).$$

Here the missing data are not survey nonresponses. The missing data are the **latent class labels** Z_i.

Example 3: time-to-event data

Suppose patients are followed for time to relapse.

Patient	Observed time	Relapse observed?	Meaning
1	4.2	Yes	relapse at 4.2 months
2	6.0	No	relapse time exceeds 6.0 months
3	2.7	Yes	relapse at 2.7 months
4	8.5	No	relapse time exceeds 8.5 months

A censored observation is not “missing” in the same way as a blank cell. It is partial information about an event time.

Three examples, one common structure

Problem	What is incomplete?	Later method
Survey income	Missing response values	Imputation / likelihood / EM
Mixture model	Latent group labels	EM algorithm
Relapse study	Exact failure time for censored subjects	Survival likelihood / KM

Can we write a principled likelihood using only what was observed?

Here **principled likelihood** means a likelihood for the actual random object we observed, Y_{obs}, rather than a likelihood that pretends guessed/imputed values are true.

The answer is yes, but the resulting observed-data likelihood is often harder to work with, because we must mathematically account for all plausible values of the unobserved part.

Complete data likelihood

Imagine that the full data were available:

$$Y_{\text{com}} = (Y_{\text{obs}}, Y_{\text{mis}}).$$

The complete-data likelihood is

$$L_{\text{com}}(\theta) = f(Y_{\text{obs}}, Y_{\text{mis}}; \theta), \qquad \ell_{\text{com}}(\theta) = \log L_{\text{com}}(\theta).$$

In many models, the complete-data likelihood has a simple form. That is why we often introduce latent or missing variables deliberately.

Observed data likelihood

In practice, we only observe Y_{obs}.

The observed-data likelihood is obtained by summing or integrating out the missing part:

$$L_{\text{obs}}(\theta) = f(Y_{\text{obs}};\theta) = \int f(Y_{\text{obs}}, Y_{\text{mis}};\theta)\, dY_{\text{mis}}.$$

For discrete missing data:

$$L_{\text{obs}}(\theta) = \sum_{y_{\text{mis}}} f(Y_{\text{obs}}, y_{\text{mis}};\theta).$$

The observed likelihood is principled, but often algebraically or computationally difficult.

The central difficulty

If complete data were known

$$\ell_{\text{com}}(\theta)$$

may be easy to maximize.

With observed data only

$$\ell_{\text{obs}}(\theta) = \log \int \exp\{\ell_{\text{com}}(\theta)\}\, dY_{\text{mis}}$$

may be hard to maximize.

The EM algorithm will use the simpler complete-data log-likelihood while respecting the fact that some data are unobserved.

A Normal example

Suppose

$$Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2),$$

but some Y_i values are missing.

If the complete data were observed, the MLEs are familiar:

$$\hat{\mu} = \bar{Y}, \qquad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (Y_i - \bar{Y})^2.$$

With missing values, a tempting thought is:

"Can we somehow replace the missing sufficient statistics?"

This is the idea that will later become the E-step of EM.

Sufficient statistics viewpoint

For the Normal model, the complete-data log-likelihood depends on

$$\sum_{i=1}^{n} Y_i, \qquad \sum_{i=1}^{n} Y_i^2.$$

If some observations are missing, then these sufficient statistics are also partly missing.

Preview of an EM-style idea

At iteration t, replace missing sufficient statistics by their conditional expectations given the observed data and the current parameter estimate $\theta^{(t)}$.

Do not worry about the full EM derivation today. The point is to see why the method is natural.

Introduce a missingness indicator

For each observation, define a response indicator:

$$R_i = \begin{cases} 1, & \text{if } Y_i \text{ is observed,} \\ 0, & \text{if } Y_i \text{ is missing.} \end{cases}$$

Missingness is now treated as part of the data-generating process.

The observed values alone do not tell us everything. We also need assumptions about how the missingness occurred.

The missingness mechanism

A missingness mechanism is a probabilistic model for the response pattern (Rubin, 1976; Little and Rubin, 2019):

$$\mathbb{P}(R \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi),$$

where ψ denotes parameters governing the missingness process.

- The data model is about Y and parameter θ.
- The missingness model is about R and parameter ψ.
- The central question is how R depends on Y_{obs} and Y_{mis}.

The classical terminology MCAR, MAR, and MNAR comes from the missing-data framework of Rubin (1976); see also Little and Rubin (2019).

Three missingness mechanisms: preview

Name	Meaning	Example intuition
MCAR	Missingness does not depend on observed or unobserved values	A lab machine randomly fails
MAR	Missingness may depend on observed values, but not on missing values after conditioning on observed values	Income missingness depends on age/education recorded in the data
MNAR	Missingness depends on the missing value itself, even after conditioning on observed values	Very high-income people are less likely to report income

These terms were formalized in the missing-data literature, especially Rubin (1976) and Little and Rubin (2019).

> The distinction is about the data-generating mechanism, not just the pattern of blank cells in a spreadsheet.

Patterns versus mechanisms

Two datasets may show the same missingness pattern but have different mechanisms.

Pattern

Which entries are missing?

- monotone missingness;
- intermittent missingness;
- block missingness.

Mechanism

Why are they missing?

- random accident;
- depends on recorded covariates;
- depends on unreported values.

A statistical method is justified by assumptions about the mechanism, not by the pattern alone.

What is imputation?

Imputation means filling in missing values with plausible values so that standard complete-data methods can be applied.

Common first ideas:

- mean or median imputation;
- regression imputation;

Imputation is not the same as recovering the true missing value. It is a statistical device for inference and prediction.

Why mean imputation can be dangerous

Suppose several missing values are replaced by the observed sample mean.

What changes?

- The mean may look reasonable.
- The variance is usually underestimated.
- Correlations with other variables may be distorted.
- Standard errors are too optimistic if imputed values are treated as real observations.

A good method should reflect uncertainty about what the missing values could have been.

Conditional imputation idea

A more statistical approach uses the conditional distribution

$$Y_{\text{mis}} \mid Y_{\text{obs}}, \theta.$$

Instead of asking:

"What single value should replace the missing entry?"

we ask:

"Given what we observed, what values are plausible under the model?"

This view connects imputation, likelihood, Bayesian thinking, and EM.

The EM algorithm in one slide

The EM algorithm was introduced by Dempster et al. (1977) for maximum likelihood estimation from incomplete data.

Given current parameter estimate $\theta^{(t)}$:

1. **E-step:** compute the expected complete-data log-likelihood

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}} \left[\ell_{\text{com}}(\theta) \mid Y_{\text{obs}} \right].$$

2. **M-step:** maximize this surrogate quantity:

$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta \mid \theta^{(t)}).$$

(Details begin in Lecture 4; today only the motivation).

Why EM feels natural

If the missing data were known	When the missing data are unknown
Write complete-data log-likelihood	Take its conditional expectation
Maximize directly	Maximize the expected complete-data log-likelihood
Obtain standard MLE update	Iterate until convergence

EM alternates between "filling in" missing information probabilistically and updating the parameter estimate.

This is not the same as single imputation; EM works at the level of expected log-likelihood or expected sufficient statistics.

Today's core ideas

1. Incomplete data appear in many forms: missing responses, latent variables, and censored observations.
2. Complete-data likelihoods are often simpler than observed-data likelihoods.
3. Observed-data likelihoods are obtained by integrating or summing over the unobserved part.
4. Missingness mechanisms matter: MCAR, MAR, and MNAR are assumptions about why data are missing.
5. Imputation should be treated as a statistical device, not as truth recovery.
6. EM will provide a principled way to do likelihood estimation under missing or latent data.

For next lecture

Next lecture: types of missingness and first imputation methods

We will study MCAR, MAR, and MNAR more carefully and discuss consequences for bias, efficiency, and valid inference.

Suggested reading before the next class:

- Little and Rubin (2019, Chapter 1);
- van Buuren (2018, Chapter 2);
- Josse (2018), Section on MCAR/MAR/MNAR.

Think about one real dataset you have seen where missingness may not be random.

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Lecture II: Types of Missingness and Their Implications

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Outline of Today's Lecture

A Brief Recap

Incomplete data can affect the statistical methodologies.

In particular, we saw that:

- a complete-data likelihood can be easy to write;
- an observed-data likelihood may require summing or integrating over what is unobserved;
- naive deletion or naive filling-in can change both bias and uncertainty;
- the reason for missingness matters as much as the amount missing.

Today's guiding question

When can we safely ignore the missingness process?

It asks whether the observed part of the data is statistically representative enough for the inferential task at hand.

> If two datasets have the same percentage of missing values, can one be harmless and the other seriously biased?

Running example: exam performance data

Suppose we are studying the relationship between study hours and exam score.

Student	Study hours	Attendance	Stress score	Exam score
1	4.5	82%	3	71
2	7.0	93%	2	88
3	1.5	45%	8	?
4	5.5	76%	5	79
5	2.0	50%	7	?

Possible stories:

- score missing due to a random data-entry failure;
- score missing because low-attendance students missed the exam;
- score missing because students who performed poorly refused to report it.

Same table. Different statistical implications.

Full data, observed data, missing data

Let the full data be denoted by $Y = (Y_{\text{obs}}, Y_{\text{mis}})$ where

- Y_{obs} is the part we actually observe;
- Y_{mis} is the part that exists conceptually but is not observed;
- inference is about a model parameter θ in the distribution of Y.

Observed-data likelihood

Following the incomplete-data likelihood notation of Rubin (1976) and van Buuren (2018, Chapter 2, Sections 2.2.3–2.2.4), for continuous missing values,

$$L_{\text{obs}}(\theta) = f(Y_{\text{obs}} \mid \theta) = \int f(Y_{\text{obs}}, Y_{\text{mis}} \mid \theta)\, dY_{\text{mis}}.$$

For discrete missing values, the integral is replaced by a sum.

The missingness indicator

For each unit and variable, define an indicator

$$R = \begin{cases} 1, & \text{if the value is observed,} \\ 0, & \text{if the value is missing.} \end{cases}$$

The missingness mechanism

In Rubin's theory every data point has some likelihood of being missing.

The process that governs these probabilities is called the *missing data mechanism* or *response mechanism*.

A missingness mechanism is a probabilistic model for the response pattern (Rubin, 1976; Little and Rubin, 2019):

$$\mathbb{P}(R \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi),$$

where ψ denotes parameters governing the missingness process.

- The data model is about Y and parameter θ.
- The missingness model is about R and parameter ψ.
- The central question is how R depends on Y_{obs} and Y_{mis}.

A warning about notation

Different books use slightly different notation. In these slides, we use Rubin's response-indicator convention with $R = 1$ for observed and $R = 0$ for missing, as in the standard likelihood formulation of Rubin (1976); see also van Buuren (2018, Section 2.2.3).

Sometimes we also have fully observed covariates X, such as age, sex, treatment arm, school, or sampling stratum.

Then one may write

$$\mathbb{P}(R \mid Y_{\text{obs}}, Y_{\text{mis}}, X, \psi).$$

Whether X is considered part of Y_{obs} or written separately is a matter of notation. The key issue is whether missingness depends on information we have observed or on information we have not observed.

Three Types of Missingness

Rubin (1976) classified missing data problems into three categories:

1. MCAR
2. MAR
3. MNAR

MCAR: Missing Completely At Random

Definition

The data are **missing completely at random** if the probability of missingness does not depend on either observed or missing values:

$$\mathbb{P}(R \mid Y_{\mathrm{obs}}, Y_{\mathrm{mis}}, \psi) = \mathbb{P}(R \mid \psi).$$

Intuition:

- the missing cases are like a random subsample of the full dataset;
- the missingness process is unrelated to the variables. The causes of the missing data are unrelated to the data.
- the observed sample remains representative, apart from random loss of information.

MCAR is a strong condition. The definition follows the response-mechanism formulation of Rubin (1976); see also van Buuren (2018, Sections 1.2 and 2.2.4) and Little and Rubin (2019, Chapter 1).

MCAR examples

Plausible MCAR examples

- A batch of paper questionnaires is lost due to accidental water damage.
- A lab machine fails for one randomly chosen afternoon.

Probably not MCAR

- Low-income respondents are less likely to report income.
- Sicker patients are less likely to attend follow-up visits.
- Students with poor performance are more likely to skip an optional test.

In real datasets, MCAR should usually be treated as a claim that needs justification, not as the default assumption.

Implications of MCAR

Under MCAR, complete-case analysis may be unbiased for many simple targets.

For estimating a population mean, if R is independent of Y,

$$\mathbb{E}(Y \mid R = 1) = \mathbb{E}(Y).$$

So the mean among observed units targets the same population mean.

> MCAR does not mean “no problem”. We still lose information (reducing sample size), so standard errors usually increase, reducing efficiency.

MCAR primarily protects against bias due to missingness; it does not protect against loss of precision.

Complete-case analysis

Complete case analysis (listwise deletion) is the default way of handling incomplete data in many statistical packages, including SPSS, SAS and Stata.

The function `na.omit()` does the same in S-PLUS and R. The procedure eliminates all cases with one or more missing values on the analysis variables.

Advantage: *Convenience*. If the data are MCAR, listwise deletion produces unbiased estimates of means, variances and regression weights.

Under MCAR, listwise deletion produces standard errors and significance levels that are correct for the reduced subset of data, but that are often larger relative to all available data

MCAR and complete-case analysis

What may be okay

- estimating simple means;
- estimating regression slopes when complete cases are representative;
- using standard likelihood methods on complete cases, with reduced sample size.

What still goes wrong

- lower effective sample size;
- larger standard errors;
- possible loss of power;
- wasted information from partially observed units.

Complete-case analysis is simplest under MCAR, but it might not be the most efficient use of the data.

Little's MCAR Test: (Little, 1988)

A formal statistical test to assess whether data are MCAR.

A significant result suggests deviation from MCAR.

MAR: Missing At Random

Definition

The data are **missing at random** if, after conditioning on the observed data, missingness does not depend on the missing values:

$$\mathbb{P}(R \mid Y_{\mathrm{obs}}, Y_{\mathrm{mis}}, \psi) = \mathbb{P}(R \mid Y_{\mathrm{obs}}, \psi).$$

Intuition:

- missingness may depend on what we have observed;
- it should not additionally depend on the values we failed to observe;
- observed variables can explain the missingness pattern.

MAR is weaker than MCAR, but still an assumption. This conditional definition follows Rubin (1976); see also van Buuren (2018, Sections 1.2 and 2.2.4).

MAR examples

Example: income nonresponse

Income is missing more often among older respondents and respondents from certain occupations. If age and occupation are observed, and after conditioning on them missingness does not depend on the actual income value, this is MAR.

Example: exam score

Exam score is missing more often for students with low attendance. If attendance is observed, and after conditioning on attendance missingness does not depend on the unobserved score, this is MAR.

MAR means the randomness in missingness can be explained by observed information.

MAR is conditional

A common misunderstanding:

"The missingness is related to age, so it is not random."

Statistically, this can still be MAR if age is observed.

For example,

$$\mathbb{P}(R = 1 \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi) = \mathbb{P}(R = 1 \mid \text{age}, \psi).$$

Another example: Women are (probably) less likely to disclose their weight, but their gender is recorded. In this case, weight is MAR.

The phrase "at random" in MAR is technical. It means conditionally random given the observed data.

Why MAR can still cause bias under deletion

Suppose low-attendance students are more likely to have missing exam scores.

If attendance is associated with exam score, then complete cases have a different attendance distribution from the full class.

Complete-case mean score estimates

$$\mathbb{E}(Y \mid R = 1),$$

not necessarily

$$\mathbb{E}(Y).$$

Under MAR, deletion can be biased if the observed variables related to missingness are also related to the outcome.

A small numerical MAR example

Group	True mean score	Fraction in class	Observation probability
High attendance	80	0.5	0.9
Low attendance	60	0.5	0.3

True class mean:

$$0.5(80) + 0.5(60) = 70.$$

Observed mean among complete cases:

$$\frac{0.5(0.9)(80) + 0.5(0.3)(60)}{0.5(0.9) + 0.5(0.3)} = 75.$$

Even though missingness depends only on observed group membership, complete-case analysis over-represents the high-attendance group.

the same bias principle under differential response is discussed in Little and Rubin (2019, Chapter 1) and van Buuren (2018, Section 1.2).

Implications of MAR

Under MAR, good analyses should use the observed variables that explain missingness.

Typical approaches include:

- likelihood based on the observed data;
- EM algorithm for maximum likelihood estimation;
- multiple imputation using a model for missing values;
- inverse-probability weighting when response probabilities are modelled.

> MAR is the main working assumption behind many standard missing-data methods.

The quality of the analysis depends strongly on whether the imputation or likelihood model includes the right observed predictors (Sterne et al., 2009; van Buuren, 2018).

MNAR: Missing Not At Random

Definition

The data are **missing not at random** if the probability of missingness depends on the missing values even after conditioning on observed data:

$$\mathbb{P}(R \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi) \text{ depends on } Y_{\text{mis}}.$$

Intuition:

- the unobserved value helps determine whether it is observed;
- the missingness process is informative;
- the observed data alone cannot fully explain who is missing.

MNAR is also called **nonignorable missingness** in many texts (Rubin, 1976; Little and Rubin, 2019; Molenberghs and Kenward, 2007). See also van Buuren (2018, Sections 1.2 and 3.8).

MNAR examples

- Patients whose symptoms worsen are more likely to drop out of a clinical trial, even after conditioning on previous measurements.
- Students with low unobserved exam scores are less likely to submit optional self-reported scores.
- A sensor fails precisely when the value exceeds its operating range.

> In MNAR settings, the missing part of the data is not just absent; it is systematically different in a way that the observed data may not reveal.

clinical-trial dropout and nonignorable missingness are discussed in Molenberghs and Kenward (2007) and National Research Council (2010).

A small numerical MNAR example

Let Y be binary:

$$Y = 1 \text{ means high risk,} \qquad Y = 0 \text{ means low risk.}$$

Assume

$$\mathbb{P}(Y = 1) = 0.5.$$

But suppose observation probabilities are

$$\mathbb{P}(R = 1 \mid Y = 1) = 0.2, \qquad \mathbb{P}(R = 1 \mid Y = 0) = 0.8.$$

Then among observed cases,

$$\mathbb{P}(Y = 1 \mid R = 1) = \frac{0.5(0.2)}{0.5(0.2) + 0.5(0.8)} = 0.2.$$

The true high-risk fraction is 0.5, but the observed high-risk fraction is only 0.2.

This synthetic numerical example illustrates the selection-bias logic behind nonignorable mechanisms (Little and Rubin, 2019; Molenberghs and Kenward, 2007).

Implications of MNAR

Under MNAR, standard observed-data methods may fail unless the missingness process is explicitly modelled.

Possible strategies:

- collect more data on why values are missing;
- use subject-matter knowledge about the missingness process;
- specify a joint model for outcomes and response indicators;
- conduct sensitivity analyses over plausible MNAR mechanisms;
- report conclusions transparently as assumption-dependent.

MNAR assumptions are usually not identifiable from the observed data alone. They require modelling choices and sensitivity analysis.

MCAR, MAR, MNAR side by side

Type	Mechanism	Everyday interpretation
MCAR	$\mathbb{P}(R \mid Y_{obs}, Y_{mis}) = \mathbb{P}(R)$	Missingness unrelated to the data values.
MAR	$\mathbb{P}(R \mid Y_{obs}, Y_{mis}) = \mathbb{P}(R \mid Y_{obs})$	Missingness related only to observed information.
MNAR	depends on Y_{mis}	Missingness related to the unobserved value itself.

MCAR is a special case of MAR, but MAR is not the same as MCAR.

Which methods are naturally justified?

Type	Naive deletion?	Usually better approach
MCAR	Often unbiased, but inefficient	Use all available information; likelihood or imputation may improve precision.
MAR	Often biased unless special conditions hold	Observed-data likelihood, EM, multiple imputation, weighting.
MNAR	Generally risky	Model missingness, collect auxiliary information, sensitivity analysis.

How to classify an example

For any missing-data problem, ask three questions.

1. **What is missing?** Outcome, covariate, exposure, label, follow-up time, etc.
2. **What variables are observed?** Are predictors of missingness available in the dataset?
3. **Could missingness depend on the missing value itself?** Is there a plausible mechanism by which the unobserved value affects whether it is observed?

Classification depends on the data-collection story, not only on the data table.

An exercise

Classify each as MCAR, MAR, or MNAR.

1. A computer crash deletes a random 5% of lab measurements.
2. Income is missing more often for younger people; age is observed.
3. Income is missing more often for high-income individuals, even after conditioning on observed variables.
4. Blood pressure follow-up is missing more often for patients whose previous blood pressure was high.
5. A satisfaction score is missing because dissatisfied customers refuse to answer the survey.

Classroom classification exercise: answers

1. **MCAR:** a random computer crash is unrelated to either the observed or unobserved lab values.
2. **MAR:** missingness depends on age, and age is observed.
3. **MNAR:** missingness still depends on the unobserved income value after conditioning on observed variables.
4. **MAR:** missingness depends on previous blood pressure, which is observed, not necessarily on the unobserved follow-up value.
5. **MNAR:** dissatisfied customers refuse to answer because of the unreported satisfaction value itself.

These answers are model-based classifications: a different data-collection story or extra observed variables could change the classification.

Full joint model with missingness

A full probabilistic model may factor as

$$f(Y_{\mathrm{obs}}, Y_{\mathrm{mis}}, R \mid \theta, \psi) = f(Y_{\mathrm{obs}}, Y_{\mathrm{mis}} \mid \theta) \cdot \mathbb{P}(R \mid Y_{\mathrm{obs}}, Y_{\mathrm{mis}}, \psi).$$

The first factor is the "data" model.

The second factor is the missingness mechanism.

> If we care only about θ, do we always need to model ψ?

This is where the notion of **ignorability** appears.

Ignorability: the basic idea

For likelihood-based inference, the missingness mechanism is often called **ignorable** when two conditions hold (Rubin, 1976; Little and Rubin, 2019):

1. the data are MAR;
2. the parameters θ and ψ are distinct, meaning the parameter space separates as a product space.

Then likelihood inference for θ can be based on

$$L_{\mathrm{obs}}(\theta) = f(Y_{\mathrm{obs}} \mid \theta),$$

without explicitly modelling $\mathbb{P}(R \mid Y, \psi)$. (For more details, see Theorem 7.1 of Rubin (1976)).

> Ignorable does not mean unimportant. It means the missingness model can be omitted for a particular likelihood calculation under specific assumptions.

A Remark

Ignorable missingness is not the same as listwise deletion.

Do not confuse these two different ideas.

Ignorable mechanism

Under MAR plus parameter distinctness, the missingness model need not be explicitly included in the likelihood for θ.

Listwise deletion

Throw away all incomplete records and analyze only complete cases.

The mechanism may be ignorable, but incomplete records may still contain useful information that should not be discarded.

Where EM will enter

Under MAR and an ignorable mechanism, one natural target is the observed-data likelihood

$$L_{\text{obs}}(\theta) = f(Y_{\text{obs}} \mid \theta).$$

But this likelihood may be computationally hard because it involves sums or integrals over the missing data (Rubin, 1976; Schafer, 1997).

The EM algorithm helps by alternating between:

- estimating the missing sufficient information conditionally on what was observed;
- maximizing an expected complete-data log-likelihood.

The MCAR/MAR/MNAR distinction tells us when observed-data likelihood methods are conceptually justified; EM will help us compute them.

Can we test the missingness assumption?

Some aspects of missingness can be explored from the observed data.

Examples:

- compare observed covariates between complete and incomplete cases;
- regress the missingness indicator R on observed variables;
- visualize missingness patterns;
- use formal tests for MCAR in some multivariate settings, such as Little's test (Little, 1988).

Testing MAR vs MNAR

MNAR vs. MAR cannot be tested simply because data that are MNAR involve unobserved data. The deciding information is in Y_{mis}.

MAR itself is an unverifiable assumption, and must be justified in the context of each problem.

Practical workflow for missing data

1. Describe the amount and pattern of missingness.
2. Ask how each variable became missing.
3. Identify observed variables related to missingness.
4. Decide which assumption is plausible: MCAR, MAR, or MNAR.
5. Choose a method consistent with that assumption.
6. Report the assumption and, when needed, sensitivity analyses.

Bias and efficiency: two different concerns

Bias

Does the estimator target the correct population quantity?

Efficiency

How much uncertainty remains around the estimator?

Missing data can affect both:

- MCAR mainly causes loss of efficiency, though careless analysis can still waste information;
- MAR can cause bias under deletion but can often be handled by model-based methods;
- MNAR can cause bias unless the missingness mechanism is addressed.

"No bias" and "no loss of information" are not the same statement.

Why single imputation is not a full solution

A tempting approach is to fill in missing values once and then analyze the completed dataset.

Examples:

- replace missing income by mean income;
- replace missing score by predicted score from regression;
- fill missing lab value by the previous observed value.

> Single imputation often treats imputed values as if they were truly observed. This usually understates uncertainty.

Multiple imputation was developed to reflect both sampling uncertainty and imputation uncertainty (Schafer, 1997; Carpenter and Kenward, 2013).

Complete-case analysis: when to be suspicious

Be suspicious of complete-case analysis when:

- the missingness rate is large;
- missingness differs strongly by observed covariates;
- variables related to missingness are also related to the outcome;
- the missing variable is a key outcome or exposure;
- the reasons for nonresponse suggest discomfort, severity, risk, or cost;
- incomplete records still contain useful information.

Key Takeaway Messages

1. Missingness is described by both the observed data Y_{obs} and a response indicator R.
2. MCAR: missingness is unrelated to observed and missing values.
3. MAR: missingness may depend on observed values, but not on missing values after conditioning on what is observed.
4. MNAR: missingness depends on the unobserved value itself, even after conditioning on observed information.
5. Complete-case analysis is not automatically valid; its validity depends on the missingness mechanism and the target of inference.
6. MAR plus parameter distinctness gives ignorability for likelihood inference, not permission to throw away useful incomplete records.

The one-line summary

The danger of missing data is not only that values are absent, but that their absence may be informative.

Before choosing an imputation, likelihood, weighting, or EM-based method, first ask: why are these values missing?

Next lecture: conditional imputation, uncertainty, and why a single filled-in dataset is usually not enough.

References

The notation in these slides primarily follows Rubin (1976) and van Buuren (2018, Chapter 2, Sections 2.2.3–2.2.4). All numerical tables are synthetic teaching examples unless explicitly stated otherwise.

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Advanced Statistical Methods II

Lecture III: Conditional Imputation and Uncertainty

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Outline of Today's Lecture

Recap and motivation

From marginal thinking to conditional thinking

1. Conditional mean imputation

2. Regression-based imputation

3. Stochastic imputation

Why single imputation underestimates uncertainty

4. Full conditional distributions

Imputation as prediction, not truth recovery

Key Takeaway Messages

Missingness can affect inference, the validity of complete-case analysis, and the likelihood we should use.

Today we ask a more constructive question:

If some values are missing, can we fill in plausible values?

The word plausible is crucial.

- Not every filled-in value is statistically meaningful.
- A good imputation should use information in the observed data.
- It should also acknowledge that the missing value is uncertain.

See Little and Rubin (2019), Chs. 1–4; van Buuren (2018), Chs. 1–2.

A small motivating example

Suppose a clinical study records:

Patient	Age	Baseline severity	6-month outcome
1	43	Low	12
2	67	High	?
3	51	Medium	19
4	70	High	?
5	38	Low	10

What should replace the two question marks?

Possible answers:

- the overall mean of observed outcomes;
- the mean among patients with similar severity;
- a predicted value from a regression model;
- a random draw from a predictive distribution.

Focus of Today's Lecture

Imputation means replacing missing values by one or more plausible values, constructed using a statistical methodology/philosophy.

Today we focus on four ideas:

1. conditional mean imputation;
2. regression-based imputation;
3. stochastic imputation;
4. full conditional distributions and the beginning of iterative imputation.

Recap: notations

Let the full data be

$$Y = (Y_{\text{obs}}, Y_{\text{mis}}),$$

where:

- Y_{obs} denotes the actually observed part;
- Y_{mis} denotes the missing part;
- R denotes the response/missingness indicator;
- θ denotes parameters of the data model.

The central conditional distribution is

$$p(Y_{\text{mis}} \mid Y_{\text{obs}}, R, \theta).$$

Under *ignorable missingness assumptions*, we work with

$$p(Y_{\text{mis}} \mid Y_{\text{obs}}, \theta).$$

A bad default: unconditional mean imputation

A common first attempt is:

$$\tilde{Y}_i = \bar{Y}_{\mathrm{obs}} \quad \text{whenever } Y_i \text{ is missing.}$$

It uses only the marginal distribution of the observed values of Y.

Disadvantages:

- ignores predictors that may be related to Y;
- can distort correlations and regression slopes;
- treats imputed values as if they were known exactly;
- artificially reduces variation.

See Little and Rubin (2019), Ch. 4; Enders (2010), Ch. 3.

Conditional thinking

Instead of asking:

What is a typical value of Y?

we ask:

What is a plausible value of Y given what we observed?

That is,

$$p(Y_{\text{mis}} \mid Y_{\text{obs}}, \theta).$$

Imputation should preserve relationships among variables as much as possible.

For example:

- income may depend on age, education, and occupation;
- health outcome may depend on baseline severity;
- exam score may depend on attendance and previous scores.

Suppose X is fully observed and Y has missing values.

If X and Y are positively associated, then two people with different X values should not receive the same imputed Y value.

Case	Observed X	Reasonable imputed Y?
A	Low X	Lower plausible range
B	Medium X	Middle plausible range
C	High X	Higher plausible range

The conditional distribution contains two pieces

For a missing scalar value Y_i, conditional on observed information X_i, think of

$$Y_i \mid X_i = x_i.$$

This distribution has:

1. a conditional center:

$$\mathbb{E}(Y_i \mid X_i = x_i),$$

2. a conditional spread:

$$\mathbb{V}\mathrm{ar}(Y_i \mid X_i = x_i).$$

> Many simple imputation methods use the conditional center but ignore the conditional spread.

1. Conditional mean imputation

The most direct conditional imputation rule is

$$\tilde{Y}_{\text{mis}} = \mathbb{E}\left(Y_{\text{mis}} \mid Y_{\text{obs}}, \widehat{\theta}\right).$$

Replace each missing value by the conditional expectation under a fitted model.

This is better than unconditional mean imputation when the conditioning variables carry information.

But it is still deterministic:

$$\text{same observed information} \ \Rightarrow \ \text{same imputed value.}$$

Conditional mean imputation still can not capture the uncertainty fully.

Suppose

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix} \right).$$

Then

$$Y \mid X = x \sim N\left(\mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X), \sigma_Y^2(1 - \rho^2) \right).$$

Conditional mean imputation uses

$$\tilde{Y} = \widehat{\mu}_Y + \widehat{\rho}\frac{\widehat{\sigma}_Y}{\widehat{\sigma}_X}(x - \widehat{\mu}_X).$$

Standard multivariate normal conditioning; see Schafer (1997), Ch. 5.

A few remarks.

- If $\rho = 0$, then X gives no information about Y.
- If $|\rho|$ is large, then X gives strong information about Y.
- The conditional variance is

$$\sigma_Y^2(1-\rho^2).$$

Even when $|\rho|$ is large, the conditional variance is usually not zero. A missing value remains uncertain.

A basic calculation

Suppose the fitted conditional distribution is

$$Y \mid X = x \sim N(10 + 2x, 4).$$

For a missing Y with $X = 3$:

$$\mathbb{E}(Y \mid X = 3) = 10 + 2(3) = 16.$$

So conditional mean imputation gives

$$\tilde{Y} = 16.$$

But the conditional distribution is

$$Y \mid X = 3 \sim N(16, 4).$$

The imputed value 16 is only the center of plausible values. Values like 13.8, 15.2, 17.5, or 19.1 are also plausible.

Strengths	Weaknesses
Uses observed information	Understates variability
Simple to explain	Treats predictions as observed truth
Often improves over mean imputation	Can distort standard errors
Related to regression ideas	Does not propagate parameter uncertainty

It may be useful for prediction of missing entries, but it is generally not enough for valid statistical inference.

2. Regression imputation: the basic model

Suppose Y has missing values and X is fully observed.

Fit the model using complete cases:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2).$$

For a missing Y_i, impute

$$\tilde{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_i.$$

This is just conditional mean imputation under a linear regression model.

See Little and Rubin (2019), Ch. 4; van Buuren (2018), Secs. 1.3–1.4.

Regression imputation with several predictors

More generally,

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip} + \varepsilon_i .$$

Then

$$\tilde{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_{i1} + \cdots + \widehat{\beta}_p X_{ip}.$$

Useful predictors might include:

- variables related to the missing variable;
- variables related to the missingness mechanism;
- design variables, group indicators, or baseline measurements.

> Good imputation models should be at least as rich as the analysis model, and often richer.

This principle is emphasized in Rubin (1987), Schafer (1997), and Carpenter and Kenward (2013).

Why include variables related to missingness?

Recall MAR from Lecture II:

$$\mathbb{P}(R \mid Y_{\text{obs}}, Y_{\text{mis}}) = \mathbb{P}(R \mid Y_{\text{obs}}).$$

If missingness depends on a fully observed variable X, then conditioning on X can make the missingness more manageable.

Example

High-severity patients are more likely to miss follow-up, and baseline severity is observed.

Then an imputation model for the follow-up outcome should include baseline severity.

Ignoring such variables can turn an apparently harmless imputation into a biased analysis.

Regression imputation is model-based

Regression imputation depends on assumptions:

- the regression form is approximately correct;
- the residual distribution is reasonable;
- the fitted relationship for observed cases applies to missing cases;
- relevant predictors are included;

A common mistake

After deterministic regression imputation, one might analyze the completed dataset as if nothing were missing.

For example:

$$\text{Fit } Y \sim X \text{ after replacing missing } Y \text{ by } \widehat{Y}.$$

This is usually invalid for inference because the imputed values have no residual noise and no parameter uncertainty.

Consequences:

- standard errors may be too small;
- confidence intervals may be too narrow;
- fitted associations can be exaggerated or distorted.

See Little and Rubin (2019), Ch. 4; Sterne et al. (2009).

3. Stochastic Imputation

A simple way to respect conditional spread is stochastic regression imputation.

Fit

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2).$$

For missing Y_i, draw

$$\varepsilon_i^* \sim N(0, \widehat{\sigma}^2), \quad \text{(adding residual noise)}$$

and impute

$$\tilde{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_i + \varepsilon_i^*.$$

Now imputed values have realistic scatter around the regression line.

Conditional mean vs stochastic imputation

Conditional mean imputation	Stochastic imputation
Uses $\mathbb{E}(Y \mid X)$	Draws from $Y \mid X$
No residual noise	Adds residual noise
Creates too-smooth completed data	Preserves scatter better
Good for simple prediction	Better for representing uncertainty in missing values
Still ignores parameter uncertainty	May still ignore parameter uncertainty unless parameters are also drawn

Stochastic imputation is an improvement, but one randomly completed dataset is still not the whole uncertainty story.

Predictive distribution viewpoint

The most principled target is not simply

$$\mathbb{E}(Y_{\text{mis}} \mid Y_{\text{obs}}, \widehat{\theta}).$$

It is the predictive distribution

$$p(Y_{\text{mis}} \mid Y_{\text{obs}}).$$

If θ is unknown, then formally

$$p(Y_{\text{mis}} \mid Y_{\text{obs}}) = \int p(Y_{\text{mis}} \mid Y_{\text{obs}}, \theta)\, p(\theta \mid Y_{\text{obs}})\, d\theta.$$

Uncertainty comes from two sources: residual uncertainty about missing values and parameter uncertainty about the model.

This predictive-distribution view is central in Rubin (1987) and Schafer (1997).

Two sources of uncertainty

Source	Meaning
Residual uncertainty	Even if the model parameters were known, a missing value is not exactly predictable.
Parameter uncertainty	We estimate the model parameters from incomplete data, so $\widehat{\theta}$ itself is uncertain.

Conditional mean imputation ignores both. Simple stochastic imputation handles residual uncertainty but usually not parameter uncertainty.

This motivates multiple imputation.

Mini example: why one imputation is unstable

Suppose

$$Y \mid X = 3 \sim N(16, 4).$$

A stochastic imputation might draw

$$\tilde{Y} = 15.1.$$

Another equally valid draw might be

$$\tilde{Y} = 18.4.$$

A third might be

$$\tilde{Y} = 13.6.$$

If our final conclusion changes drastically depending on one random draw, we should not base inference on only one completed dataset.

The completed-data illusion

After imputation, the dataset visually appears complete.

ID	X	Y	Missing?	Status
1	2.1	13.4	No	observed
2	3.0	16.0	Yes	imputed
3	4.4	21.2	No	observed
4	1.8	14.3	Yes	imputed

But the two imputed values are not the same kind of information as the observed values.

Why variance is underestimated: intuition

If we replace a missing value by a single number, then downstream analysis treats that number as fixed.

But before imputation, it should be a distribution:

$$Y_{\mathrm{mis}} \mid Y_{\mathrm{obs}}.$$

Single imputation collapses that distribution to a point:

$$Y_{\mathrm{mis}} \mid Y_{\mathrm{obs}} \quad \longrightarrow \quad \tilde{Y}_{\mathrm{mis}}.$$

A preview of multiple imputation

Multiple imputation creates several completed datasets:

$$Y^{(1)}, Y^{(2)}, \ldots, Y^{(m)}.$$

Each has different plausible values for Y_{mis}.

Analyze each dataset, then combine the results.

Imputation	Estimate	Standard error
1	$\widehat{Q}^{(1)}$	$U^{(1)}$
2	$\widehat{Q}^{(2)}$	$U^{(2)}$
$\vdots$	$\vdots$	$\vdots$
m	$\widehat{Q}^{(m)}$	$U^{(m)}$

Multiple imputation was formalized by Rubin (1987); see also van Buuren (2018).

Rubin's combining rules: only a preview

Let $\widehat{Q}^{(j)}$ be the estimate from imputed dataset j.

The combined estimate is

$$\bar{Q} = \frac{1}{m}\sum_{j=1}^{m} \widehat{Q}^{(j)}.$$

The total variance combines:

- within-imputation variance;
- between-imputation variance.

We will not develop multiple imputation fully today, but this formula shows why several plausible datasets are useful.

When several variables have missing values

So far, we imagined one variable Y with missing values and predictors X fully observed.

But in real data, several variables may be incomplete:

$$Y_1, Y_2, \ldots, Y_p.$$

Example:

ID	Age	Income	Health score	Follow-up
1	43	52	78	82
2	67	?	61	?
3	?	44	70	73
4	70	69	?	?

Now there is no single obvious response variable and no single obvious predictor set.

4. Full conditional distributions

A full conditional distribution describes one variable given all the others.

For variable Y_j:

$$p(Y_j \mid Y_{-j}, \theta_j),$$

where Y_{-j} denotes all variables except Y_j.

Examples:

$$\begin{gathered}
Y_1 \mid Y_2, Y_3, \ldots, Y_p, \\
Y_2 \mid Y_1, Y_3, \ldots, Y_p, \\
\vdots \\
Y_p \mid Y_1, Y_2, \ldots, Y_{p-1}.
\end{gathered}$$

The full conditional idea lets us build imputation models variable by variable.

Different variables need different models

Full conditional imputation is flexible because each variable can use a suitable model.

Variable type	Possible conditional model
Continuous	Linear regression
Binary	Logistic regression
Categorical	Multinomial regression
Count	Poisson or negative binomial model
Skewed continuous	Predictive mean matching or transformations

See van Buuren (2018), Chs. 3–4; van Buuren and Groothuis-Oudshoorn (2011).

The chained-equations idea

A common practical strategy is:

1. initialize missing values crudely;
2. impute Y_1 given all other variables;
3. impute Y_2 given all other variables;
4. continue through Y_p;
5. repeat the cycle several times.

This is often called fully conditional specification or multivariate imputation by chained equations.

Each step is simple. The overall algorithm can handle complex missingness patterns.

See van Buuren (2018); implementation in R is described by van Buuren and Groothuis-Oudshoorn (2011).

A conceptual algorithm

One cycle of chained imputation

1. Regress income on age, health score, and follow-up.
2. Draw plausible missing income values.
3. Regress health score on age, income, and follow-up.
4. Draw plausible missing health scores.
5. Regress follow-up on age, income, and health score.
6. Draw plausible missing follow-up values.

After several cycles, we retain one completed dataset.

Repeat the whole procedure m times to obtain multiple imputed datasets.

Why the word "conditional" matters again

Full conditional imputation says:

Use everything currently known to impute one variable at a time.

This protects important relationships such as:

- association between income and education;
- association between baseline and follow-up measurements;
- association between risk score and disease outcome;
- group differences due to treatment, region, or cohort.

If a scientifically important relationship is absent from the imputation model, the completed data may weaken or erase it.

Scientific caution

Imputation is most credible when:

- missingness is plausibly MCAR or MAR after conditioning on observed variables;
- the imputation model includes important predictors;
- the imputation model respects the form of the data;
- sensitivity analyses are considered when MNAR is plausible;
- uncertainty due to missingness is reported honestly.

> In public health, clinical trials, survey science, and policy evaluation, overconfident imputation can become overconfident decision-making.

See National Research Council (2010) and Sterne et al. (2009) for applied cautions.

Connecting to the EM algorithm

The next major topic is the EM algorithm.

It also begins with missing or latent quantities.

But instead of filling in a single completed dataset, EM repeatedly computes conditional expectations of missing-data quantities.

Imputation viewpoint	EM viewpoint
Fill in plausible missing values	Average over missing values
Work with completed datasets	Work with expected complete-data log-likelihood
Represents uncertainty through repeated draws	Uses conditional expectation to optimize likelihood

Both ideas are built on $Y_{\text{mis}} \mid Y_{\text{obs}}, \theta$.

A few examples

For each statement, decide whether it is reasonable or flawed.

1. "I imputed missing incomes by the overall mean, then used the completed data for a regression."
2. "I imputed missing follow-up outcomes using baseline severity, age, and treatment group."
3. "I created one stochastic imputed dataset and reported ordinary standard errors."
4. "I used several imputed datasets and combined estimates to reflect between-imputation variation."

Which choices respect conditional information? Which choices respect uncertainty?

Answers

1. Flawed: overall mean imputation ignores conditional structure and understates variability.
2. Reasonable: the imputation uses observed predictors that are scientifically relevant to the missing outcome.
3. Flawed for inference: stochastic values add residual scatter, but one completed dataset does not fully represent imputation uncertainty.
4. Reasonable: multiple completed datasets can represent missing-data uncertainty when the imputation model is appropriate.

Good imputation needs both conditional information and honest uncertainty accounting.

Key Takeaway Messages

1. Imputation should be conditional on observed information, not merely based on marginal averages.
2. Conditional mean imputation uses the center of $Y_{\text{mis}} \mid Y_{\text{obs}}, \widehat{\theta}$.
3. Regression imputation is conditional mean imputation under a regression model.
4. Stochastic imputation draws from a predictive distribution and better preserves scatter.
5. Single imputation usually underestimates uncertainty.
6. Full conditional distributions allow variable-by-variable imputation in multivariate data.

Suggested reading after Lecture III

For this lecture, the most relevant readings are:

- Little and Rubin (2019), Chapters 1–4: core missing-data framework and simple methods.
- van Buuren (2018), Chapters 1–4: practical imputation, full conditional specification, and chained equations.
- van Buuren and Groothuis-Oudshoorn (2011): R implementation of chained equations.

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Advanced Statistical Methods II

Lecture IV: EM Algorithm — First Principles

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Outline of Today's Lecture

Imputation: "filling in values". using the conditional distribution of missing quantities given observed information to construct plausible completed data.

Today's focus: likelihood-based estimation with missing or latent data.

- The complete-data likelihood may be simple.
- The observed-data likelihood may require summing or integrating over what we did not observe.
- EM turns that difficult optimization problem into a sequence of easier conditional-expectation and maximization steps.

Expectation $\longrightarrow$ Maximization $\longrightarrow$ Repeat

A brief history of the EM algorithm

The idea of alternating between estimating missing information and updating parameters appeared in several earlier special cases, especially in incomplete-data likelihood problems.

The modern general formulation was given by Dempster–Laird–Rubin in their seminal paper Dempster et al. (1977).

- It introduced the name **EM algorithm**: **E**xpectation followed by **M**aximization.
- It showed that many apparently different procedures could be understood through one incomplete-data likelihood framework. It turned a collection of problem-specific tricks into a general statistical algorithm.
- The paper also established the fundamental likelihood-ascent idea that makes EM statistically attractive.

See historical comments in McLachlan and Krishnan (2008) Ch. 1.

A simple motivating example

Suppose items are produced by one of two machines, but for some reason the machine label is not recorded.

For item i:

$$Y_i = \text{number of defects}, \qquad Z_i = \begin{cases} 1, & \text{Machine 1,} \\ 2, & \text{Machine 2.} \end{cases}$$

Assume

$$\mathbb{P}(Z_i = 1) = \pi, \qquad Y_i \mid Z_i = k \sim \mathrm{Poisson}(\lambda_k), \quad k = 1, 2.$$

We observe only

$$y_1, \ldots, y_n,$$

but not $z_1, \ldots, z_n$.

> Can we estimate $\pi, \lambda_1, \lambda_2$ without knowing which machine produced which item?

Why this example is an EM problem

If the machine labels Z_i's were known, estimation would be easy.

But they are not known, so the likelihood contribution of observation i is

$$\pi \frac{e^{-\lambda_1} \lambda_1^{y_i}}{y_i!} + (1-\pi) \frac{e^{-\lambda_2} \lambda_2^{y_i}}{y_i!}.$$

Hence the observed-data log-likelihood is

$$\ell(\pi, \lambda_1, \lambda_2) = \sum_{i=1}^{n} \log \left\{ \pi f_1(y_i; \lambda_1) + (1-\pi) f_2(y_i; \lambda_2) \right\}.$$

The difficulty is the *log of a sum*. It prevents the likelihood from separating into simple machine-specific pieces.

Complete-data version: what would be easy?

Let

$$Z_{i1} = \mathbb{I}(Z_i = 1), \qquad Z_{i2} = \mathbb{I}(Z_i = 2) = 1 - Z_{i1}.$$

If these indicators were observed, the complete-data log-likelihood would be, up to constants,

$$\ell_c(\theta) = \sum_{i=1}^{n} \left[Z_{i1}\{\log \pi + y_i \log \lambda_1 - \lambda_1\} + Z_{i2}\{\log(1-\pi) + y_i \log \lambda_2 - \lambda_2\} \right],$$

where $\theta = (\pi, \lambda_1, \lambda_2)$.

With the labels known, the problem separates into two ordinary Poisson MLE problems plus an estimate of the mixing proportion.

Deriving the complete-data update for π

Focus on the terms in ℓ_c involving π:

$$\ell_c(\pi) = \sum_{i=1}^{n} \{Z_{i1} \log \pi + Z_{i2} \log(1-\pi)\} + C.$$

Differentiate:

$$\frac{\partial \ell_c}{\partial \pi} = \sum_{i=1}^{n} \frac{Z_{i1}}{\pi} - \sum_{i=1}^{n} \frac{Z_{i2}}{1-\pi}.$$

Set the derivative equal to zero:

$$\frac{\sum_i Z_{i1}}{\pi} = \frac{\sum_i Z_{i2}}{1-\pi}.$$

Since $Z_{i1} + Z_{i2} = 1$, this gives

$$\widehat{\pi} = \frac{1}{n} \sum_{i=1}^{n} Z_{i1}.$$

Deriving the complete-data updates for λ_1, λ_2

For λ_1, keep only the relevant terms:

$$\ell_c(\lambda_1) = \sum_{i=1}^{n} Z_{i1}\{y_i \log \lambda_1 - \lambda_1\} + C.$$

Differentiate and set equal to zero:

$$\frac{\partial \ell_c}{\partial \lambda_1} = \sum_{i=1}^{n} Z_{i1}\left(\frac{y_i}{\lambda_1} - 1\right) = 0.$$

Thus

$$\widehat{\lambda}_1 = \frac{\sum_i Z_{i1} y_i}{\sum_i Z_{i1}}.$$

Similarly,

$$\widehat{\lambda}_2 = \frac{\sum_i Z_{i2} y_i}{\sum_i Z_{i2}}.$$

Complete-data MLEs if labels were known

From the complete-data log-likelihood, the MLEs would be

$$\widehat{\pi} = \frac{1}{n} \sum_{i=1}^{n} Z_{i1},$$

$$\widehat{\lambda}_1 = \frac{\sum_{i=1}^{n} Z_{i1} y_i}{\sum_{i=1}^{n} Z_{i1}}, \qquad \widehat{\lambda}_2 = \frac{\sum_{i=1}^{n} Z_{i2} y_i}{\sum_{i=1}^{n} Z_{i2}}.$$

So the only obstacle is that the Z's are missing.

EM handles this by (in its M step) replacing the missing indicators (Z_{ik}) by their conditional expectations and then use the revised formulas.

E-step for the motivating example

At iteration t, suppose the current values are

$$\theta^{(t)} = (\pi^{(t)}, \lambda_1^{(t)}, \lambda_2^{(t)}).$$

Compute the posterior probability that observation i came from Machine 1:

$$\tau_i^{(t)} = \mathbb{P}(Z_i = 1 \mid Y_i = y_i, \theta^{(t)}) = \frac{\pi^{(t)} f_1(y_i; \lambda_1^{(t)})}{\pi^{(t)} f_1(y_i; \lambda_1^{(t)}) + (1 - \pi^{(t)}) f_2(y_i; \lambda_2^{(t)})}.$$

Then

$$\mathbb{E}[Z_{i1} \mid y_i, \theta^{(t)}] = \tau_i^{(t)}, \qquad \mathbb{E}[Z_{i2} \mid y_i, \theta^{(t)}] = 1 - \tau_i^{(t)}.$$

The M-step uses the complete-data MLE formulas, but replaces missing labels by responsibilities.

Thus

$$\pi^{(t+1)} = \frac{1}{n}\sum_{i=1}^{n} \tau_i^{(t)},$$

$$\lambda_1^{(t+1)} = \frac{\sum_{i=1}^{n} \tau_i^{(t)} y_i}{\sum_{i=1}^{n} \tau_i^{(t)}}, \qquad \lambda_2^{(t+1)} = \frac{\sum_{i=1}^{n}(1-\tau_i^{(t)}) y_i}{\sum_{i=1}^{n}(1-\tau_i^{(t)})}.$$

The update is exactly what we would do with known labels, except that each observation receives fractional membership in the two machines.

One numerical EM iteration

Suppose the observed defect counts are

$$0, \quad 1, \quad 2, \quad 5, \quad 6,$$

and start with

$$\pi^{(0)} = 0.5, \qquad \lambda_1^{(0)} = 1, \qquad \lambda_2^{(0)} = 5.$$

The E-step gives approximately:

y_i	0	1	2	5	6
$\tau_i^{(0)}$	0.982	0.916	0.686	0.017	0.003

Small counts are assigned mostly to Machine 1; large counts are assigned mostly to Machine 2.

Completing the numerical update

Using the values from the previous slide,

$$\sum_i \tau_i^{(0)} \approx 2.605, \qquad \sum_i \tau_i^{(0)} y_i \approx 2.395.$$

Therefore

$$\pi^{(1)} \approx \frac{2.605}{5} = 0.521,$$

$$\lambda_1^{(1)} \approx \frac{2.395}{2.605} = 0.919, \qquad \lambda_2^{(1)} \approx 4.845.$$

Starting from

$$(\pi^{(0)}, \lambda_1^{(0)}, \lambda_2^{(0)}) = (0.500, 1.000, 5.000),$$

one EM iteration gave

$$(\pi^{(1)}, \lambda_1^{(1)}, \lambda_2^{(1)}) \approx (0.521, 0.919, 4.845).$$

Repeating the same E-step and M-step once more gives

$$(\pi^{(2)}, \lambda_1^{(2)}, \lambda_2^{(2)}) \approx (0.515, 0.903, 4.816),$$

and a third iteration gives

$$(\pi^{(3)}, \lambda_1^{(3)}, \lambda_2^{(3)}) \approx (0.511, 0.894, 4.793).$$

Estimates and likelihood values over iterations

For this example, the observed-data log-likelihood is

$$\ell(\pi, \lambda_1, \lambda_2) = \sum_{i=1}^{5} \log \left\{ \pi f(y_i; \lambda_1) + (1-\pi) f(y_i; \lambda_2) \right\}.$$

Iteration	π	λ_1	λ_2	$\ell(\theta)$
0, start	0.500	1.000	5.000	-10.318
1	0.521	0.919	4.845	-10.298
2	0.515	0.903	4.816	-10.296
3	0.511	0.894	4.793	-10.295

The log-likelihood becomes less negative, hence larger, after each EM iteration shown here.

What the motivating example teaches

This example already contains the full EM logic:

1. The observed likelihood is hard because latent labels have been summed out.
2. The complete-data likelihood would be easy if the labels were known.
3. The E-step computes conditional expectations of the missing labels.
4. The M-step maximizes the expected complete-data log-likelihood.
5. Repeating the two steps produces a sequence of likelihood-improving estimates.

The rest of the lecture formalizes this pattern.

Finite-mixture EM examples are treated in McLachlan and Krishnan (2008), Ch. 1.

Complete-data versus observed-data likelihood

Let Z denote all unobserved or latent quantities.

The complete-data likelihood is

$$L_c(\theta; y, z) = p(y, z \mid \theta).$$

The observed-data likelihood is obtained by eliminating Z:

$$L(\theta; y) = p(y \mid \theta) = \sum_z p(y, z \mid \theta)$$

for discrete Z, or

$$L(\theta; y) = \int p(y, z \mid \theta)\, dz$$

for continuous Z.

See Little and Rubin (2019), Ch. 8; McLachlan and Krishnan (2008), Ch. 1.

The difficulty is computational and algebraic:

- **Marginalization creates sums or integrals.** These may not simplify analytically.
- **A log of a sum is awkward.** Even if $p(y, z \mid \theta)$ factors nicely, $\log \sum_z p(y, z \mid \theta)$ usually does not inherit that simple factorization.
- **Direct derivatives may become nonlinear.** Closed-form MLEs can disappear even when complete-data MLEs are easy.

EM does not change the likelihood we want to maximize. It changes *how* we maximize it.

The EM idea in one sentence

Given a current parameter value $\theta^{(t)}$, replace the missing complete-data information by its conditional expectation under $\theta^{(t)}$, then maximize the resulting expected complete-data log-likelihood to obtain $\theta^{(t+1)}$.

Symbolically:

$$\theta^{(t)} \xrightarrow{\text{E-step}} Q(\theta \mid \theta^{(t)}) \xrightarrow{\text{M-step}} \theta^{(t+1)}.$$

This is repeated until the parameter estimates or likelihood values stabilize.

Notation for the algorithm

We write:

- Y: observed data,
- Z: missing or latent data,
- θ: parameter vector,
- $\ell_c(\theta; Y, Z) = \log p(Y, Z \mid \theta)$: complete-data log-likelihood,
- $\ell(\theta; Y) = \log p(Y \mid \theta)$: observed-data log-likelihood.

At iteration t, the current estimate is $\theta^{(t)}$.

> The distinction between ℓ_c and ℓ is essential: EM works with the first to improve the second.

The central object: the Q-function

Define

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}[\ell_c(\theta; Y, Z) \mid Y] .$$

Interpretation:

- Treat θ inside ℓ_c as the parameter we want to optimize.
- Use the current value $\theta^{(t)}$ to determine the conditional distribution of $Z \mid Y$.
- Average the complete-data log-likelihood over that conditional distribution.

The two appearances of θ play different roles: θ is the candidate parameter; $\theta^{(t)}$ defines the expectation.

E-step: what exactly is computed?

Expectation step:

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}[\ell_c(\theta; Y, Z) \mid Y].$$

In practice, this often reduces to computing conditional expectations such as

$$\mathbb{E}[Z_i \mid Y, \theta^{(t)}], \qquad \mathbb{E}[Z_i^2 \mid Y, \theta^{(t)}],$$

or posterior class probabilities

$$\mathbb{P}(Z_i = k \mid Y_i, \theta^{(t)}).$$

The E-step does *not* necessarily fill in one value for each missing quantity. It averages over uncertainty in the missing quantities.

M-step: what exactly is maximized?

Maximization step:

$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta \mid \theta^{(t)}).$$

The M-step treats the expectations computed in the E-step as fixed quantities.

- If the complete-data MLE has a closed form, the M-step often inherits a simple update.
- If exact maximization is difficult, a generalized EM step may merely increase Q; we will discuss that later in the course.

EM and generalized EM: Dempster et al. (1977); McLachlan and Krishnan (2008).

Algorithm

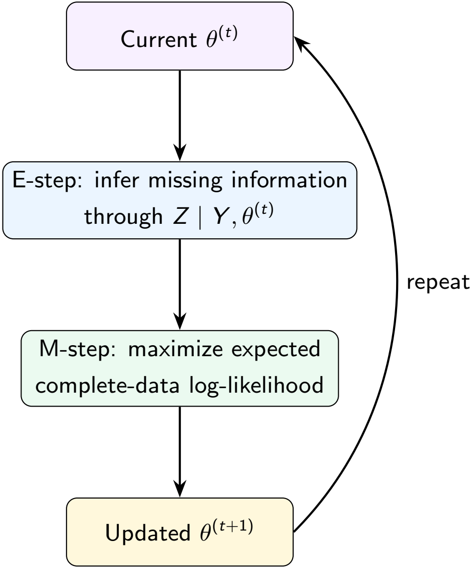
1. Choose a starting value $\theta^{(0)}$.
2. For $t = 0, 1, 2, \ldots$:
   - E: Compute $Q(\theta \mid \theta^{(t)})$.
   - M: Set
$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta \mid \theta^{(t)}).$$
3. Stop when a convergence criterion is satisfied.

Typical stopping rules use changes in

$$\|\theta^{(t+1)} - \theta^{(t)}\|, \qquad \ell(\theta^{(t+1)}) - \ell(\theta^{(t)}),$$

or both.

EM as a feedback loop



We have defined an algorithm, but why should it help?

> If we maximize $Q(\theta \mid \theta^{(t)})$, why should the *observed-data* log-likelihood $\ell(\theta; Y)$ increase?

The answer comes from a lower-bound argument based on Jensen's inequality.

We will use the discrete latent-variable case for clarity; the same idea extends to integrals.

Jensen's Inequality

Because $\log x$ is concave,

$$\log \mathbb{E}(X) \geq \mathbb{E}(\log X)$$

for a positive random variable X.

Equivalently, for probabilities r_z with $\sum_z r_z = 1$,

$$\log\left(\sum_z r_z a_z\right) \geq \sum_z r_z \log a_z.$$

Jensen lets us replace a difficult log of a weighted sum by a weighted average of simpler log terms — at the cost of obtaining a lower bound.

Constructing the EM lower bound

Let

$$r_z^{(t)} = \mathbb{P}(Z = z \mid Y = y, \theta^{(t)}).$$

Then

$$\begin{aligned}
\ell(\theta; y) = \log \sum_z p(y, z \mid \theta) &= \log \sum_z r_z^{(t)} \frac{p(y, z \mid \theta)}{r_z^{(t)}} \\
&\geq \sum_z r_z^{(t)} \log \frac{p(y, z \mid \theta)}{r_z^{(t)}} \quad \text{(using Jensen)} \\
&= \sum_z r_z^{(t)} \log p(y, z \mid \theta) - \sum_z r_z^{(t)} \log r_z^{(t)}. \\
&= Q(\theta \mid \theta^{(t)}) - [\text{a quantity not depending on } \theta].
\end{aligned}$$

Therefore maximizing Q maximizes a lower bound to the observed-data log-likelihood, with the bound fixed during that M-step.

Theorem: Ascent Property of EM Algorithm

Theorem (Ascent Property of EM Algorithm)

Suppose the conditional distribution of $Z \mid Y = y, \theta^{(t)}$ is well-defined and the M-step is exact, that is,

$$Q(\theta^{(t+1)} \mid \theta^{(t)}) \geq Q(\theta^{(t)} \mid \theta^{(t)}).$$

Then the observed-data log-likelihood is nondecreasing:

$$\ell(\theta^{(t+1)}; y) \geq \ell(\theta^{(t)}; y).$$

This ascent property is the usual monotonicity statement for EM: the observed-data likelihood does not decrease from one exact EM iteration to the next.

See McLachlan and Krishnan (2008), Ch. 3, section 3.2.

Why is the bound tight at the current iterate?

At $\theta = \theta^{(t)}$,

$$\frac{p(y, z \mid \theta^{(t)})}{r_z^{(t)}} = \frac{p(y, z \mid \theta^{(t)})}{p(z \mid y, \theta^{(t)})} = p(y \mid \theta^{(t)}),$$

which does not depend on z.

Therefore Jensen's inequality becomes an equality at the current parameter value.

The lower bound touches the log-likelihood at $\theta^{(t)}$.

This "touch and improve" picture is closely related to the MM principle that we will study later.

Convergence Statement I

Theorem (Convergence of likelihood values)

Under standard regularity conditions, if the observed-data likelihood is bounded above and exact EM iterations are performed, then the sequence

$$\ell(\theta^{(0)}; y),\ \ell(\theta^{(1)}; y),\ \ell(\theta^{(2)}; y), \dots$$

is nondecreasing and converges to a finite limit.

The ascent property gives monotonicity; boundedness of the likelihood turns monotonicity into convergence of the likelihood values.

Compare Section 3.4 of McLachlan and Krishnan (2008); see also Wu (1983).

Convergence Statement II

Theorem (Limit points are stationary, informally stated)

Under suitable smoothness, compactness, and identifiability-type conditions, every limit point of an EM sequence is a stationary point of the observed-data likelihood. That is, if $\theta^{(t_j)} \to \theta^*$*, then typically*

$$\nabla \ell(\theta^*; y) = 0,$$

or the appropriate constrained/boundary version of this condition holds.

Stationary point does not mean global maximum. It may be a local maximum, saddle point, or boundary point in irregular problems.

See Section 3.5 of McLachlan and Krishnan (2008); Wu (1983).

What monotonicity does not guarantee

Likelihood ascent is valuable, but it is not the same as global optimality.

- EM can converge to a local maximum.
- It can converge to a saddle point or boundary point in irregular problems.
- Different starting values can lead to different solutions.
- Monotonicity says nothing about speed; EM can be slow near the optimum.

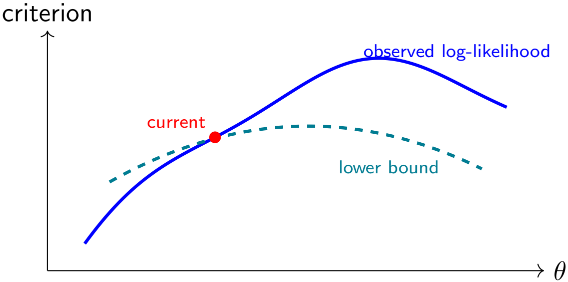
Convergence theory: Wu (1983); McLachlan and Krishnan (2008), Ch. 3.

A geometric picture of EM

Imagine the observed log-likelihood as a difficult curve.

At iteration t:

1. E-step constructs a tractable lower bound touching the likelihood at $\theta^{(t)}$.
2. M-step moves to the maximizer of that bound.
3. Repeating the construction creates a sequence of nondecreasing likelihood values.



An example: Poisson data with missing counts

Suppose

$$X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Poisson}(\lambda),$$

but only the first r counts are observed.

The remaining $m = n - r$ counts are missing.

Let

$$S_{\mathrm{obs}} = \sum_{i=1}^{r} X_i.$$

> How would EM estimate λ if we deliberately treat the unobserved counts as missing data?

This example is intentionally simple: it lets us see every EM step algebraically.

If all n counts were observed,

$$\ell_c(\lambda) = \sum_{i=1}^{n} \left[X_i \log \lambda - \lambda - \log(X_i!)\right].$$

Ignoring terms that do not involve λ,

$$\ell_c(\lambda) = \left(\sum_{i=1}^{n} X_i\right) \log \lambda - n\lambda + C.$$

The complete-data MLE is therefore

$$\widehat{\lambda}_{\text{comp}} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

> The only missing complete-data sufficient statistic is the sum of the missing counts.

E-step for the Poisson example

At iteration t, each missing count has conditional expectation

$$\mathbb{E}[X_i \mid X_{\text{obs}}, \lambda^{(t)}] = \lambda^{(t)},$$

because the counts are independent.

Hence the expected missing total is

$$\mathbb{E}\left[\sum_{i=r+1}^{n} X_i \middle| X_{\text{obs}}, \lambda^{(t)}\right] = m\lambda^{(t)}.$$

So the E-step replaces the missing sufficient statistic by

$$S_{\text{mis}}^{(t)} = m\lambda^{(t)}.$$

M-step for the Poisson example

The complete-data MLE formula was the full sample mean.

Replace the missing total by its E-step expectation:

$$\lambda^{(t+1)} = \frac{S_{\text{obs}} + m\lambda^{(t)}}{n}.$$

Thus EM produces the update

$$\boxed{\lambda^{(t+1)} = \frac{S_{\text{obs}}}{n} + \frac{m}{n}\lambda^{(t)}}.$$

Numerical iteration

Suppose

$$n = 10, \qquad r = 6, \qquad m = 4, \qquad S_{\text{obs}} = 18.$$

Start with $\lambda^{(0)} = 5$.

Then

$$\lambda^{(1)} = \frac{18 + 4(5)}{10} = 3.8,$$

$$\lambda^{(2)} = \frac{18 + 4(3.8)}{10} = 3.32,$$

$$\lambda^{(3)} = \frac{18 + 4(3.32)}{10} = 3.128.$$

The sequence keeps moving toward 3.

What is the limiting estimate?

At a fixed point λ^*,

$$\lambda^* = \frac{S_{\text{obs}} + m\lambda^*}{n}.$$

Therefore

$$(n - m)\lambda^* = S_{\text{obs}},$$

so

$$\boxed{\lambda^* = \frac{S_{\text{obs}}}{r}}.$$

That is simply the mean of the observed Poisson counts.

The missing counts contain no additional information about λ here, so EM converges to the observed-data MLE.

An useful strategy

This problem is not where EM is needed computationally; direct maximization is easier.

But it cleanly reveals the architecture:

1. Identify missing sufficient statistics.
2. Compute their conditional expectations under current parameters.
3. Plug those expectations into the complete-data estimation formula.
4. Iterate.

In more interesting models, exactly the same logic survives even when the algebra becomes much richer.

Choosing starting values

The starting value $\theta^{(0)}$ matters because the likelihood may have multiple stationary points.

Practical strategies include:

- method-of-moments or simple preliminary estimates;
- random starts;
- clustering-based initialization for mixture models;
- several dispersed starting values, retaining the best final likelihood.

A single EM run from a single arbitrary start can create false confidence.

Monitoring convergence

Useful diagnostics include:

- plot $\ell(\theta^{(t)})$ against iteration;
- monitor $\|\theta^{(t+1)} - \theta^{(t)}\|$;
- inspect whether different starts converge to the same solution;
- check for parameters approaching boundaries or degeneracy.

A common rule is

$$\frac{|\ell^{(t+1)} - \ell^{(t)}|}{1 + |\ell^{(t)}|} < \varepsilon.$$

Common misconceptions

1. **“E-step means replacing each missing value by its mean.”**
   Not always. It means taking the conditional expectation of the *complete-data log-likelihood* or the sufficient quantities appearing in it.
2. **“EM maximizes the complete-data likelihood.”**
   No. Its target is the observed-data likelihood.
3. **“Monotone likelihood means global maximum.”**
   No. Local maxima remain possible.
4. **“EM automatically gives standard errors.”**
   No. Additional calculations are needed for uncertainty quantification.

Some exercises

For each statement, decide whether it is correct.

1. The E-step chooses the most likely value of every missing quantity.
2. The M-step maximizes $Q(\theta \mid \theta^{(t)})$, not necessarily the observed log-likelihood directly.
3. An exact EM iteration can decrease the observed-data likelihood slightly because of numerical noise in theory.
4. Different starting values may lead to different EM solutions.
5. EM is useful only when literal database entries are missing.

Try to justify each answer in one sentence.

Answers

1. **False.** The E-step averages over the conditional distribution of missing/latent quantities; it does not generally make hard assignments.
2. **True.** The M-step optimizes the expected complete-data log-likelihood Q, which is constructed to improve the observed likelihood.
3. **False in exact theory.** Exact EM has nondecreasing observed likelihood; numerical implementations can show tiny violations due to approximation or roundoff.
4. **True.** Nonconcave likelihoods may have multiple local maxima or stationary points.
5. **False.** Latent labels, hidden states, random effects, and other unobserved quantities also create EM problems.

Key Takeaway Messages

1. EM is designed for likelihood problems with missing or latent information.
2. The observed-data likelihood is obtained by summing/integrating over unobserved quantities and may be harder to optimize than the complete-data likelihood.
3. The E-step computes
$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}[\ell_c(\theta; Y, Z) \mid Y].$$
4. The M-step maximizes Q to obtain $\theta^{(t+1)}$.
5. Jensen's inequality explains the lower-bound construction and the nondecreasing likelihood property.
6. Monotone ascent does not guarantee a global maximum or fast convergence.
7. EM is best understood as exploiting a simpler complete-data problem to solve a harder observed-data optimization problem.

Suggested reading after Lecture IV

For first-principles understanding:

- McLachlan and Krishnan (2008), Chapters 1–3: systematic treatment of EM, likelihood ascent, convergence, and examples.
- Little and Rubin (2019), Chapter 8: likelihood methods with missing data and the EM algorithm.
- Dempster et al. (1977): the foundational EM paper; especially the general framework and monotonicity idea.

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Advanced Statistical Methods II
Lecture V: EM for Normal Mixtures

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Outline of Today's Lecture

Recap and objective

Why normal mixtures?

Observed and complete data likelihoods

E-step: responsibilities

M-step: update formulas

Practical issues

Key takeaway messages

A problem

Summary of Lecture IV

In Lecture IV, we introduced EM through the incomplete-data likelihood viewpoint.

- Observed data: Y.
- Missing or latent data: Z.
- Complete-data log-likelihood:
$$\ell_c(\theta; Y, Z) = \log p(Y, Z \mid \theta).$$
- Observed-data log-likelihood:
$$\ell(\theta; Y) = \log p(Y \mid \theta) = \log \sum_z p(Y, z \mid \theta),$$
with an integral instead of a sum when Z is continuous.

EM replaces the missing complete-data information by its conditional expectation under the current parameter value.

See Dempster et al. (1977); McLachlan and Krishnan (2008), Chs. 1–3.

The EM template from Lecture IV

At iteration t, define

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}\left[\ell_c(\theta; Y, Z) \mid Y\right].$$

E-step

Compute the conditional distribution of the missing information:

$$p(Z \mid Y, \theta^{(t)}),$$

and hence compute $Q(\theta \mid \theta^{(t)})$.

M-step

Update

$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta \mid \theta^{(t)}).$$

Today's Goal

How do we fit a model when the population is made of several hidden subpopulations, but we do not observe the subpopulation labels?

For mixtures, EM is a standard way to turn hidden subgroup membership into an estimable likelihood problem.

See McLachlan and Peel (2000); McLachlan and Krishnan (2008).

Finite Mixtures

A finite mixture model can be described by two stages.

For each observation $i = 1, \dots, n$:

1. Choose a hidden component label

$$Z_i \in \{1, \dots, K\}, \qquad \mathbb{P}(Z_i = k) = \pi_k.$$

2. Generate the response from the component-specific distribution

$$Y_i \mid Z_i = k \sim f_k(y; \theta_k).$$

The observed density is then

$$f(y; \theta) = \sum_{k=1}^{K} \pi_k f_k(y; \theta_k), \qquad \pi_k > 0, \quad \sum_{k=1}^{K} \pi_k = 1.$$

Normal mixture model

In a **normal mixture**, the components are normal distributions.

For univariate data:

$$Y_i \mid Z_i = k \sim \mathcal{N}(\mu_k, \sigma_k^2), \qquad k = 1, \ldots, K.$$

The marginal density of Y_i is

$$f(y_i; \theta) = \sum_{k=1}^{K} \pi_k \, \phi(y_i; \mu_k, \sigma_k^2),$$

where $\phi(y; \mu, \sigma^2)$ denotes the $\mathcal{N}(\mu, \sigma^2)$ density.

The parameter vector is

$$\theta = (\pi_1, \ldots, \pi_K, \mu_1, \ldots, \mu_K, \sigma_1^2, \ldots, \sigma_K^2).$$

Mixture density: intuition

For $K = 2$,

$$f(y;\theta) = \pi\phi(y;\mu_1,\sigma_1^2) + (1-\pi)\phi(y;\mu_2,\sigma_2^2).$$

- π controls the relative size of component 1.
- μ_1, μ_2 control the centers of the two components.
- σ_1^2, σ_2^2 control the within-component spreads.

A mixture can produce skewness, multimodality and heavy-tailed-looking shapes even when every component is normal.

The word *mixture* means that each observation has one hidden component label, but we do not observe it.

Notations

We follow the same notation as Lecture IV.

Symbol	Meaning
Y_i	observed response for unit i
Z_i	unobserved component label
Z_{ik}	indicator $\mathbb{I}(Z_i = k)$
θ	full parameter vector
$\theta^{(t)}$	current parameter value in EM iteration t
$\tau_{ik}^{(t)}$	posterior responsibility of component k for observation i

The main new object today is

$$\tau_{ik}^{(t)} = \mathbb{P}_{\theta^{(t)}}(Z_i = k \mid Y_i = y_i),$$

called the **responsibility**.

The observed-data likelihood

We observe only $y_1, \ldots, y_n$, not $z_1, \ldots, z_n$.

The observed-data likelihood is

$$L(\theta) = \prod_{i=1}^{n} \left\{ \sum_{k=1}^{K} \pi_k \phi(y_i; \mu_k, \sigma_k^2) \right\}.$$

Therefore the observed-data log-likelihood is

$$\ell(\theta) = \sum_{i=1}^{n} \log \left\{ \sum_{k=1}^{K} \pi_k \phi(y_i; \mu_k, \sigma_k^2) \right\}.$$

The log of a sum is the source of computational difficulty. It prevents the likelihood from separating cleanly by component.

Why direct maximization is awkward

If the component labels were known, each observation would belong to one normal sample.

But without labels,

$$\ell(\theta) = \sum_{i=1}^{n} \log \left\{ \pi_1 \phi_{i1} + \cdots + \pi_K \phi_{iK} \right\}.$$

This creates several difficulties:

- the log-likelihood is usually non-concave;
- there may be multiple local maxima;
- closed-form maximizers are generally not available from the observed likelihood directly.

Introduce latent allocation indicators

Define

$$Z_{ik} = \mathbb{I}(Z_i = k), \qquad k = 1, \ldots, K.$$

For each i, exactly one component label is active:

$$Z_{ik} \in \{0, 1\}, \qquad \sum_{k=1}^{K} Z_{ik} = 1.$$

The complete data are

$$\{(y_i, z_{i1}, \ldots, z_{iK}) : i = 1, \ldots, n\}.$$

The missing data are missing **classification labels**.

Complete-data likelihood

Given Z_{ik}'s, the joint density contribution of observation i can be written as

$$\prod_{k=1}^{K} \left[\pi_k \phi(y_i; \mu_k, \sigma_k^2)\right]^{Z_{ik}} .$$

Thus the complete-data likelihood is

$$L_c(\theta) = \prod_{i=1}^{n} \prod_{k=1}^{K} \left[\pi_k \phi(y_i; \mu_k, \sigma_k^2)\right]^{Z_{ik}} .$$

The complete-data log-likelihood is

$$\ell_c(\theta) = \sum_{i=1}^{n} \sum_{k=1}^{K} Z_{ik} \left\{\log \pi_k + \log \phi(y_i; \mu_k, \sigma_k^2)\right\} .$$

Complete-data log-likelihood for normal components

For the univariate normal density,

$$\log \phi(y_i; \mu_k, \sigma_k^2) = -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log \sigma_k^2 - \frac{(y_i - \mu_k)^2}{2\sigma_k^2}.$$

Ignoring the constant $-\frac{n}{2}\log(2\pi)$,

$$\ell_c(\theta) = \sum_{i=1}^{n}\sum_{k=1}^{K} Z_{ik}\left[\log \pi_k - \frac{1}{2}\log \sigma_k^2 - \frac{(y_i - \mu_k)^2}{2\sigma_k^2}\right] + \text{constant}.$$

This expression is easy to maximize if we know the labels Z_{ik}. EM will replace Z_{ik} by its conditional expectation.

If labels were known: complete-data MLEs

If Z_{ik}'s were observed, define

$$n_k = \sum_{i=1}^{n} Z_{ik}.$$

The complete-data MLEs would be

$$\widehat{\pi}_k = \frac{n_k}{n},$$

$$\widehat{\mu}_k = \frac{\sum_{i=1}^n Z_{ik} y_i}{\sum_{i=1}^n Z_{ik}},$$

$$\widehat{\sigma}_k^2 = \frac{\sum_{i=1}^n Z_{ik} (y_i - \widehat{\mu}_k)^2}{\sum_{i=1}^n Z_{ik}}.$$

The M-step will have exactly the same form, except that hard labels Z_{ik} will be replaced by soft labels $\tau_{ik}^{(t)}$.

The part of ℓ_c involving $\pi_1, \ldots, \pi_K$ is

$$\ell_{c,\pi} = \sum_{k=1}^{K} n_k \log \pi_k, \qquad n_k = \sum_{i=1}^{n} Z_{ik},$$

subject to the constraint

$$\sum_{k=1}^{K} \pi_k = 1.$$

Use the Lagrangian

$$\mathcal{L}(\pi_1, \ldots, \pi_K, \lambda) = \sum_{k=1}^{K} n_k \log \pi_k + \lambda \left(\sum_{k=1}^{K} \pi_k - 1 \right).$$

$$\implies \frac{\partial \mathcal{L}}{\partial \pi_k} = \frac{n_k}{\pi_k} + \lambda = 0 \quad \implies \quad \pi_k = -\frac{n_k}{\lambda}.$$

Summing over k,

$$1 = \sum_{k=1}^{K} \pi_k = -\frac{1}{\lambda} \sum_{k=1}^{K} n_k = -\frac{n}{\lambda},$$

Derivation II: component means

For a fixed component k, the terms involving μ_k are

$$-\frac{1}{2\sigma_k^2}\sum_{i=1}^{n} Z_{ik}(y_i-\mu_k)^2.$$

Thus maximizing with respect to μ_k is equivalent to minimizing

$$S_k(\mu_k)=\sum_{i=1}^{n} Z_{ik}(y_i-\mu_k)^2.$$

Differentiating and setting the derivative equal to zero gives:

$$\sum_{i=1}^{n} Z_{ik}(y_i-\widehat{\mu}_k)=0.$$

Therefore

$$\boxed{\widehat{\mu}_k=\frac{\sum_{i=1}^{n} Z_{ik}y_i}{\sum_{i=1}^{n} Z_{ik}}}=\frac{1}{n_k}\sum_{i:Z_i=k} y_i.$$

Derivation III: component variances

Let $v_k = \sigma_k^2$. For a fixed k, after substituting $\widehat{\mu}_k$, focus on

$$\begin{aligned} g(v_k) &= -\frac{n_k}{2}\log v_k - \frac{1}{2v_k}\sum_{i=1}^{n} Z_{ik}(y_i - \widehat{\mu}_k)^2 \\ &= -\frac{n_k}{2}\log v_k - \frac{A_k}{2v_k} \\ \implies g'(v_k) &= -\frac{n_k}{2v_k} + \frac{A_k}{2v_k^2}. \end{aligned}$$

Set $g'(v_k) = 0$:

$$-n_k v_k + A_k = 0 \quad \implies \quad \boxed{\widehat{\sigma}_k^2 = \widehat{v}_k = \frac{A_k}{n_k}} = \frac{\sum_{i=1}^{n} Z_{ik}(y_i - \widehat{\mu}_k)^2}{\sum_{i=1}^{n} Z_{ik}}.$$

What must be computed in the E-step?

The E-step computes

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}[\ell_c(\theta; Y, Z) \mid Y].$$

In the complete-data log-likelihood, the only random missing quantities are the indicators Z_{ik}.

So we need

$$\mathbb{E}_{\theta^{(t)}}[Z_{ik} \mid Y_i = y_i].$$

Since Z_{ik} is an indicator,

$$\mathbb{E}_{\theta^{(t)}}[Z_{ik} \mid Y_i = y_i] = \mathbb{P}_{\theta^{(t)}}(Z_i = k \mid Y_i = y_i).$$

This conditional probability is called the **responsibility**:

$$\tau_{ik}^{(t)} = \mathbb{P}_{\theta^{(t)}}(Z_i = k \mid Y_i = y_i).$$

Derivation of the responsibility formula

By Bayes' rule,

$$\mathbb{P}(Z_i = k \mid Y_i = y_i, \theta^{(t)}) = \frac{\mathbb{P}(Z_i = k \mid \theta^{(t)}) f(y_i \mid Z_i = k, \theta^{(t)})}{f(y_i \mid \theta^{(t)})}.$$

Under the mixture model,

$$\mathbb{P}(Z_i = k \mid \theta^{(t)}) = \pi_k^{(t)},$$

$$f(y_i \mid Z_i = k, \theta^{(t)}) = \phi(y_i; \mu_k^{(t)}, (\sigma_k^2)^{(t)}),$$

and

$$f(y_i \mid \theta^{(t)}) = \sum_{\ell=1}^{K} \pi_\ell^{(t)} \phi(y_i; \mu_\ell^{(t)}, (\sigma_\ell^2)^{(t)}).$$

E-step formula

Therefore,

$$\boxed{\tau_{ik}^{(t)} = \frac{\pi_k^{(t)}\phi(y_i; \mu_k^{(t)}, (\sigma_k^2)^{(t)})}{\sum_{\ell=1}^{K} \pi_\ell^{(t)}\phi(y_i; \mu_\ell^{(t)}, (\sigma_\ell^2)^{(t)})}}$$

for $i = 1, \ldots, n$ and $k = 1, \ldots, K$.

$\tau_{ik}^{(t)}$ measures how much component k is responsible for observation i, given the current parameter estimates.

For each observation,

$$0 \le \tau_{ik}^{(t)} \le 1, \qquad \sum_{k=1}^{K} \tau_{ik}^{(t)} = 1.$$

Responsibilities are soft labels

A hard clustering method might assign

$$Y_i \longrightarrow \text{one component only.}$$

EM instead assigns probabilities:

$$Y_i \longrightarrow (\tau_{i1}^{(t)}, \dots, \tau_{iK}^{(t)}).$$

Observation	Hard label	EM responsibility vector
clearly near component 1	1	approximately $(1, 0, \dots, 0)$
overlap region	ambiguous	fractional probabilities
clearly near component 2	2	approximately $(0, 1, 0, \dots, 0)$

This is why EM for mixtures is often described as **soft clustering**.

The Q-function for normal mixtures

Replace Z_{ik} in the complete-data log-likelihood by $\tau_{ik}^{(t)}$.

Thus,

$$Q(\theta \mid \theta^{(t)}) = \sum_{i=1}^{n} \sum_{k=1}^{K} \tau_{ik}^{(t)} \left[\log \pi_k - \frac{1}{2} \log \sigma_k^2 - \frac{(y_i - \mu_k)^2}{2\sigma_k^2} \right] + \text{constant}.$$

The E-step has converted the missing-label problem into a weighted normal likelihood problem.

The weights are not arbitrary; they are posterior probabilities under the current parameters.

M-step overview

The M-step maximizes

$$Q(\theta \mid \theta^{(t)})$$

with respect to

$$\pi_1, \ldots, \pi_K, \quad \mu_1, \ldots, \mu_K, \quad \sigma_1^2, \ldots, \sigma_K^2.$$

The resulting updates are:

$$\pi_k^{(t+1)} = \frac{1}{n} \sum_{i=1}^{n} \tau_{ik}^{(t)},$$

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^{n} \tau_{ik}^{(t)} y_i}{\sum_{i=1}^{n} \tau_{ik}^{(t)}},$$

$$(\sigma_k^2)^{(t+1)} = \frac{\sum_{i=1}^{n} \tau_{ik}^{(t)} (y_i - \mu_k^{(t+1)})^2}{\sum_{i=1}^{n} \tau_{ik}^{(t)}}.$$

Deriving the update for mixing proportions

The part of Q involving π_k's is

$$Q_\pi = \sum_{i=1}^{n} \sum_{k=1}^{K} \tau_{ik}^{(t)} \log \pi_k, \qquad \sum_{k=1}^{K} \pi_k = 1.$$

Use a Lagrange multiplier:

$$\mathcal{L} = Q_\pi + \lambda \left(\sum_{k=1}^{K} \pi_k - 1 \right).$$

Differentiating,

$$\frac{\partial \mathcal{L}}{\partial \pi_k} = \frac{\sum_{i=1}^{n} \tau_{ik}^{(t)}}{\pi_k} + \lambda = 0.$$

Thus,

$$\pi_k = -\frac{\sum_i \tau_{ik}^{(t)}}{\lambda}.$$

Completing the derivation for π_k

Use the constraint $\sum_k \pi_k = 1$:

$$\sum_{k=1}^{K} \left(-\frac{\sum_i \tau_{ik}^{(t)}}{\lambda} \right) = 1.$$

Since

$$\sum_{k=1}^{K} \sum_{i=1}^{n} \tau_{ik}^{(t)} = \sum_{i=1}^{n} \sum_{k=1}^{K} \tau_{ik}^{(t)} = n,$$

we get $-n/\lambda = 1$, hence $\lambda = -n$.

Therefore,

$$\boxed{\pi_k^{(t+1)} = \frac{1}{n} \sum_{i=1}^{n} \tau_{ik}^{(t)}}$$

The updated mixing proportion is the average responsibility of component k.

Deriving the update for μ_k

For a fixed component k, the part of Q involving μ_k is

$$-\frac{1}{2\sigma_k^2}\sum_{i=1}^{n}\tau_{ik}^{(t)}(y_i-\mu_k)^2.$$

Maximizing this is equivalent to minimizing

$$\sum_{i=1}^{n}\tau_{ik}^{(t)}(y_i-\mu_k)^2.$$

Differentiate with respect to μ_k:

$$\frac{\partial}{\partial\mu_k}\sum_{i=1}^{n}\tau_{ik}^{(t)}(y_i-\mu_k)^2=-2\sum_{i=1}^{n}\tau_{ik}^{(t)}(y_i-\mu_k).$$

Setting this equal to zero gives

$$\boxed{\mu_k^{(t+1)}=\frac{\sum_{i=1}^{n}\tau_{ik}^{(t)}y_i}{\sum_{i=1}^{n}\tau_{ik}^{(t)}}}$$

For a fixed component k, the part involving σ_k^2 is

$$-\frac{1}{2}\sum_{i=1}^{n}\tau_{ik}^{(t)}\left[\log\sigma_k^2+\frac{(y_i-\mu_k)^2}{\sigma_k^2}\right].$$

Let $v_k=\sigma_k^2$. Differentiate with respect to v_k:

$$\frac{\partial Q}{\partial v_k}=-\frac{1}{2}\sum_{i=1}^{n}\tau_{ik}^{(t)}\left[\frac{1}{v_k}-\frac{(y_i-\mu_k)^2}{v_k^2}\right].$$

Set the derivative equal to zero:

$$\sum_{i=1}^{n}\tau_{ik}^{(t)}v_k=\sum_{i=1}^{n}\tau_{ik}^{(t)}(y_i-\mu_k)^2.$$

Variance update

Using the updated mean $\mu_k^{(t+1)}$, we get

$$\boxed{(\sigma_k^2)^{(t+1)} = \frac{\sum_{i=1}^n \tau_{ik}^{(t)}(y_i - \mu_k^{(t+1)})^2}{\sum_{i=1}^n \tau_{ik}^{(t)}}}$$

The M-step is simply a weighted version of the usual MLEs for normal data.

Quantity	Complete-data MLE	EM M-step
sample size in component k	$\sum_i Z_{ik}$	$\sum_i \tau_{ik}^{(t)}$
component mean	hard average	weighted average
component variance	hard variance	weighted variance

The EM algorithm for univariate normal mixtures

Algorithm

Choose starting values

$$\theta^{(0)} = (\pi_k^{(0)}, \mu_k^{(0)}, (\sigma_k^2)^{(0)})_{k=1}^K.$$

For $t = 0, 1, 2, \ldots$:

1. **E-step:** compute responsibilities $\tau_{ik}^{(t)}$.
2. **M-step:** update π_k, μ_k, σ_k^2 using weighted MLE formulas.
3. Compute the observed log-likelihood

$$\ell(\theta^{(t)}) = \sum_i \log \left\{ \sum_k \pi_k^{(t)} \phi(y_i; \mu_k^{(t)}, (\sigma_k^2)^{(t)}) \right\}.$$

4. Stop when the parameter changes or likelihood improvement is small.

Initialization matters

EM is an ascent algorithm, but the normal-mixture likelihood may have several local maxima.

Starting values can be chosen using:

- random initial allocations;
- quantile-based splitting of the data;
- several random starts followed by selecting the solution with largest observed log-likelihood.

See McLachlan and Peel (2000), Chs. 2–3.

Choosing the number of components

So far, we have treated K as known.

In practice, K may be chosen using:

- scientific knowledge about the number of subpopulations;
- likelihood-based criteria such as AIC or BIC;
- validation and interpretability.

A larger K can fit the data better, but may overfit or produce components with no meaningful interpretation.

Summary

1. A normal mixture model assumes hidden component labels Z_i.
2. The observed likelihood contains $\log \sum_k$ terms and is hard to maximize directly.
3. The complete-data likelihood is easy because it separates by component.
4. The E-step computes responsibilities

$$\tau_{ik}^{(t)} = \mathbb{P}_{\theta^{(t)}}(Z_i = k \mid Y_i = y_i).$$

5. The M-step computes weighted normal MLEs.

A Question

Suppose it is known

$$\mu_1 = \mu_2 = \mu \quad \& \quad \sigma_1^2 = \sigma_2^2 = \sigma^2.$$

Then what would be the E-step and the M-step?

Bivariate Normal with one missing value

Suppose

$$(x_1, y_1), (x_2, y_2), \ldots, (x_{n-1}, y_{n-1}), (x_n, *)$$

are treated as observations from a Bivariate Normal distribution, where $*$ denotes a missing value and

$$x_n = \bar{x}_{n-1} = \frac{1}{n-1}\sum_{i=1}^{n-1} x_i.$$

Use the MLEs based on the first $n-1$ complete observations as initial values.

Show that the corresponding EM algorithm converges in two cycles of iteration.

Notation and initial values

Let $m = n - 1$. From the first m complete observations define the MLEs

$$\bar{x} = \frac{1}{m}\sum_{i=1}^{m} x_i, \qquad \bar{y} = \frac{1}{m}\sum_{i=1}^{m} y_i,$$

$$s_{xx} = \frac{1}{m}\sum_{i=1}^{m}(x_i - \bar{x})^2, \quad s_{yy} = \frac{1}{m}\sum_{i=1}^{m}(y_i - \bar{y})^2, \quad s_{xy} = \frac{1}{m}\sum_{i=1}^{m}(x_i - \bar{x})(y_i - \bar{y}).$$

The initial values are therefore

$$\mu_x^{(0)} = \bar{x}, \quad \mu_y^{(0)} = \bar{y}, \quad \sigma_{xx}^{(0)} = s_{xx}, \quad \sigma_{yy}^{(0)} = s_{yy}, \quad \sigma_{xy}^{(0)} = s_{xy}.$$

But the problem states that

$$x_n = \bar{x}.$$

E-step in the first cycle

For a bivariate normal distribution,

$$Y \mid X = x \sim N\left(\mu_y + \frac{\sigma_{xy}}{\sigma_{xx}}(x - \mu_x), \sigma_{yy} - \frac{\sigma_{xy}^2}{\sigma_{xx}}\right).$$

At the initial value, $x_n = \mu_x^{(0)} = \bar{x}$. Hence

$$\widehat{y}_n^{(1)} = \mathbb{E}_{\theta^{(0)}}(Y_n \mid X_n = x_n) = \mu_y^{(0)} = \bar{y}.$$

Also,

$$\mathbb{E}_{\theta^{(0)}}(Y_n^2 \mid X_n = x_n) = \bar{y}^2 + \tau^2,$$

where

$$\tau^2 = s_{yy} - \frac{s_{xy}^2}{s_{xx}}.$$

M-step in the first cycle

Using the expected completed data,

$$\mu_x^{(1)} = \frac{m\bar{x} + x_n}{n} = \bar{x}, \qquad \mu_y^{(1)} = \frac{m\bar{y} + \bar{y}}{n} = \bar{y}.$$

For the covariance terms,

$$\sigma_{xx}^{(1)} = \frac{1}{n}\left\{\sum_{i=1}^{m}(x_i - \bar{x})^2 + (x_n - \bar{x})^2\right\} = \frac{m}{n}s_{xx},$$

$$\sigma_{xy}^{(1)} = \frac{1}{n}\left\{\sum_{i=1}^{m}(x_i - \bar{x})(y_i - \bar{y}) + (x_n - \bar{x})\mathbb{E}(Y_n - \bar{y} \mid x_n)\right\} = \frac{m}{n}s_{xy}.$$

For the Y-variance,

$$\sigma_{yy}^{(1)} = \frac{1}{n}\left\{ms_{yy} + \mathbb{E}[(Y_n - \bar{y})^2 \mid x_n]\right\} = \frac{1}{n}\left\{ms_{yy} + s_{yy} - \frac{s_{xy}^2}{s_{xx}}\right\}.$$

Thus

$$\sigma_{yy}^{(1)} = s_{yy} - \frac{1}{n}\frac{s_{xy}^2}{s_{xx}}.$$

Why the next cycle changes nothing

In the second E-step, still

$$x_n = \mu_x^{(1)} = \bar{x},$$

so the conditional mean remains

$$\mathbb{E}_{\theta^{(1)}}(Y_n \mid X_n = x_n) = \mu_y^{(1)} = \bar{y}.$$

The conditional variance under $\theta^{(1)}$ is

$$\sigma_{yy}^{(1)} - \frac{(\sigma_{xy}^{(1)})^2}{\sigma_{xx}^{(1)}}.$$

To prove convergence, it is enough to show that this conditional variance is also the same as in the first E-step.

Why the next cycle changes nothing, continued

Substitute the first-cycle values:

$$\left(s_{yy} - \frac{1}{n}\frac{s_{xy}^2}{s_{xx}}\right) - \frac{\left(\frac{m}{n}s_{xy}\right)^2}{\frac{m}{n}s_{xx}} = \left(s_{yy} - \frac{1}{n}\frac{s_{xy}^2}{s_{xx}}\right) - \frac{m}{n}\frac{s_{xy}^2}{s_{xx}}.$$

Since $m = n - 1$,

$$\frac{1}{n} + \frac{m}{n} = 1.$$

Therefore

$$\sigma_{yy}^{(1)} - \frac{(\sigma_{xy}^{(1)})^2}{\sigma_{xx}^{(1)}} = s_{yy} - \frac{s_{xy}^2}{s_{xx}} = \tau^2.$$

The second E-step gives exactly the same first and second conditional moments as the first E-step. Hence the next M-step reproduces $\theta^{(1)}$.

Problem 1: conclusion

The EM sequence is

$$\theta^{(0)} \quad \longrightarrow \quad \theta^{(1)} \quad \longrightarrow \quad \theta^{(2)} = \theta^{(1)}.$$

Thus the algorithm has reached a fixed point after the first update; equivalently, after checking the second cycle, it has converged in two cycles of iteration.

The result depends crucially on $x_n = \bar{x}$. Because the missing observation is located at the current mean of X, the conditional mean imputation for Y_n remains $\bar{y}$, and the conditional variance stabilizes after the first EM update.

Do not replace Y_n only by its conditional mean when estimating variances. EM uses the conditional second moment, which includes the conditional variance term.

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Advanced Statistical Methods II

Lecture VI: EM for Incomplete Tables and General Missing-Data Likelihoods

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Outline of Today's Lecture

Where we are after Lecture V

In Lecture V, we applied EM to finite normal mixtures.

- Observed data: numerical observations $Y = (Y_1, \ldots, Y_n)$.
- Missing data: latent component labels $Z = (Z_1, \ldots, Z_n)$.
- E-step: compute responsibilities

$$\tau_{ik}^{(t)} = \mathbb{P}_{\theta^{(t)}}(Z_i = k \mid Y_i).$$

- M-step: replace missing indicators by responsibilities and use complete-data MLE formulas.

missing complete-data sufficient statistics are replaced by conditional expectations.

Today's lecture

Today we move from hidden labels in mixtures to incomplete categorical data and general missing-data likelihoods.

> If the observed table is only a partial view of the full table, can we still do likelihood-based estimation as if the table were complete?

What makes a table incomplete?

A categorical table becomes incomplete when we do not know the exact category for every unit.

Examples

- In a contingency table, some records have row variable observed but column variable missing.
- In a classification problem, we may know that an observation belongs to a subset of categories, but not the exact category.

For multinomial models, the complete-data sufficient statistics are cell counts. If some cell memberships are unknown, EM estimates the missing parts of those counts.

Complete multinomial counts

Suppose each unit belongs to one of K categories.

$$\mathbb{P}(\text{category } k) = p_k, \qquad p_k > 0, \qquad \sum_{k=1}^{K} p_k = 1.$$

If the full cell counts are observed,

$$N_k = \#\{\text{units in category } k\}, \qquad \sum_{k=1}^{K} N_k = N,$$

then

$$(N_1, \ldots, N_K) \sim \mathrm{Multinomial}(N; p_1, \ldots, p_K).$$

Ignoring constants, the complete-data log-likelihood is

$$\ell_c(\boldsymbol{p}) = \sum_{k=1}^{K} N_k \log p_k.$$

See, e.g., Agresti (2013), Ch. 1.

Complete-data MLE for the multinomial model

Maximize

$$\sum_{k=1}^{K} N_k \log p_k \quad \text{s.t} \quad \sum_{k=1}^{K} p_k = 1.$$

Set

$$\mathcal{L}(\boldsymbol{p}, \lambda) = \sum_{k=1}^{K} N_k \log p_k + \lambda \left(1 - \sum_{k=1}^{K} p_k \right).$$

Then

$$\frac{\partial \mathcal{L}}{\partial p_k} = \frac{N_k}{p_k} - \lambda = 0 \quad \Rightarrow \quad p_k = \frac{N_k}{\lambda}.$$

Using $\sum_k p_k = 1$, we get $\lambda = N$, hence $\widehat{p}_k = N_k / N$.

A simple incomplete table

Consider three categories with probabilities

$$\boldsymbol{p} = (p_1, p_2, p_3), \qquad p_1 + p_2 + p_3 = 1.$$

Observed information:

Observed information	**Meaning**	**Count**
Category 1 exactly	definitely in cell 1	n_1
Category 2 exactly	definitely in cell 2	n_2
Category 3 exactly	definitely in cell 3	n_3
Category 1 or 2	exact cell unknown	m

Total sample size:

$$N = n_1 + n_2 + n_3 + m.$$

The m ambiguous observations are known to be in $\{1, 2\}$, but their exact split is missing.

Observed-data likelihood for the simple table

For an observation known exactly to be in category k, the probability contribution is p_k.

For an observation known only to be in category 1 or 2, the probability contribution is

$$p_1 + p_2.$$

Therefore, ignoring multinomial constants, the observed-data likelihood is

$$L(\boldsymbol{p}; Y) \propto p_1^{n_1} p_2^{n_2} p_3^{n_3} (p_1 + p_2)^m.$$

The observed log-likelihood is

$$\ell(\boldsymbol{p}; Y) = n_1 \log p_1 + n_2 \log p_2 + n_3 \log p_3 + m \log(p_1 + p_2).$$

Complete-data formulation

Let

$U =$ number of ambiguous observations that truly belong to category 1.

Then

$m - U =$ number of ambiguous observations that truly belong to category 2.

If U were known, the complete cell counts would be

$$N_1 = n_1 + U, \qquad N_2 = n_2 + (m - U), \qquad N_3 = n_3.$$

The complete-data log-likelihood would be

$$\ell_c(\boldsymbol{p}; Y, U) = (n_1 + U) \log p_1 + (n_2 + m - U) \log p_2 + n_3 \log p_3.$$

The complete-data MLE would simply be complete cell count divided by N.

Distribution of the missing split

Given that an ambiguous observation lies in $\{1, 2\}$, its conditional probability of being in category 1 is

$$\mathbb{P}_{\boldsymbol{p}}(1 \mid 1 \text{ or } 2) = \frac{p_1}{p_1 + p_2}.$$

Therefore, at iteration t,

$$U \mid Y, \boldsymbol{p}^{(t)} \sim \text{Binomial}\left(m, \frac{p_1^{(t)}}{p_1^{(t)} + p_2^{(t)}}\right).$$

Hence the E-step gives

$$\widehat{U}^{(t)} = \mathbb{E}_{\boldsymbol{p}^{(t)}}(U \mid Y) = m \frac{p_1^{(t)}}{p_1^{(t)} + p_2^{(t)}}.$$

Similarly,

$$\mathbb{E}_{\boldsymbol{p}^{(t)}}(m - U \mid Y) = m \frac{p_2^{(t)}}{p_1^{(t)} + p_2^{(t)}}.$$

E-step for the simple table

The expected complete counts at iteration t are:

$$\widetilde{N}_1^{(t)} = n_1 + m\frac{p_1^{(t)}}{p_1^{(t)} + p_2^{(t)}},$$

$$\widetilde{N}_2^{(t)} = n_2 + m\frac{p_2^{(t)}}{p_1^{(t)} + p_2^{(t)}},$$

$$\widetilde{N}_3^{(t)} = n_3.$$

Thus

$$Q(\boldsymbol{p} \mid \boldsymbol{p}^{(t)}) = \sum_{k=1}^{3} \widetilde{N}_k^{(t)} \log p_k.$$

This is exactly the complete multinomial log-likelihood, with cell counts replaced by expected cell counts.

M-step for the simple table

The M-step maximizes

$$Q(\boldsymbol{p} \mid \boldsymbol{p}^{(t)}) = \sum_{k=1}^{3} \widetilde{N}_k^{(t)} \log p_k$$

subject to $p_1 + p_2 + p_3 = 1$.

Therefore, using the same multinomial MLE argument,

$$p_1^{(t+1)} = \frac{\widetilde{N}_1^{(t)}}{N},$$

$$p_2^{(t+1)} = \frac{\widetilde{N}_2^{(t)}}{N},$$

$$p_3^{(t+1)} = \frac{\widetilde{N}_3^{(t)}}{N} = \frac{n_3}{N}.$$

The M-step treats the expected completed table as if it were the completed table.

Numerical example: data and initialization

Suppose

$$n_1 = 18, \qquad n_2 = 12, \qquad n_3 = 20, \qquad m = 10.$$

Then

$$N = 18 + 12 + 20 + 10 = 60.$$

Start from

$$\boldsymbol{p}^{(0)} = (1/3, 1/3, 1/3).$$

The observed log-likelihood, ignoring constants, is

$$\ell(\boldsymbol{p}) = 18 \log p_1 + 12 \log p_2 + 20 \log p_3 + 10 \log(p_1 + p_2).$$

How does EM split the 10 ambiguous observations between categories 1 and 2?

Numerical example: first EM iteration

At $\boldsymbol{p}^{(0)} = (1/3, 1/3, 1/3)$,

$$\widehat{U}^{(0)} = 10\frac{1/3}{1/3 + 1/3} = 5.$$

So the expected complete counts are

$$\widetilde{N}_1^{(0)} = 18 + 5 = 23,$$

$$\widetilde{N}_2^{(0)} = 12 + 5 = 17,$$

$$\widetilde{N}_3^{(0)} = 20.$$

The M-step gives

$$\boldsymbol{p}^{(1)} = \frac{1}{60}(23, 17, 20) = (0.3833, 0.2833, 0.3333).$$

Numerical example: EM trace

Iteration	$\widehat{U}^{(t)}$	$p_1^{(t+1)}$	$p_2^{(t+1)}$	$p_3^{(t+1)}$	$\ell(\boldsymbol{p}^{(t+1)})$
0	5.0000	0.3833	0.2833	0.3333	-64.420
1	5.7500	0.3958	0.2708	0.3333	-64.380
2	5.9375	0.3990	0.2677	0.3333	-64.377
3	5.9844	0.3997	0.2669	0.3333	-64.377

EM monotonically increases the observed-data log-likelihood and approaches $(0.4000, 0.2667, 0.3333)$.

Likelihood values ignore multinomial constants.

Why the limiting answer is intuitive

The observed data tell us two things.

1. Category 3 is exactly observed:

$$\widehat{p}_3 = \frac{20}{60} = \frac{1}{3}.$$

2. Among observations known to be in categories 1 or 2, the exact observed split is

$$\frac{n_1}{n_1 + n_2} = \frac{18}{30} = 0.6.$$

Since

$$\widehat{p}_1 + \widehat{p}_2 = 1 - \widehat{p}_3 = \frac{40}{60},$$

we get

$$\widehat{p}_1 = 0.6 \times \frac{40}{60} = 0.4, \qquad \widehat{p}_2 = 0.4 \times \frac{40}{60} = 0.2667.$$

The EM algorithm reaches the same maximum likelihood answer through repeated expected completions.

A Problem: overlapping ambiguous categories

Suppose there are three categories with probabilities

$$\boldsymbol{p} = (p_1, p_2, p_3), \qquad p_1 + p_2 + p_3 = 1.$$

The observed data are:

Observed information	**Count**
exactly category 1	n_1
exactly category 2	n_2
exactly category 3	n_3
category 1 or category 2, but not known which	m_{12}
category 2 or category 3, but not known which	m_{23}

Derive one EM update for (p_1, p_2, p_3).

Solution to the problem

At iteration t, split each ambiguous group conditionally using the current probabilities.

$$\widetilde{u}_{12}^{(t)} = \mathbb{E}_{\boldsymbol{p}^{(t)}}(\text{number from } m_{12} \text{ in category } 1 \mid Y) = m_{12} \frac{p_1^{(t)}}{p_1^{(t)} + p_2^{(t)}}.$$

Hence the expected number from m_{12} in category 2 is

$$m_{12} - \widetilde{u}_{12}^{(t)} = m_{12} \frac{p_2^{(t)}}{p_1^{(t)} + p_2^{(t)}}.$$

Similarly,

$$\widetilde{u}_{23}^{(t)} = m_{23} \frac{p_2^{(t)}}{p_2^{(t)} + p_3^{(t)}}$$

for the expected number from m_{23} in category 2, and

$$m_{23} - \widetilde{u}_{23}^{(t)} = m_{23} \frac{p_3^{(t)}}{p_2^{(t)} + p_3^{(t)}}$$

for category 3.

Solution to the problem

The expected complete counts are

$$\widetilde{N}_1^{(t)} = n_1 + \widetilde{u}_{12}^{(t)},$$

$$\widetilde{N}_2^{(t)} = n_2 + \{m_{12} - \widetilde{u}_{12}^{(t)}\} + \widetilde{u}_{23}^{(t)},$$

$$\widetilde{N}_3^{(t)} = n_3 + \{m_{23} - \widetilde{u}_{23}^{(t)}\}.$$

Let

$$N = n_1 + n_2 + n_3 + m_{12} + m_{23}.$$

Then the M-step is just normalization:

$$p_k^{(t+1)} = \frac{\widetilde{N}_k^{(t)}}{N}, \qquad k = 1, 2, 3.$$

The trick is that category 2 receives expected contributions from *two different ambiguous groups*.

From one-way to two-way tables

Now suppose each unit has two categorical variables.

For example:

	Test +	**Test −**
Disease +	n_{11}	n_{12}
Disease −	n_{21}	n_{22}

Let

$$p_{ij} = \mathbb{P}(\text{row } i, \text{ column } j), \qquad \sum_{i,j} p_{ij} = 1.$$

With complete counts,

$$\ell_c(\boldsymbol{p}) = \sum_{i,j} n_{ij} \log p_{ij}, \qquad \widehat{p}_{ij} = \frac{n_{ij}}{N}.$$

Different kinds of incomplete records

For a two-way table, a record may be incomplete in several ways.

Record type	What is observed?
Complete	row and column are both observed
Row-only	row observed, column missing
Column-only	column observed, row missing
Neither observed	both row and column missing

Let

- c_{ij}: fully observed count in cell (i, j),
- a_i: row i observed but column missing,
- b_j: column j observed but row missing,
- r: neither row nor column observed.

Each incomplete record contributes to several possible cells. EM assigns fractional expected counts to those cells.

E-step: row-only records

Consider the a_i records for which row i is known, but column is missing.

Given row i, the conditional probability of column j is

$$\mathbb{P}_{\boldsymbol{p}}(\text{column } j \mid \text{row } i) = \frac{p_{ij}}{p_{i+}},$$

where

$$p_{i+} = \sum_j p_{ij}.$$

Therefore, at iteration t, the expected contribution to cell (i, j) from row-only records is

$$a_i \frac{p_{ij}^{(t)}}{p_{i+}^{(t)}}.$$

Row-only records are distributed across the row according to the current conditional column probabilities.

E-step: column-only and fully missing records

For the b_j records with column j observed but row missing,

$$\mathbb{P}_{\boldsymbol{p}}(\text{row } i \mid \text{column } j) = \frac{p_{ij}}{p_{+j}},$$

where

$$p_{+j} = \sum_i p_{ij}.$$

So their expected contribution to cell (i, j) is

$$b_j \frac{p_{ij}^{(t)}}{p_{+j}^{(t)}}.$$

For the r records with neither row nor column observed, the expected contribution to cell (i, j) is simply

$$r p_{ij}^{(t)}.$$

Fully missing records add very little information about the relative cell probabilities; they mostly affect total sample size.

Expected completed table

Putting the pieces together, the expected complete count in cell (i, j) is

$$\widetilde{n}_{ij}^{(t)} = c_{ij} + a_i \frac{p_{ij}^{(t)}}{p_{i+}^{(t)}} + b_j \frac{p_{ij}^{(t)}}{p_{+j}^{(t)}} + r p_{ij}^{(t)}.$$

Then

$$Q(\boldsymbol{p} \mid \boldsymbol{p}^{(t)}) = \sum_{i,j} \widetilde{n}_{ij}^{(t)} \log p_{ij}.$$

The M-step is therefore

$$p_{ij}^{(t+1)} = \frac{\widetilde{n}_{ij}^{(t)}}{N}.$$

> The table version of EM is especially transparent: incomplete records are redistributed into cells using current conditional probabilities.

A small two-way numerical setup

Suppose the complete observed records are

	Column 1	**Column 2**
Row 1	12	8
Row 2	6	14

Additionally:

- $a_1 = 5$: row 1 observed, column missing;
- $a_2 = 3$: row 2 observed, column missing;
- $b_1 = 4$: column 1 observed, row missing;
- $b_2 = 2$: column 2 observed, row missing;
- $r = 0$: no records with both variables missing.

Total sample size:

$$N = 12 + 8 + 6 + 14 + 5 + 3 + 4 + 2 = 54.$$

One E-step from proportional starting values

Use the complete observed table as a starting point:

$$\boldsymbol{p}^{(0)} = \frac{1}{40}\begin{pmatrix} 12 & 8 \\ 6 & 14 \end{pmatrix} = \begin{pmatrix} 0.30 & 0.20 \\ 0.15 & 0.35 \end{pmatrix}.$$

Then

$$p_{1+}^{(0)} = 0.50, \qquad p_{2+}^{(0)} = 0.50,$$
$$p_{+1}^{(0)} = 0.45, \qquad p_{+2}^{(0)} = 0.55.$$

For example, the row-1 missing-column records are split as

$$5 \times \frac{0.30}{0.50} = 3.0 \quad \text{and} \quad 5 \times \frac{0.20}{0.50} = 2.0.$$

The same idea is applied to every incomplete group of records.

Expected counts in the small two-way example

Using the formula

$$\widetilde{n}_{ij}^{(0)} = c_{ij} + a_i \frac{p_{ij}^{(0)}}{p_{i+}^{(0)}} + b_j \frac{p_{ij}^{(0)}}{p_{+j}^{(0)}},$$

we get approximately

	Column 1	**Column 2**
Row 1	$12 + 3.00 + 2.67 = 17.67$	$8 + 2.00 + 0.73 = 10.73$
Row 2	$6 + 0.90 + 1.33 = 8.23$	$14 + 2.10 + 1.27 = 17.37$

The M-step gives

$$p_{ij}^{(1)} = \frac{\widetilde{n}_{ij}^{(0)}}{54}.$$

Thus

$$\boldsymbol{p}^{(1)} \approx \begin{pmatrix} 0.3272 & 0.1988 \\ 0.1525 & 0.3217 \end{pmatrix}.$$

Problem 2: mixed missingness in a 2×2 table

Let the complete cell probabilities be p_{ij}, $i, j = 1, 2$, with

$$\sum_{i=1}^{2} \sum_{j=1}^{2} p_{ij} = 1.$$

Observed data contain:

- fully observed counts n_{ij};
- row-only records: r_i, row i observed but column missing;
- column-only records: c_j, column j observed but row missing;
- fully missing records: f, neither row nor column observed.

Write the E-step expected complete count $\tilde{n}_{ij}^{(t)}$, and then the M-step.

Solution to Problem 2: general formula

At iteration t, define the current row and column margins

$$p_{i+}^{(t)} = p_{i1}^{(t)} + p_{i2}^{(t)}, \qquad p_{+j}^{(t)} = p_{1j}^{(t)} + p_{2j}^{(t)}.$$

Then

$$\widetilde{n}_{ij}^{(t)} = n_{ij} + r_i \frac{p_{ij}^{(t)}}{p_{i+}^{(t)}} + c_j \frac{p_{ij}^{(t)}}{p_{+j}^{(t)}} + f p_{ij}^{(t)}.$$

The four terms have clear meanings:

- observed complete count;
- expected allocation of row-only records across columns;
- expected allocation of column-only records across rows;
- expected allocation of fully missing records across all cells.

EM for exponential-family complete data

Many complete-data models can be written as

$$\ell_c(\theta; Y, Z) = \eta(\theta)^T S(Y, Z) - A(\theta) + C(Y, Z),$$

where

- $S(Y, Z)$ is the complete-data sufficient statistic,
- $\eta(\theta)$ is the natural parameter,
- $A(\theta)$ is the log-partition function.

Then the EM Q-function is

$$Q(\theta \mid \theta^{(t)}) = \eta(\theta)^T \mathbb{E}_{\theta^{(t)}}\{S(Y, Z) \mid Y\} - A(\theta) + \text{constant}.$$

In exponential-family problems, the E-step often reduces to computing the expected complete-data sufficient statistic.

A general EM recipe

1. **Choose the complete data.**
$$X = (Y, Z).$$
2. **Write the complete-data log-likelihood.**
$$\ell_c(\theta; Y, Z).$$
3. **Identify complete-data sufficient statistics.**
$$S(Y, Z).$$
4. **Compute their conditional expectations.**
$$\widetilde{S}^{(t)} = \mathbb{E}_{\theta^{(t)}}\{S(Y, Z) \mid Y\}.$$
5. **Maximize as if $\widetilde{S}^{(t)}$ were observed.**
$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta \mid \theta^{(t)}).$$

The same structure across examples

Problem	Missing part	E-step computes
Poisson hidden source	source labels	expected labels
Normal mixture	component labels	responsibilities
Incomplete multinomial	missing cell split	expected cell counts
Incomplete contingency table	missing row/column	expected cell counts
Bivariate normal with missing values	missing coordinates	conditional moments

The examples look different, but the mathematical pattern is the same: conditional expectation creates a surrogate completed dataset.

EM for incomplete tables: algorithm

Input

Observed incomplete table and an initial probability table $\boldsymbol{p}^{(0)}$.

For $t = 0, 1, 2, \ldots$:

1. **E-step:** compute expected complete cell counts

$$\widetilde{n}_{ij}^{(t)} = \mathbb{E}_{\boldsymbol{p}^{(t)}}(n_{ij}^{\text{complete}} \mid Y).$$

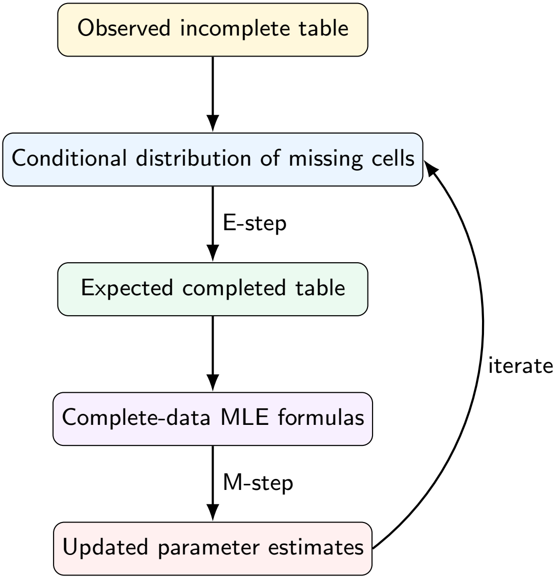
2. **M-step:** update probabilities

$$p_{ij}^{(t+1)} = \frac{\widetilde{n}_{ij}^{(t)}}{N}.$$

3. **Check convergence:** stop when parameter changes or observed likelihood changes are small.

The whole algorithm is "fill expected counts, then normalize".

Summary



Observed incomplete table
Conditional distribution of missing cells
E-step
Expected completed table
Complete-data MLE formulas
M-step
Updated parameter estimates
iterate

Key takeaway messages

1. In incomplete tables, the complete-data sufficient statistics are usually cell counts.
2. The E-step computes expected missing cell counts using the current parameter estimates.
3. The M-step uses the ordinary complete-data MLE formulas with expected counts.
4. The same principle extends to general missing-data likelihoods through expected sufficient statistics.
5. EM gives maximum likelihood estimates, but standard errors require additional observed-information calculations or resampling.
6. EM is principled only relative to the assumed missing-data model and missingness mechanism.

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Advanced Statistical Methods II

Lecture VII: MM Algorithm

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Outline of Today's Lecture

A broader view

In Lectures IV–VI, we studied EM.

- EM was introduced as a method for incomplete or latent-data likelihoods.
- The E-step computed conditional expectations of missing sufficient statistics.
- The M-step maximized an expected complete-data log-likelihood.
- The ascent property made the observed-data likelihood nondecreasing.

This lecture takes a broader view: many useful iterative algorithms work by replacing a hard objective by an easier surrogate objective. This is the main idea behind MM algorithms (Hunter and Lange, 2004; Lange, 2016).

Why do we need another optimization principle?

Suppose we want to solve

$$\widehat{\theta} = \arg\max_{\theta\in\Theta} \ell(\theta) \qquad \text{or} \qquad \widehat{\theta} = \arg\min_{\theta\in\Theta} f(\theta).$$

In many statistical problems:

- the objective is not quadratic;
- the likelihood contains sums inside logarithms;
- the objective may be nondifferentiable;
- direct numerical optimization may be unstable;
- a closed-form update may not be available for the original objective.

> Can we replace the hard objective by an easier objective at each iteration, while still guaranteeing improvement in the original objective?

The central MM idea

At iteration t, instead of optimizing the original objective directly, construct a surrogate objective that is easier to optimize.

Goal	**Surrogate relation**	**Update**
Minimize $f(\theta)$	majorize f by $g(\theta \mid \theta^{(t)})$	minimize g
Maximize $\ell(\theta)$	minorize ℓ by $q(\theta \mid \theta^{(t)})$	maximize q

MM means: do not attack the mountain directly. Build a local, easy-to-climb approximation whose improvement forces improvement in the original objective.

Two names, one principle

The letters "MM" are used in two closely related ways.

- Majorization-minimization: for minimizing an objective f.
- Minorization-maximization: for maximizing an objective ℓ.

Same logic, different direction

- If loss is bad and small is good, build an upper bound and push the upper bound down.
- If log-likelihood is good and large is good, build a lower bound and push the lower bound up.

The approach is often called *optimization transfer*: transfer the optimization problem to a simpler surrogate problem (Lange et al., 2000).

Definition: majorization

Let $f : \Theta \to \mathbb{R}$ be the objective we want to minimize. A function $g(\theta \mid \theta^{(t)})$ is said to majorize f at $\theta^{(t)}$ if

$$\boxed{g(\theta \mid \theta^{(t)}) \geq f(\theta) \quad \text{for all } \theta \in \Theta}$$

and

$$\boxed{g(\theta^{(t)} \mid \theta^{(t)}) = f(\theta^{(t)}).}$$

The first condition says the surrogate lies above the objective. The second says it touches the objective at the current iterate.

The MM update for minimization

Once a majorizer is available, define

$$\boxed{\theta^{(t+1)} \in \arg\min_{\theta\in\Theta} g(\theta \mid \theta^{(t)}).}$$

This update is useful when g is much easier to minimize than f.

- f may be nonlinear, nondifferentiable, or awkward.
- g may be quadratic, separable, or have a closed-form minimizer.
- The surrogate changes from iteration to iteration.

The method is not "approximate optimization" in a vague sense. The majorization relation gives a formal descent guarantee.

Descent property of MM

Theorem: Descent Property of the MM Algorithm.

Suppose $g(\theta \mid \theta^{(t)})$ majorizes $f(\theta)$ at $\theta^{(t)}$, and suppose

$$\theta^{(t+1)} \in \arg\min_{\theta \in \Theta} g(\theta \mid \theta^{(t)}).$$

Then

$$\boxed{f(\theta^{(t+1)}) \leq f(\theta^{(t)}).}$$

This is the minimization analogue of the EM ascent property.

Proof of descent property

The proof consists of three steps.

1. By majorization,

$$f(\theta^{(t+1)}) \leq g(\theta^{(t+1)} \mid \theta^{(t)}).$$

2. Since $\theta^{(t+1)}$ minimizes the surrogate,

$$g(\theta^{(t+1)} \mid \theta^{(t)}) \leq g(\theta^{(t)} \mid \theta^{(t)}).$$

3. By the touching condition,

$$g(\theta^{(t)} \mid \theta^{(t)}) = f(\theta^{(t)}).$$

Combining the three displays gives

$$\boxed{f(\theta^{(t+1)}) \leq f(\theta^{(t)}).}$$

Intuition

At iteration t:

1. The surrogate surface sits above the true objective.
2. At the current point, the surrogate and objective agree.
3. We move to a point where the surrogate is smaller.
4. Since the true objective lies below the surrogate, the true objective cannot increase.

The surrogate is allowed to be crude away from $\theta^{(t)}$, but it must be tight at $\theta^{(t)}$ and must bound the objective in the correct direction.

Definition: minorization

Let $\ell : \Theta \to \mathbb{R}$ be the objective we want to maximize. A function $q(\theta \mid \theta^{(t)})$ is said to minorize ℓ at $\theta^{(t)}$ if

$$\boxed{q(\theta \mid \theta^{(t)}) \leq \ell(\theta) \quad \text{for all } \theta \in \Theta}$$

and

$$\boxed{q(\theta^{(t)} \mid \theta^{(t)}) = \ell(\theta^{(t)}).}$$

For likelihood problems, we usually maximize log-likelihoods. Therefore, the relevant MM version is often minorization-maximization.

The MM update for maximization

Given a minorizer q, define

$$\boxed{\theta^{(t+1)} \in \arg\max_{\theta \in \Theta} q(\theta \mid \theta^{(t)}).}$$

This is especially useful when q has a simpler form than ℓ:

- separable terms instead of coupled terms;
- quadratic lower bounds instead of complicated log-likelihoods;
- complete-data style likelihoods instead of observed-data likelihoods.

This is exactly the kind of logic we saw in EM, although EM constructs its lower bound in a special probabilistic way.

Ascent property of MM

Theorem: Ascent Property of the MM Algorithm.

Suppose $q(\theta \mid \theta^{(t)})$ minorizes $\ell(\theta)$ at $\theta^{(t)}$, and suppose

$$\theta^{(t+1)} \in \arg\max_{\theta \in \Theta} q(\theta \mid \theta^{(t)}).$$

Then

$$\boxed{\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)}).}$$

Thus the objective sequence is monotone nondecreasing.

Proof of ascent property

The proof mirrors the minimization case.

1. By minorization,

$$\ell(\theta^{(t+1)}) \geq q(\theta^{(t+1)} \mid \theta^{(t)}).$$

2. Since $\theta^{(t+1)}$ maximizes the surrogate,

$$q(\theta^{(t+1)} \mid \theta^{(t)}) \geq q(\theta^{(t)} \mid \theta^{(t)}).$$

3. By the touching condition,

$$q(\theta^{(t)} \mid \theta^{(t)}) = \ell(\theta^{(t)}).$$

Hence

$$\boxed{\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)}).}$$

Where do surrogates come from?

MM is useful only if we can construct good surrogates. Common tools include:

- Jensen's inequality;
- tangent-line inequalities for convex or concave functions;
- quadratic upper or lower bounds;
- Cauchy–Schwarz and related inequalities.

The art of MM is to choose a surrogate that is easy enough to optimize but tight enough to converge reasonably fast.

Surrogates from tangent inequalities

A basic source of MM algorithms is the supporting hyperplane property.

If h is concave, then

$$h(u) \leq h(u_0) + h'(u_0)(u - u_0)$$

for appropriate u, u_0. The tangent line lies above a concave function.

If h is convex, then

$$h(u) \geq h(u_0) + h'(u_0)(u - u_0).$$

The tangent line lies below a convex function.

Interpretation

These inequalities can turn complicated nonlinear terms into linear or quadratic surrogate terms.

Surrogates from quadratic bounds

Suppose f is differentiable and its gradient is Lipschitz with constant L:

$$\|\nabla f(\theta) - \nabla f(\eta)\| \leq L\|\theta - \eta\|.$$

Then

$$\boxed{f(\theta) \leq f(\theta^{(t)}) + \nabla f(\theta^{(t)})^\top (\theta - \theta^{(t)}) + \frac{L}{2}\|\theta - \theta^{(t)}\|^2.}$$

This is a quadratic majorizer.

Minimizing this quadratic majorizer gives the gradient-descent update with step size $1/L$.

Gradient descent as an MM algorithm

The quadratic majorizer is

$$g(\theta \mid \theta^{(t)}) = f(\theta^{(t)}) + \nabla f(\theta^{(t)})^\top (\theta - \theta^{(t)}) + \frac{L}{2}\|\theta - \theta^{(t)}\|^2.$$

Differentiate with respect to θ:

$$\nabla_\theta g(\theta \mid \theta^{(t)}) = \nabla f(\theta^{(t)}) + L(\theta - \theta^{(t)}).$$

Setting this equal to zero gives

$$\boxed{\theta^{(t+1)} = \theta^{(t)} - \frac{1}{L}\nabla f(\theta^{(t)}).}$$

Thus a familiar algorithm can be understood as MM.

Surrogates from Jensen's inequality

Jensen's inequality often removes a log of a sum:

$$\log\left(\sum_{k=1}^{K} a_k\right) = \log\left(\sum_{k=1}^{K} w_k \frac{a_k}{w_k}\right) \geq \sum_{k=1}^{K} w_k \log\left(\frac{a_k}{w_k}\right),$$

where $w_k > 0$ and $\sum_k w_k = 1$.

This is the key inequality behind the EM lower bound for mixture likelihoods.

Equality occurs when

$$w_k \propto a_k.$$

For a finite mixture,

$$\ell(\theta) = \sum_{i=1}^{n} \log \left\{ \sum_{k=1}^{K} \pi_k f_k(y_i; \psi_k) \right\}.$$

Introduce weights w_{ik}, with $w_{ik} \geq 0$ and $\sum_k w_{ik} = 1$. Then

$$\log \left\{ \sum_k \pi_k f_k(y_i; \psi_k) \right\} \geq \sum_k w_{ik} \log \left\{ \frac{\pi_k f_k(y_i; \psi_k)}{w_{ik}} \right\}.$$

At $\theta^{(t)}$, equality is achieved by

$$\boxed{w_{ik} = \tau_{ik}^{(t)} = \mathbb{P}_{\theta^{(t)}}(Z_i = k \mid Y_i = y_i).}$$

This is why EM can be viewed as a special MM algorithm.

Observed log-likelihood & EM

Recall the EM notation from Lecture IV.

- Observed data: $Y_{\rm obs}$.
- Missing data: $Y_{\rm mis}$.
- Complete data: $Y_{\rm com} = (Y_{\rm obs}, Y_{\rm mis})$.

The observed-data likelihood is

$$L(\theta; Y_{\rm obs}) = \int L_c(\theta; Y_{\rm obs}, y_{\rm mis})\, dy_{\rm mis}.$$

The EM algorithm constructs a lower bound for

$$\ell(\theta) = \log L(\theta; Y_{\rm obs}).$$

The EM surrogate

Define

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}\left[\ell_c(\theta; Y_{\mathrm{com}}) \mid Y_{\mathrm{obs}}\right].$$

For monotonicity, the important object is not just Q, but the lower-bound relation behind it.

One can write

$$\ell(\theta) = Q(\theta \mid \theta^{(t)}) + [\text{a positive quantity not depending on } \theta].$$

EM increases Q, and the decomposition ensures that this produces ascent in the observed-data log-likelihood.

EM versus general MM

Feature	EM	General MM
Surrogate source	conditional expectation	any valid inequality
Typical objective	likelihood	likelihood or loss
Missing data needed?	yes, conceptually	no
Guarantee	likelihood ascent	ascent/descent
Update style	E-step/M-step	construct/optimize surrogate

EM is therefore a particularly important member of the larger MM family, but MM can be used even when there is no natural missing-data interpretation.

Why use MM for robust regression?

Consider LAD regression:

$$\widehat{\beta}_{\mathrm{LAD}} = \arg\min_{\beta \in \mathbb{R}^p} \sum_{i=1}^{n} |y_i - x_i^\top \beta|.$$

This is statistically useful because absolute loss is less sensitive to large residuals than squared loss.

But the objective is nondifferentiable when

$$y_i - x_i^\top \beta = 0.$$

MM can convert the absolute-value objective into a sequence of weighted least-squares problems.

A useful inequality for absolute value

For any $u \in \mathbb{R}$ and any $u^{(t)} \neq 0$,

$$\boxed{|u| \leq \frac{u^2}{2|u^{(t)}|} + \frac{|u^{(t)}|}{2}.}$$

Why is this true?

$$0 \leq (|u| - |u^{(t)}|)^2 = u^2 - 2|u||u^{(t)}| + |u^{(t)}|^2.$$

Rearranging gives the inequality.

Equality holds when $|u| = |u^{(t)}|$, so in particular at the current residual.

Majorizing the LAD objective

Let

$$r_i(\beta) = y_i - x_i^\top \beta, \qquad r_i^{(t)} = r_i(\beta^{(t)}).$$

Using the inequality,

$$|r_i(\beta)| \le \frac{r_i(\beta)^2}{2|r_i^{(t)}|} + \frac{|r_i^{(t)}|}{2}.$$

Therefore, a majorizer of the LAD objective is

$$g(\beta \mid \beta^{(t)}) = \sum_{i=1}^{n} \left[\frac{(y_i - x_i^\top \beta)^2}{2|r_i^{(t)}|} + \frac{|r_i^{(t)}|}{2} \right].$$

The second term does not depend on β.

The resulting weighted least-squares update

Minimizing the majorizer is equivalent to minimizing

$$\sum_{i=1}^{n} w_i^{(t)}(y_i - x_i^\top \beta)^2, \qquad w_i^{(t)} = \frac{1}{2|r_i^{(t)}|}.$$

Thus

$$\boxed{\beta^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} y,}$$

where

$$W^{(t)} = \mathrm{diag}(w_1^{(t)}, \ldots, w_n^{(t)}).$$

In practice, replace $|r_i^{(t)}|$ by $\sqrt{(r_i^{(t)})^2 + \varepsilon}$ or $|r_i^{(t)}| + \varepsilon$ to avoid division by zero.

A scalar example with a nonsmooth penalty

Consider the objective

$$f(\theta) = \frac{1}{2}(y - \theta)^2 + \lambda|\theta|, \qquad \lambda > 0.$$

This is a one-parameter version of a penalized least-squares problem.

The absolute-value term is nondifferentiable at $\theta = 0$.

Using the same inequality,

$$|\theta| \leq \frac{\theta^2}{2|\theta^{(t)}|} + \frac{|\theta^{(t)}|}{2}.$$

Hence a majorizer is

$$g(\theta \mid \theta^{(t)}) = \frac{1}{2}(y - \theta)^2 + \lambda \left\{ \frac{\theta^2}{2|\theta^{(t)}|} + \frac{|\theta^{(t)}|}{2} \right\}.$$

Solving the surrogate problem

Ignoring constants not involving θ, minimize

$$g(\theta \mid \theta^{(t)}) = \frac{1}{2}(y-\theta)^2 + \frac{\lambda}{2|\theta^{(t)}|}\theta^2 + \text{constant}.$$

Differentiate:

$$\frac{\partial g}{\partial \theta} = -(y-\theta) + \frac{\lambda}{|\theta^{(t)}|}\theta.$$

Set equal to zero:

$$-y + \theta + \frac{\lambda}{|\theta^{(t)}|}\theta = 0.$$

Therefore,

$$\boxed{\theta^{(t+1)} = \frac{|\theta^{(t)}|}{|\theta^{(t)}| + \lambda} y.}$$

A few observations

The update

$$\theta^{(t+1)} = \frac{|\theta^{(t)}|}{|\theta^{(t)}| + \lambda} y$$

shrinks y toward zero.

- If λ is large, the shrinkage is stronger.
- If $|\theta^{(t)}|$ is small, the shrinkage is stronger.
- The update arises from replacing a nonsmooth penalty by a quadratic surrogate.

> This is not the most efficient way to solve the scalar LASSO problem, but it shows how MM turns nonsmooth optimization into repeated smooth optimization.

Likelihood maximization through minorization

Suppose we want to maximize a log-likelihood $\ell(\theta)$. A common MM strategy is:

1. At $\theta^{(t)}$, find a lower bound

$$q(\theta \mid \theta^{(t)}) \leq \ell(\theta).$$

2. Make sure the bound touches:

$$q(\theta^{(t)} \mid \theta^{(t)}) = \ell(\theta^{(t)}).$$

3. Maximize the lower bound:

$$\theta^{(t+1)} = \arg\max_{\theta} q(\theta \mid \theta^{(t)}).$$

> This is a direct analogue of EM, but the lower bound may come from analytical inequalities rather than missing-data conditioning.

Quadratic minorization for concave likelihoods

For many GLM-type likelihoods, $\ell(\beta)$ is concave but not quadratic.

A quadratic minorizer has the form

$$q(\beta \mid \beta^{(t)}) = \ell(\beta^{(t)}) + \nabla\ell(\beta^{(t)})^\top(\beta - \beta^{(t)}) - \frac{M}{2}\|\beta - \beta^{(t)}\|^2,$$

where M is chosen large enough.

Maximizing this gives

$$\boxed{\beta^{(t+1)} = \beta^{(t)} + \frac{1}{M}\nabla\ell(\beta^{(t)}).}$$

For maximization, this is gradient ascent viewed as a minorization-maximization algorithm.

Newton's method uses local curvature:

$$\theta^{(t+1)} = \theta^{(t)} - \left\{\nabla^2 f(\theta^{(t)})\right\}^{-1} \nabla f(\theta^{(t)}).$$

It can converge very fast near the optimum, but may fail to decrease f globally unless modified.

MM instead prioritizes monotonicity:

$$f(\theta^{(t+1)}) \leq f(\theta^{(t)}) \quad \text{or} \quad \ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)}).$$

Newton methods are often fast but need safeguards. MM methods are often stable but may be slow if the surrogate is too conservative.

What makes a good surrogate?

A good surrogate should satisfy two requirements.

1. **Computational simplicity**: optimizing it should be easier than optimizing the original objective.
2. The surrogate should not be too loose. A very loose surrogate may give tiny steps and slow convergence.

Tightness matters

Two valid majorizers can produce very different algorithms.

- A tight majorizer follows the original objective closely near $\theta^{(t)}$.
- A loose majorizer may guarantee descent but take overly cautious steps.
- A surrogate that is easy to optimize but extremely loose may be computationally inefficient.

MM design is a trade-off between easy updates and fast convergence.

This mirrors EM: EM is stable and monotone, but in some problems it converges slowly.

Common stopping criteria for MM algorithms include:

- small change in parameter:

$$\|\theta^{(t+1)} - \theta^{(t)}\| < \varepsilon;$$

- small relative change in objective:

$$\frac{|f(\theta^{(t+1)}) - f(\theta^{(t)})|}{1 + |f(\theta^{(t)})|} < \varepsilon;$$

- small norm of gradient or score;
- maximum number of iterations.

Monotonic improvement does not by itself guarantee that the algorithm has reached the global optimum.

Local maxima and local minima

MM inherits the usual difficulties of nonconvex optimization.

- For nonconvex minimization, MM may converge to a local minimum or stationary point.
- For nonconcave likelihood maximization, MM may converge to a local maximum or saddle-type stationary point.
- Different starting values may lead to different solutions.

> The monotonicity theorem says that each step improves the objective. It does not say that the final answer is globally best.

This point is familiar from EM for finite mixtures.

Exercise 1: verify a majorizer

Let

$$f(\theta) = |\theta|.$$

For a current value $\theta^{(t)} \neq 0$, define

$$g(\theta \mid \theta^{(t)}) = \frac{\theta^2}{2|\theta^{(t)}|} + \frac{|\theta^{(t)}|}{2}.$$

Show that g majorizes f at $\theta^{(t)}$. Then find the MM update for minimizing

$$\frac{1}{2}(y-\theta)^2 + \lambda|\theta|.$$

Exercise 2: minorization and ascent

Suppose $q(\theta \mid \theta^{(t)})$ satisfies

$$q(\theta \mid \theta^{(t)}) \leq \ell(\theta) \quad \text{for all } \theta,$$

and

$$q(\theta^{(t)} \mid \theta^{(t)}) = \ell(\theta^{(t)}).$$

Suppose $\theta^{(t+1)}$ only improves the surrogate:

$$q(\theta^{(t+1)} \mid \theta^{(t)}) \geq q(\theta^{(t)} \mid \theta^{(t)}),$$

but does not necessarily maximize it.

Does $\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})$ still follow?

Solution to Exercise 2

Yes. Exact maximization is not needed.

The same inequality chain works:

$$\ell(\theta^{(t+1)}) \geq q(\theta^{(t+1)} \mid \theta^{(t)})$$

by minorization.

By assumed improvement,

$$q(\theta^{(t+1)} \mid \theta^{(t)}) \geq q(\theta^{(t)} \mid \theta^{(t)}).$$

By tangency,

$$q(\theta^{(t)} \mid \theta^{(t)}) = \ell(\theta^{(t)}).$$

Therefore,

$$\boxed{\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)}).}$$

MM algorithm: the full recipe

For minimization:

1. Start with $\theta^{(0)}$.
2. At iteration t, construct $g(\theta \mid \theta^{(t)})$ such that

$$g(\theta \mid \theta^{(t)}) \geq f(\theta), \qquad g(\theta^{(t)} \mid \theta^{(t)}) = f(\theta^{(t)}).$$

3. Set

$$\theta^{(t+1)} = \arg\min_{\theta} g(\theta \mid \theta^{(t)}).$$

4. Repeat until convergence.

For maximization, reverse the inequalities and maximize the minorizer.

Key comparisons

Method	Core idea	Strength
EM	conditional expectation	natural for missing data
MM	surrogate optimization	very general
Newton	local quadratic approximation	fast near optimum
Gradient descent	first-order quadratic bound	simple, scalable

Key Takeaway Messages

1. MM algorithms replace a hard objective by an easier surrogate.
2. For minimization, use a majorizer and minimize it.
3. For maximization, use a minorizer and maximize it.
4. The touching condition is essential for monotonicity.
5. EM is a special and important MM-type algorithm for incomplete-data likelihoods.
6. MM can also handle robust regression, penalized estimation, GLMs, and many other optimization problems.

Bound the difficult objective, optimize the easier bound, and inherit monotone improvement.

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Advanced Statistical Methods II

Lecture 8: Time-to-event Data, Hazard, and Censoring

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Outline of Today's Lecture

Motivation

Time-to-event data

Survival function and related quantities

Censoring

Independent censoring

Likelihood contributions

Risk sets and first glimpse of Kaplan–Meier

Common pitfalls and interpretation

Practice questions

Key takeaway messages

The story so far...

In the first part of the course, we have repeatedly seen that the data we wish we had and the data we actually observe may be different.

- Missing covariate/response values led us to missing-data mechanisms.
- Latent labels led us to EM for mixture models.
- Incomplete cell counts led us to EM for tables.
- Difficult objective functions led us to MM algorithms.

Today we meet another major kind of incomplete observation: **time-to-event data with censoring**.
Survival analysis is a systematic likelihood-based framework for such data (Kleinbaum and Klein, 2012; Kalbfleisch and Prentice, 2002).

Today's focus

Suppose the response variable is a waiting time:

$$T = \text{time from a well-defined origin to a well-defined event.}$$

Examples:

- time from admission to discharge;
- time from treatment to death or recovery;
- time from installation to machine failure.

If some individuals have not yet experienced the event when the study ends, replacing their event time by the last follow-up time is wrong: the last follow-up time is not the true event time.

A simple motivating dataset

Consider a small clinical follow-up study.

Patient	Follow-up time	Status	Meaning
1	4	1	relapse observed at time 4
2	6	0	relapse-free until time 6
3	7	1	relapse observed at time 7
4	9	0	relapse-free until time 9
5	10	1	relapse observed at time 10

Here status is

$$\delta_i = \begin{cases} 1, & \text{event observed,} \\ 0, & \text{right censored.} \end{cases}$$

What should we do with patients 2 and 4?

What censored observations tell us

For patient 2, the recorded value is

$$(U_2, \delta_2) = (6, 0).$$

This does **not** mean

$$T_2 = 6.$$

It means

$$T_2 > 6.$$

Similarly, for patient 4,

$$(U_4, \delta_4) = (9, 0) \quad \Longrightarrow \quad T_4 > 9.$$

A censored observation is incomplete but still informative. It gives an inequality rather than an exact value.

Today's main questions

1. What exactly is time-to-event data?
2. What are survival, hazard, and cumulative hazard functions?
3. How should we interpret the hazard function?
4. What are right censoring, left censoring, and interval censoring?
5. What does independent or noninformative censoring mean?
6. How do censored observations contribute to likelihoods?

Time-to-event variable?

A time-to-event variable is a nonnegative random variable

$$T \geq 0,$$

where T measures the time until a specified event occurs.

A survival-analysis problem must specify three ingredients clearly:

1. the time origin;
2. the event of interest;
3. the time scale.

Example

In a cancer study, the time origin may be date of surgery, the event may be recurrence, and the time scale may be months after surgery.

Complete event-time data

If every subject is followed until the event occurs, we observe

$$T_1, T_2, \ldots, T_n.$$

Then standard distributional ideas apply:

- empirical distribution of event times;
- mean or median event time;
- parametric models such as exponential or Weibull;
- ordinary likelihood based on densities.

> Difficulties in survival analysis: the response is positive, might be even skewed; and some event times are only partially observed.

Let T be a nonnegative event time.

The distribution function is

$$F(t) = \mathbb{P}(T \leq t), \qquad t \geq 0.$$

The survival function is

$$\boxed{S(t) = \mathbb{P}(T > t) = 1 - F(t).}$$

$S(t)$ is the probability that an individual survives beyond time t, or equivalently, remains event-free beyond time t.

Typical properties:

- $S(0) \leq 1$, often $S(0) = 1$;
- $S(t)$ is nonincreasing in t;
- $S(t) \to 0$ as $t \to \infty$.

Density and survival: two equivalent views

If T is continuous with density $f(t)$, then

$$F(t) = \int_0^t f(u)\,du.$$

Hence

$$S(t) = 1 - F(t) = \int_t^\infty f(u)\,du.$$

Also,

$$f(t) = F'(t) = -S'(t).$$

Interpretation

The density describes where event times tend to fall. The survival function describes how many individuals remain event-free beyond each time point.

The hazard function

The hazard function is defined as

$$\boxed{h(t) = \lim_{\Delta t \downarrow 0} \frac{\mathbb{P}(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}.}$$

For continuous T, this becomes

$$\boxed{h(t) = \frac{f(t)}{S(t)}.}$$

The hazard is the instantaneous event rate at time t, among those who have survived up to time t (Cox and Oakes, 1984; Kleinbaum and Klein, 2012).

A hazard is not a probability. It is a rate and may exceed 1 depending on the time scale.

Deriving $h(t) = f(t)/S(t)$

Start with the conditional probability over a short interval:

$$\mathbb{P}(t \le T < t + \Delta t \mid T \ge t) = \frac{\mathbb{P}(t \le T < t + \Delta t)}{\mathbb{P}(T \ge t)}.$$

For small Δt,

$$\mathbb{P}(t \le T < t + \Delta t) \approx f(t)\Delta t.$$

Also, for continuous T,

$$\mathbb{P}(T \ge t) = \mathbb{P}(T > t) = S(t).$$

Therefore,

$$\frac{\mathbb{P}(t \le T < t + \Delta t \mid T \ge t)}{\Delta t} \approx \frac{f(t)}{S(t)}.$$

Taking $\Delta t \downarrow 0$ gives

$$\boxed{h(t) = f(t)/S(t).}$$

How to read a hazard function

Suppose $h(12) = 0.04$ per month.

- This does **not** mean a probability of 0.04 over the whole study.
- It means that near month 12, among those still event-free, the instantaneous event rate is about 0.04 per month.
- Over a very short interval Δt,

$$\mathbb{P}(12 \leq T < 12 + \Delta t \mid T \geq 12) \approx 0.04\Delta t.$$

The phrase "among those still at risk" is essential. Hazards are conditional on survival up to that time.

Cumulative hazard

The cumulative hazard function is

$$\boxed{H(t) = \int_0^t h(u)\,du.}$$

Using $h(t) = f(t)/S(t) = -S'(t)/S(t)$,

$$h(t) = -\frac{d}{dt}\log S(t).$$

Integrating from 0 to t, assuming $S(0) = 1$,

$$\boxed{H(t) = -\log S(t).}$$

Equivalently,

$$\boxed{S(t) = \exp\{-H(t)\}.}$$

Survival vs hazard: different questions

Quantity	Question answered
$S(t) = \mathbb{P}(T > t)$	What proportion remains event-free beyond time t?
$f(t)$	Around which times do events tend to occur?
$h(t)$	Among those still at risk at time t, how fast are events occurring now?
$H(t)$	How much event risk has accumulated up to time t?

Example: exponential survival

If

$$T \sim \mathrm{Exponential}(\lambda), \qquad \lambda > 0,$$

then

$$f(t) = \lambda e^{-\lambda t}, \qquad S(t) = e^{-\lambda t}.$$

Thus

$$h(t) = \frac{f(t)}{S(t)} = \frac{\lambda e^{-\lambda t}}{e^{-\lambda t}} = \lambda.$$

The exponential distribution has a constant hazard. It is useful as a starting model, but constant hazard is often unrealistic in medical and reliability applications.

Example: Weibull survival

One common parametrization of the Weibull distribution is

$$S(t) = \exp\{-\lambda t^{\alpha}\}, \qquad \lambda > 0,\ \alpha > 0.$$

Then

$$H(t) = \lambda t^{\alpha},$$

so

$$h(t) = H'(t) = \lambda \alpha t^{\alpha - 1}.$$

Parameter value	Hazard shape
$\alpha = 1$	constant hazard
$\alpha > 1$	increasing hazard
$0 < \alpha < 1$	decreasing hazard

Reliability intuition

An increasing hazard may represent ageing or wear-out. A decreasing hazard may represent early failures among fragile units.

Quick check: survival, density, hazard

Suppose
$$S(t) = \exp(-2t^3), \qquad t \geq 0.$$

Find $H(t)$, $h(t)$, and $f(t)$.

Solution.
$$H(t) = -\log S(t) = 2t^3.$$

Hence
$$h(t) = H'(t) = 6t^2.$$

Finally,
$$f(t) = h(t)S(t) = 6t^2 \exp(-2t^3).$$

What is censoring?

Censoring occurs when the event time T is not observed exactly, but partial information about T is available.

A censored observation provides a set of possible values for T, rather than one exact value.

Typical forms:

- right censoring: $T > C$;
- left censoring: $T < C$;
- interval censoring: $L < T \leq R$.

Censoring is not the same as missingness with no information. It is partial information about an event time.

Right censoring

Right censoring is the most common case in introductory survival analysis.

We observe

$$U = \min(T, C), \qquad \delta = \mathbb{I}(T \leq C),$$

where

- T is the event time;
- C is the censoring time;
- $\delta = 1$ means the event was observed;
- $\delta = 0$ means the subject was censored.

If $\delta = 0$, then

$$U = C < T,$$

so we know

$$T > U.$$

Right-censored data are usually recorded as pairs (U_i, δ_i), not simply as observed times.

Common sources of right censoring

Right censoring can arise in several ways.

Source	Interpretation
Administrative end	study ends before event occurs
Loss to follow-up	subject leaves study before event is observed
Dropout	subject stops participating
Competing practical end	device removed before failure
No event by analysis date	event may occur later, but not yet observed

The statistical validity of standard methods depends on how censoring occurs. Loss to follow-up caused by worsening health may be informative.

Left censoring

Left censoring occurs when we only know that the event occurred before a certain time.

$$T \leq C.$$

Examples

- Infection occurred before the first positive test, but the exact infection time is unknown.
- A machine was already defective when inspected for the first time.

Left censoring gives an upper bound on T. Right censoring gives a lower bound on T.

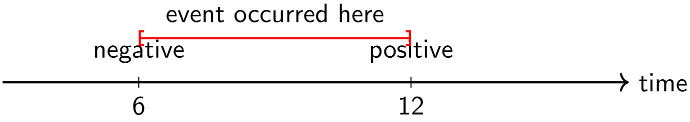
Interval censoring

Interval censoring occurs when we know that the event time lies in an interval:

$$L < T \leq R.$$

Example: periodic medical screening

A patient tests negative at month 6 and positive at month 12. If the event is infection, then the infection time is known to lie between months 6 and 12.



Interval censoring is common whenever subjects are inspected periodically rather than continuously.

Censoring vs truncation

Censoring and truncation are different concepts.

Mechanism	What happens?
Censoring	subject is in the sample, but exact event time is not fully observed
Truncation	subject is not included in the sample unless their event time satisfies a condition

Example of left truncation

A study enrolls only people who have survived long enough to attend a clinic visit. Individuals who had the event before the clinic visit are absent from the dataset.

Today we focus mainly on censoring. Truncation needs different likelihood adjustments and should not be treated as ordinary censoring.

An exercise

Classify each as right censored, left censored, interval censored, or truncated.

1. A patient is event-free at the final study visit at month 24.
2. A disease was already present at the first diagnostic test.
3. A subject tested disease-free in January and diseased in April.
4. Only individuals who survived at least 5 years after diagnosis are included in a retrospective dataset.
5. A machine is still working when the reliability experiment ends.

An exercise: answers

1. Right censored: the event time exceeds 24 months.
2. Left censored: the event time occurred before the first test.
3. Interval censored: the event occurred between January and April.
4. Left truncated: inclusion requires surviving at least 5 years.
5. Right censored: the failure time exceeds the end of the experiment.

A useful rule: censoring means partial information for a sampled unit; truncation affects whether the unit enters the sample at all.

Why censoring assumptions matter

Survival methods do not magically solve every incomplete follow-up problem.

The standard analysis of right-censored data usually assumes that censoring is independent or noninformative in an appropriate sense (Kalbfleisch and Prentice, 2002; Kleinbaum and Klein, 2012).

Informally, independent censoring means that, after accounting for modeled covariates if any, the censoring mechanism does not carry additional information about the subject's event time.

If sicker patients are more likely to drop out, treating dropout as ordinary independent censoring may bias survival estimates.

A mathematical formulation

Let

- T be the event time;
- C be the censoring time;
- X be observed covariates.

A common condition is conditional independence:

$$\boxed{T \perp C \mid X.}$$

For simple one-sample survival analysis without covariates, this reduces conceptually to

$$T \perp C.$$

The censoring time may depend on observed covariates, but after conditioning on those covariates it should not depend on the unobserved event time in an unexplained way.

Independent vs informative censoring

More plausible independent censoring	Possible informative censoring
Study ends on a fixed administrative date	Patients drop out because symptoms worsen
Device is removed due to unrelated relocation	Machine is removed because it is visibly deteriorating
Follow-up stops due to funding end date	High-risk participants become harder to contact

Relation to missing-data mechanisms

There is a strong conceptual link with Lectures I–II.

Missing data language	**Survival language**
Observed value missing	Exact event time not observed
Missingness mechanism	Censoring mechanism
Ignorable missingness under conditions	Noninformative/independent censoring under conditions
Complete-case deletion may bias estimates	Deleting censored observations may bias survival estimates

Censoring is not arbitrary missingness. It has a special structure: the censored record usually tells us that $T_\cdot$ is above or below some value.

Why likelihoods handle censoring naturally

Likelihoods are based on the probability of the data we actually observed.

For an exact observed failure at time t, we observe

$$T = t.$$

So the likelihood contribution is based on

$$f(t;\theta).$$

For a right-censored observation at time c, we observe

$$T > c.$$

So the likelihood contribution is based on

$$S(c;\theta) = \mathbb{P}_\theta(T > c).$$

The likelihood contribution must match the information contained in the observation: exact time gives a density; right censoring gives a survival probability.

Right-censored likelihood: preview

For independent observations (U_i, δ_i),

$$U_i = \min(T_i, C_i), \qquad \delta_i = \mathbb{I}(T_i \le C_i),$$

the event-time part of the likelihood is

$$\boxed{L(\theta) = \prod_{i=1}^{n} f(U_i; \theta)^{\delta_i} S(U_i; \theta)^{1-\delta_i}.}$$

Equivalently, since $f(t) = h(t)S(t)$,

$$L(\theta) = \prod_{i=1}^{n} \{h(U_i; \theta) S(U_i; \theta)\}^{\delta_i} S(U_i; \theta)^{1-\delta_i}.$$

Therefore,

$$\boxed{L(\theta) = \prod_{i=1}^{n} h(U_i; \theta)^{\delta_i} S(U_i; \theta).}$$

Likelihood contributions by censoring type

Observation type	Information observed	Contribution
Exact event	$T = t$	$f(t)$
Right censored	$T > c$	$S(c)$
Left censored	$T \leq c$	$F(c)$
Interval censored	$L < T \leq R$	$F(R) - F(L)$

This table is one of the main bridges from today's lecture to the next lecture on censored likelihoods.

Worked mini-example: write the likelihood

Suppose $T \sim \mathrm{Exponential}(\lambda)$ and we observe

$$(4,1), \quad (6,0), \quad (7,1), \quad (9,0), \quad (10,1).$$

For the exponential distribution,

$$f(t;\lambda) = \lambda e^{-\lambda t}, \qquad S(t;\lambda) = e^{-\lambda t}.$$

The likelihood is

$$L(\lambda) = \{\lambda e^{-4\lambda}\}\{e^{-6\lambda}\}\{\lambda e^{-7\lambda}\}\{e^{-9\lambda}\}\{\lambda e^{-10\lambda}\}.$$

Thus

$$\boxed{L(\lambda) = \lambda^3 \exp\{-\lambda(4+6+7+9+10)\}.}$$

What would go wrong if we ignored censoring?

For the same data:

$$(4, 1), \quad (6, 0), \quad (7, 1), \quad (9, 0), \quad (10, 1),$$

there are two common mistakes.

1. Treat 6 and 9 as event times.
2. Delete the censored observations $(6, 0)$ and $(9, 0)$.

Both strategies distort the information. Censored observations are neither exact failures nor useless records.

What would go wrong if we ignored censoring?

For the same data:

$$(4,1), \quad (6,0), \quad (7,1), \quad (9,0), \quad (10,1),$$

there are two common mistakes.

1. Treat 6 and 9 as event times.
2. Delete the censored observations $(6,0)$ and $(9,0)$.

> Both strategies distort the information. Censored observations are neither exact failures nor useless records.

For exponential data, the correct likelihood uses

$$\text{number of observed events} = 3, \qquad \text{total observed follow-up time} = 36.$$

So the MLE is

$$\widehat{\lambda} = \frac{3}{36} = 0.0833.$$

Deriving the exponential MLE with censoring: I

For right-censored exponential data,

$$L(\lambda) = \prod_{i=1}^{n} \{\lambda e^{-\lambda U_i}\}^{\delta_i} \{e^{-\lambda U_i}\}^{1-\delta_i}.$$

Combine the powers of λ and the exponential terms:

$$L(\lambda) = \lambda^{\sum_i \delta_i} \exp\left\{-\lambda \sum_{i=1}^{n} U_i\right\}.$$

Therefore the log-likelihood is

$$\ell(\lambda) = \left(\sum_i \delta_i\right) \log\lambda - \lambda \sum_i U_i.$$

From the previous slide,

$$\ell(\lambda) = \left(\sum_i \delta_i\right) \log \lambda - \lambda \sum_i U_i.$$

Differentiating,

$$\frac{d\ell}{d\lambda} = \frac{\sum_i \delta_i}{\lambda} - \sum_i U_i.$$

Setting this equal to zero gives

$$\frac{\sum_i \delta_i}{\lambda} = \sum_i U_i.$$

Thus

$$\boxed{\widehat{\lambda} = \frac{\sum_i \delta_i}{\sum_i U_i}.}$$

Deriving the exponential MLE with censoring: III

To check that the critical point is a maximum, differentiate once more:

$$\frac{d^2\ell}{d\lambda^2} = -\frac{\sum_i \delta_i}{\lambda^2}.$$

If at least one event is observed, then $\sum_i \delta_i > 0$, so

$$\frac{d^2\ell}{d\lambda^2} < 0.$$

Therefore the critical point is a maximum.

For exponential right-censored data,

$$\boxed{\widehat{\lambda} = \frac{\text{number of observed events}}{\text{total observed follow-up time}}}.$$

Risk sets

At any time t, the risk set is the set of individuals who are still under observation and event-free just before t.

We write

$$Y(t) = \#\{i : U_i \geq t\}.$$

At an observed event time t_j, define

- n_j: number at risk just before t_j;
- d_j: number of events at t_j.

Risk sets are the bookkeeping device that lets us compare events with the subjects who were actually available to experience those events.

Risk sets for the motivating dataset

For the data

$$(4,1), \quad (6,0), \quad (7,1), \quad (9,0), \quad (10,1),$$

observed event times are $4, 7, 10$.

Event time t_j	At risk n_j	Events d_j	Conditional event fraction
4	5	1	$1/5$
7	3	1	$1/3$
10	1	1	$1/1$

Why is $n_j = 3$ at time 7?

Patient 1 already had the event at 4. Patient 2 was censored at 6. Patients 3, 4, and 5 are still at risk just before 7.

A preview of the product-limit idea

At event time t_j, the estimated conditional probability of surviving past that event time is

$$1 - \frac{d_j}{n_j}.$$

Multiplying these conditional survival probabilities gives the Kaplan–Meier estimator:

$$\widehat{S}(t) = \prod_{t_j \le t} \left(1 - \frac{d_j}{n_j}\right).$$

> This is only a preview. We will derive and study the Kaplan–Meier estimator carefully in next week.

A Question

A subject has status $\delta = 0$ at time 8. Another subject has status $\delta = 1$ at time 8.

Do the two records contain the same information? Are their likelihood contributions same?

A Question

A subject has status $\delta = 0$ at time 8. Another subject has status $\delta = 1$ at time 8.

Do the two records contain the same information? Are their likelihood contributions same?

Solution. No.

- $(8, 1)$ means an event occurred exactly at time 8.
- $(8, 0)$ means the subject was event-free through time 8 and the event time is greater than 8.

Their likelihood contributions are different:

$$(8, 1) : f(8), \qquad (8, 0) : S(8).$$

The same recorded time can mean different things depending on the event indicator.

Key takeaway messages

1. Time-to-event data require a clear origin, event, and time scale.
2. The survival function is $S(t) = \mathbb{P}(T > t)$.
3. The hazard $h(t) = f(t)/S(t)$ is an instantaneous event rate among those still at risk.
4. Censoring gives partial information about T; it is not the same as observing T exactly.
5. Right-censored data are commonly recorded as (U_i, δ_i).
6. Independent/noninformative censoring is a key assumption behind standard survival methods.
7. Likelihood contributions must reflect what was actually observed.

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Advanced Statistical Methods II
Lecture 9: Likelihood for Censored Survival Data

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Outline of Today's Lecture

From censoring to likelihood

Likelihood contributions

Exponential model

Weibull model

Gamma and other parametric models

Maximum likelihood with censoring

Key takeaway messages

Where we are

Last week, we introduced the basic ingredients of time-to-event data.

- Event time: T.
- Survival function: $S(t) = \mathbb{P}(T > t)$.
- Density: $f(t)$, when it exists.
- Hazard function: $h(t) = f(t)/S(t)$.
- Right-censored observation: we know $T > c$, not $T = c$.

Today we ask: **how should exact and censored observations enter the likelihood?**

The central idea

Let T be the true event time and let θ denote the parameter of a survival model.

Observed information	Meaning	Likelihood contribution
Exact event at t	$T = t$	$f(t;\theta)$
Right censored at c	$T > c$	$S(c;\theta)$
Left censored at c	$T \leq c$	$F(c;\theta)$
Interval censored in $(a, b]$	$a < T \leq b$	$S(a;\theta) - S(b;\theta)$

A small dataset from Lecture IX

Suppose we observe the following data:

Subject	Observed time u_i	Status δ_i	Interpretation
1	4	1	event at 4
2	6	0	event-free beyond 6
3	7	1	event at 7
4	9	0	event-free beyond 9
5	10	1	event at 10

Here

$$\delta_i = 1 \text{ means exact failure,} \qquad \delta_i = 0 \text{ means right censoring.}$$

What should subject 2 contribute to the likelihood: $f(6;\theta)$ or $S(6;\theta)$?

What subject 2 contributes

For subject 2, the observation is

$$(u_2, \delta_2) = (6, 0).$$

This means

$$T_2 > 6.$$

Therefore the probability of observing this information is

$$\mathbb{P}_\theta(T_2 > 6) = S(6; \theta).$$

Observed data notation

For right-censored data, we usually observe

$$U_i = \min(T_i, C_i),$$

where C_i is the censoring time, and

$$\delta_i = \mathbb{I}(T_i \leq C_i).$$

Thus:

$$\delta_i = 1 \Longrightarrow U_i = T_i \quad \text{is an exact event time,}$$

$$\delta_i = 0 \Longrightarrow U_i = C_i \quad \text{and } T_i > C_i.$$

The observed data are (U_i, δ_i), not simply U_i.

Exact event time contribution

Suppose subject i has an exact event time:

$$\delta_i = 1, \qquad U_i = T_i = u_i.$$

The likelihood contribution is the density at the event time:

$$L_i(\theta) = f(u_i; \theta).$$

Using $f(t) = h(t)S(t)$, this can also be written as

$$L_i(\theta) = h(u_i; \theta)S(u_i; \theta).$$

Right-censored contribution

Suppose subject i is right censored:

$$\delta_i = 0, \qquad U_i = C_i = u_i.$$

The information is

$$T_i > u_i.$$

Therefore the likelihood contribution is

$$L_i(\theta) = \mathbb{P}_\theta(T_i > u_i) = S(u_i; \theta).$$

Right-censoring contributes a survival probability, not a density.

Combining exact and right-censored observations

For one observation (u_i, δ_i), the two cases can be combined as

$$L_i(\theta) = \{f(u_i;\theta)\}^{\delta_i}\{S(u_i;\theta)\}^{1-\delta_i}.$$

Equivalently, since $f = hS$,

$$L_i(\theta) = \{h(u_i;\theta)\}^{\delta_i} S(u_i;\theta).$$

For independent observations,

$$L(\theta) = \prod_{i=1}^{n}\{f(u_i;\theta)\}^{\delta_i}\{S(u_i;\theta)\}^{1-\delta_i}.$$

This is the standard likelihood for independent right-censored survival data (Cox and Oakes, 1984; Collett, 2015).

Log-likelihood form

Taking logarithms gives

$$\ell(\theta) = \sum_{i=1}^{n} \left[\delta_i \log f(u_i;\theta) + (1-\delta_i)\log S(u_i;\theta)\right].$$

Equivalently,

$$\ell(\theta) = \sum_{i=1}^{n} \left[\delta_i \log h(u_i;\theta) + \log S(u_i;\theta)\right].$$

Why is the second form often conceptually useful?

Because it separates two parts:

- everyone must survive until their observed time;
- only uncensored subjects experience the event at the observed time.

What about the censoring mechanism?

Strictly speaking, the observed data depend on both T_i and C_i.

A standard assumption is **independent censoring**:

$$T_i \perp C_i$$

with the survival-model parameter θ appearing only in the distribution of T_i.

Then the part of the full likelihood involving θ is proportional to

$$\prod_{i=1}^{n} \{f(u_i;\theta)\}^{\delta_i} \{S(u_i;\theta)\}^{1-\delta_i}.$$

If censoring depends on the unobserved event time in a way not accounted for, this likelihood can be misleading.

Why independent censoring matters

Consider two situations.

Situation A: administrative censoring

A study ends on 31 December. Anyone event-free at that date is censored. This may be plausibly independent of the biological event time.

Situation B: informative dropout

Patients with worsening symptoms leave the study early. Then censoring may be associated with imminent failure.

The exponential survival model

The exponential model assumes a constant hazard rate:

$$T \sim \mathrm{Exponential}(\lambda), \qquad \lambda > 0.$$

Its main functions are

$$f(t;\lambda) = \lambda e^{-\lambda t}, \qquad t > 0,$$

$$S(t;\lambda) = e^{-\lambda t},$$

$$h(t;\lambda) = \lambda.$$

The exponential model is simple and analytically useful, but its constant hazard assumption is often restrictive.

Exponential likelihood with censoring

For right-censored data,

$$L(\lambda) = \prod_{i=1}^{n} \{\lambda e^{-\lambda u_i}\}^{\delta_i} \{e^{-\lambda u_i}\}^{1-\delta_i}.$$

Simplifying,

$$L(\lambda) = \lambda^{\sum_i \delta_i} \exp\left(-\lambda \sum_{i=1}^{n} u_i\right).$$

Let

$$d = \sum_{i=1}^{n} \delta_i$$

be the number of observed failures. Then

$$L(\lambda) = \lambda^{d} \exp\left(-\lambda \sum_{i} u_i\right).$$

Exponential log-likelihood and score

The log-likelihood is

$$\ell(\lambda) = d \log \lambda - \lambda \sum_{i=1}^{n} u_i.$$

The score equation is

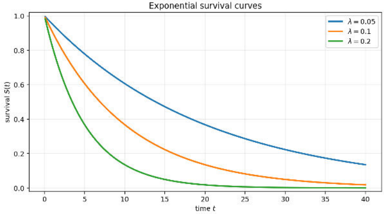
$$\frac{d\ell}{d\lambda} = \frac{d}{\lambda} - \sum_{i=1}^{n} u_i = 0.$$

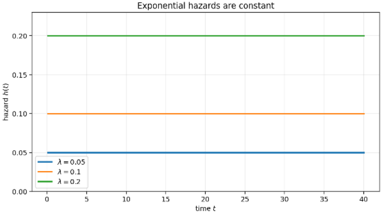
Therefore,

$$\widehat{\lambda} = \frac{d}{\sum_{i=1}^{n} u_i}.$$

The denominator uses the **total observed time on study**, including censored follow-up times.

Exponential survival and hazard: plots





Increasing λ makes survival drop faster, but the hazard is still flat over time. This is the key restriction of the exponential model.

An worked example: administrative censoring

Suppose a reliability experiment is stopped at 18 months. We observe

$$(u_i) = (3, 5, 6, 8, 11, 12, 15, 18),$$

$$(\delta_i) = (1, 0, 1, 1, 0, 1, 0, 1).$$

Here $\delta_i = 0$ means the unit was still working when last observed. Then

$$d = 5, \qquad \sum_i u_i = 78.$$

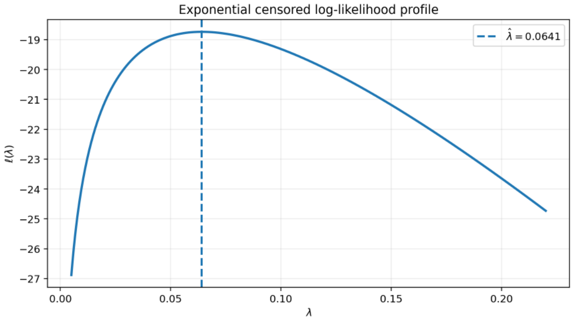
For the exponential model,

$$\widehat{\lambda} = \frac{d}{\sum_i u_i} = \frac{5}{78} = 0.0641.$$

Log-likelihood profile for the example

For the same dataset,

$$\ell(\lambda) = 5 \log \lambda - 78\lambda.$$



The peak occurs at $\widehat{\lambda} = 5/78$.

Why go beyond the exponential model?

The exponential model assumes

$$h(t) = \lambda,$$

so the instantaneous event rate is constant over time.

But many real phenomena have time-varying hazards.

- A machine may be more likely to fail as it ages.
- Early post-surgery risk may be high and then decline.
- A treatment effect may change over follow-up time.

The Weibull model is a simple parametric model that allows increasing or decreasing hazards.

The Weibull survival model

Use the shape-scale parameterization:

$$T \sim \mathrm{Weibull}(k, \eta), \qquad k > 0, \quad \eta > 0.$$

The survival function is

$$S(t; k, \eta) = \exp\left[-\left(\frac{t}{\eta}\right)^{k}\right].$$

The density is

$$f(t; k, \eta) = \frac{k}{\eta}\left(\frac{t}{\eta}\right)^{k-1} \exp\left[-\left(\frac{t}{\eta}\right)^{k}\right].$$

The hazard is

$$h(t; k, \eta) = \frac{k}{\eta}\left(\frac{t}{\eta}\right)^{k-1}.$$

For the Weibull model,

$$h(t; k, \eta) = \frac{k}{\eta}\left(\frac{t}{\eta}\right)^{k-1}.$$

Shape k	Hazard behavior	Interpretation
$k = 1$	constant	exponential special case
$k > 1$	increasing	aging, cumulative damage
$0 < k < 1$	decreasing	early high-risk period

Weibull likelihood with censoring

For right-censored data,

$$L(k,\eta) = \prod_{i=1}^{n} \{f(u_i; k,\eta)\}^{\delta_i} \{S(u_i; k,\eta)\}^{1-\delta_i}.$$

Using the hazard-survival form,

$$L(k,\eta) = \prod_{i=1}^{n} \{h(u_i; k,\eta)\}^{\delta_i} S(u_i; k,\eta).$$

Thus

$$\ell(k,\eta) = \sum_i \delta_i \log h(u_i; k,\eta) + \sum_i \log S(u_i; k,\eta).$$

Weibull log-likelihood

Substitute

$$\log h(u_i; k, \eta) = \log k - \log \eta + (k-1)\log(u_i/\eta),$$

$$\log S(u_i; k, \eta) = -\left(\frac{u_i}{\eta}\right)^k .$$

Then

$$\ell(k,\eta) = \sum_i \delta_i \left[\log k - \log \eta + (k-1)\log(u_i/\eta)\right] - \sum_i \left(\frac{u_i}{\eta}\right)^k .$$

Equivalently,

$$\ell(k,\eta) = d\log k - dk\log\eta + (k-1)\sum_i \delta_i \log u_i - \sum_i \left(\frac{u_i}{\eta}\right)^k .$$

Weibull MLE: one analytic simplification

For fixed k, differentiate $\ell(k,\eta)$ with respect to η.

The relevant terms are

$$-dk\log\eta - \sum_i \frac{u_i^k}{\eta^k}.$$

Setting the derivative equal to zero gives

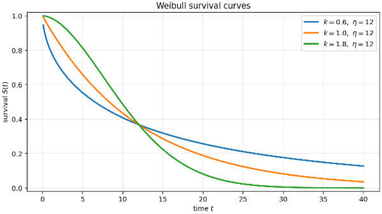
$$-\frac{dk}{\eta} + \frac{k}{\eta^{k+1}}\sum_i u_i^k = 0.$$

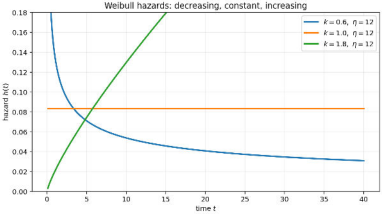
Hence

$$\widehat{\eta}(k)^k = \frac{\sum_i u_i^k}{d}.$$

For Weibull MLE, η has a closed-form update if k is fixed, but k usually requires numerical solution.

Weibull survival and hazard: plots





The same scale $\eta = 12$ can produce decreasing, constant, or increasing hazards depending on k.

Worked example: evaluating a Weibull likelihood

Use the data from the administrative-censoring example:

$$(u_i) = (3, 5, 6, 8, 11, 12, 15, 18), \quad (\delta_i) = (1, 0, 1, 1, 0, 1, 0, 1).$$

Evaluate the Weibull log-likelihood at $k = 1.5, \eta = 12$:

$$\ell(k, \eta) = \sum_i \delta_i \log h(u_i; k, \eta) + \sum_i \log S(u_i; k, \eta).$$

Using

$$\log h(u_i) = \log(1.5) - \log(12) + 0.5 \log(u_i/12),$$
$$\log S(u_i) = -(u_i/12)^{1.5},$$

we get

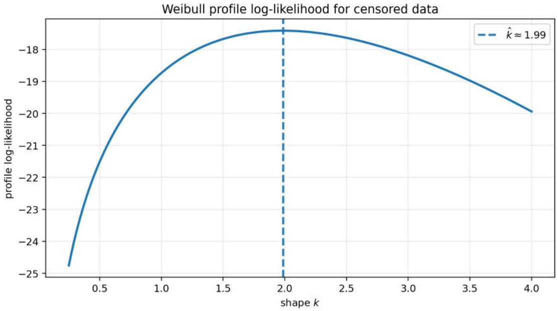
$$\ell(1.5, 12) = -17.8411.$$

Worked example: profiling the Weibull shape

For fixed k, recall that

$$\widehat{\eta}(k)^k = \frac{\sum_i u_i^k}{d}.$$

Thus we can reduce the two-dimensional optimization to a one-dimensional profile over k.



For this dataset, the profile maximum is approximately

$$\widehat{k} \approx 1.99, \qquad \widehat{\eta} \approx 13.77.$$

The gamma model

A gamma survival model may be written as

$$T \sim \mathrm{Gamma}(\alpha, \beta),$$

where $\alpha > 0$ is a shape parameter and $\beta > 0$ is a rate parameter.

The density is

$$f(t; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t}, \qquad t > 0.$$

The survival function is

$$S(t; \alpha, \beta) = \mathbb{P}_{\alpha,\beta}(T > t).$$

Unlike the exponential model, the gamma survival function generally involves the incomplete gamma function.

Gamma likelihood with censoring

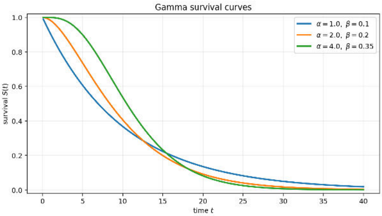
The censored likelihood is still conceptually simple:

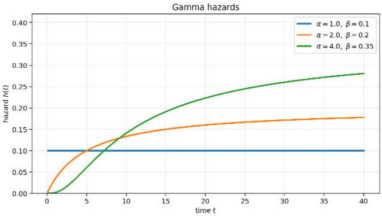
$$L(\alpha, \beta) = \prod_{i=1}^{n} \{f(u_i; \alpha, \beta)\}^{\delta_i} \{S(u_i; \alpha, \beta)\}^{1-\delta_i}.$$

The log-likelihood is

$$\ell(\alpha, \beta) = \sum_i \delta_i \log f(u_i; \alpha, \beta) + \sum_i (1 - \delta_i) \log S(u_i; \alpha, \beta).$$

Gamma survival and hazard: plots





Gamma models are more flexible than exponential models, but their survival function and MLE usually need numerical routines.

Worked example: gamma likelihood contributions

Let $T \sim \mathrm{Gamma}(\alpha, \beta)$ with $\alpha = 2$ and $\beta = 0.2$. Consider two observations:

$$(u_1, \delta_1) = (6, 1), \qquad (u_2, \delta_2) = (11, 0).$$

The first is an exact failure, so its contribution is

$$f(6; 2, 0.2) = 0.0723, \qquad \log f(6; 2, 0.2) = -2.6271.$$

The second is right-censored, so its contribution is

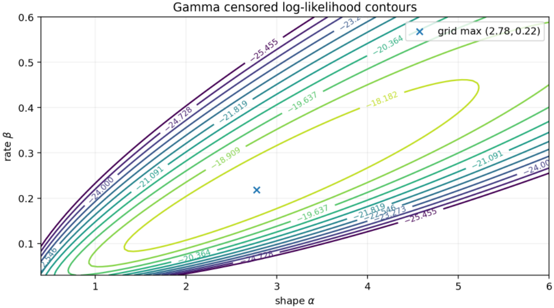
$$S(11; 2, 0.2) = 0.3546, \qquad \log S(11; 2, 0.2) = -1.0368.$$

The same observed time enters as a density if $\delta = 1$, and as a survival probability if $\delta = 0$.

Worked example: gamma likelihood surface

For the administrative-censoring dataset, the censored gamma log-likelihood is

$$\ell(\alpha, \beta) = \sum_i \delta_i \log f(u_i; \alpha, \beta) + \sum_i (1 - \delta_i) \log S(u_i; \alpha, \beta).$$



A simple grid search gives an approximate maximum near

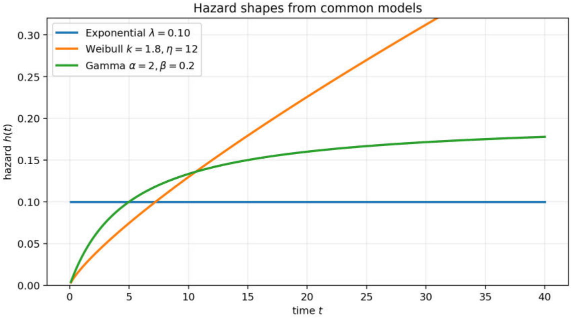
$$\widehat{\alpha} \approx 2.78, \qquad \widehat{\beta} \approx 0.218.$$

Common parametric survival models

Model	Typical use	Hazard behavior	MLE with censoring
Exponential	first simple model	constant	often analytic
Weibull	reliability, medical studies	monotone	partly analytic, partly numerical
Gamma	flexible waiting time	non-monotone	numerical
Log-normal	skewed event times	non-monotone	numerical
Log-logistic	long-tailed survival	non-monotone	numerical

Parametric survival analysis is likelihood-based: choose a distribution for T, write f and S, then maximize the censored likelihood.

Hazard shapes: comparing common models



A parametric survival model is not only a model for event times; it also encodes a scientific assumption about how instantaneous risk evolves with time.

The general MLE problem

For a parametric family indexed by θ, the MLE is

$$\widehat{\theta} = \arg\max_{\theta} \ell(\theta),$$

where

$$\ell(\theta) = \sum_{i} \left[\delta_i \log f(u_i;\theta) + (1-\delta_i) \log S(u_i;\theta)\right].$$

Equivalently,

$$\ell(\theta) = \sum_{i} \left[\delta_i \log h(u_i;\theta) + \log S(u_i;\theta)\right].$$

Score equations

The score function is

$$U(\theta) = \frac{\partial \ell(\theta)}{\partial \theta}.$$

The MLE usually solves

$$U(\widehat{\theta}) = 0.$$

For censored data,

$$U(\theta) = \sum_i \left[\delta_i \frac{\partial}{\partial \theta} \log f(u_i; \theta) + (1 - \delta_i) \frac{\partial}{\partial \theta} \log S(u_i; \theta) \right].$$

The censored observations are not absent from the score. They contribute through $\partial \log S(u_i; \theta) / \partial \theta$.

Why censoring is not MCAR

In missing data language, MCAR means that the missingness indicator is unrelated to the observed and unobserved data values.

Right censoring is different:

$$\delta_i = 0 \quad \Rightarrow \quad T_i > U_i.$$

So the observed censoring status directly tells us something about the unobserved event time.

A right-censored event time is not merely an unknown blank cell. It is constrained to lie beyond the censoring time.

This is why censored observations should not be deleted under the logic of complete-case analysis.

Key takeaway messages

1. Exact failure at t contributes $f(t;\theta)$.
2. Right censoring at c contributes $S(c;\theta)$.
3. For right-censored observations,

$$L(\theta) = \prod_i f(u_i;\theta)^{\delta_i} S(u_i;\theta)^{1-\delta_i}.$$

4. Equivalently,

$$L(\theta) = \prod_i h(u_i;\theta)^{\delta_i} S(u_i;\theta).$$

5. Exponential MLE under censoring is $\widehat{\lambda} = d / \sum_i u_i$.
6. Weibull and gamma models follow the same likelihood principle but usually require numerical optimization.

Looking ahead

In the next survival lecture, we will move from parametric likelihoods to nonparametric estimation of the survival curve.

- risk sets;
- event times;
- product-limit idea;
- Kaplan–Meier estimator;
- survival curves and median survival.

Today: likelihood for censored data.
Next: estimating $S(t)$ without specifying a parametric distribution.

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Advanced Statistical Methods II

Lecture 10: Nonparametric Estimation of Survival Curve

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First Semester, AY 2026–27

Outline of Today's Lecture

Why nonparametric survival estimation?

Risk sets and event times

Product-limit idea

Kaplan–Meier estimator

Nelson–Aalen estimator

Survival curves and median survival

KM plots for parametric goodness-of-fit

Uncertainty and practical issues

Summary

Where we are

In Lectures 8 and 9, we developed the language and likelihood of survival data.

- Event time: T.
- Survival function: $S(t) = \mathbb{P}(T > t)$.
- Censoring time: C.
- Observed data: $Y = \min(T, C)$, $\delta = \mathbb{I}(T \leq C)$.
- Likelihood contribution: $f(y_i)^{\delta_i} S(y_i)^{1-\delta_i}$, under independent censoring.

Today we do **not** assume exponential, Weibull, gamma, or any specific parametric family. We estimate $S(t)$ directly from censored data.

The central question

Suppose we observe

$$(y_1, \delta_1), \ldots, (y_n, \delta_n),$$

where $\delta_i = 1$ means an event was observed and $\delta_i = 0$ means right censoring.

How can we estimate the whole curve

$$S(t) = \mathbb{P}(T > t)$$

without specifying a distribution for T?

- We cannot simply treat censored times as event times.
- We cannot simply discard censored observations.
- We need an estimator that uses the information: *subject survived at least up to the censoring time*.

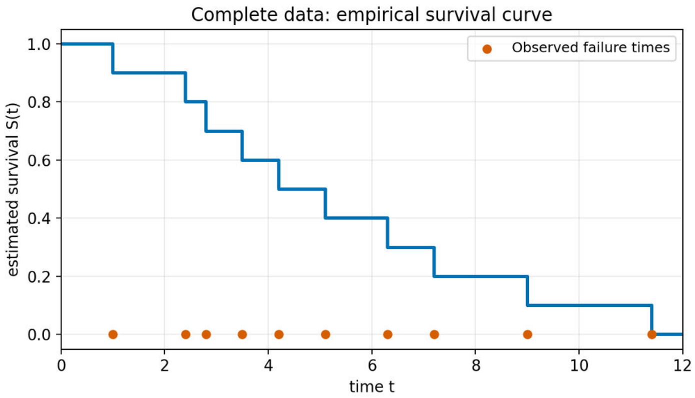
Why not just use the empirical survival curve?

If we observed complete failure times $t_1, \ldots, t_n$, a natural estimator would be

$$\widehat{S}_{\mathrm{emp}}(t) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}(t_i > t) = \frac{\#\{i : t_i > t\}}{n}.$$

This empirical survival estimator is the starting point for the nonparametric discussion; see, for example, the complete-data derivation in Wang (2005, pp. 27–30).

Complete data: empirical survival curve



With complete data, every observed time that is an event causes a jump down. With censoring, the main difficulty is deciding **who should still be counted in the denominator** at each event time.

A note

For right-censored data, the observed time Y_i means two different things:

δ_i	What we observe	What we know about T_i
1	event at y_i	$T_i = y_i$
0	censored at y_i	$T_i > y_i$

A censored time is not the same as a failure time. It tells us that the person/item was still event-free until that time.

This is why Kaplan–Meier estimation is built around **risk sets** and **conditional survival probabilities**.

Distinct event times

Let

$$y_{(1)} < y_{(2)} < \cdots < y_{(k)}$$

be the distinct **uncensored event times**.

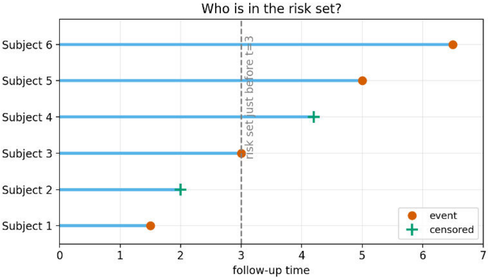
For the event time $y_{(j)}$, define

$$R_j = \{i : Y_i \geq y_{(j)}\}, \qquad N_j = |R_j|, \qquad d_j = \#\{i : Y_i = y_{(j)}, \delta_i = 1\}.$$

- We only multiply factors at observed event times.
- Censoring times affect the future denominators, but should not produce jumps in $\widehat{S}(t)$.
- If more than one failure occurs at the same time, that event time has $d_j > 1$.

Risk set

The risk set just before time $y_{(j)}$ contains individuals who have not yet failed and have not yet been censored before $y_{(j)}$.



At an event time, the estimated conditional failure probability is $\widehat{h}_j = \frac{d_j}{N_j}$. Thus the estimated conditional survival probability through that event time is

$$1 - \widehat{h}_j = 1 - \frac{d_j}{N_j}.$$

Who is in the risk set?

Suppose the next event time is $t = 3$.

- A subject who failed before 3 is not in the risk set.
- A subject censored before 3 is not in the risk set.
- A subject censored after 3 is in the risk set at 3.
- A subject failing at 3 is in the risk set just before 3.

Interpretation

The denominator N_j is not the original sample size. It is the number of individuals who were still **under observation and event-free** just before the event time.

The product-limit idea

For ordered event times, survival beyond a later time can be decomposed into a product of conditional survival probabilities.

For example,

$$\mathbb{P}(T > t_3) = \mathbb{P}(T > t_1)\mathbb{P}(T > t_2 \mid T > t_1)\mathbb{P}(T > t_3 \mid T > t_2).$$

More generally, if $y_{(j)} \leq t$, estimate each conditional factor by

$$\widehat{\mathbb{P}}(\text{survive past } y_{(j)} \mid \text{at risk just before } y_{(j)}) = 1 - \frac{d_j}{N_j}.$$

The Kaplan–Meier estimator multiplies these factors.

Why multiplication?

At each event time, we estimate a **conditional** probability.

$$\widehat{S}(t) = \widehat{\mathbb{P}}(T > y_{(1)}) \times \widehat{\mathbb{P}}(T > y_{(2)} \mid T > y_{(1)}) \times \cdots .$$

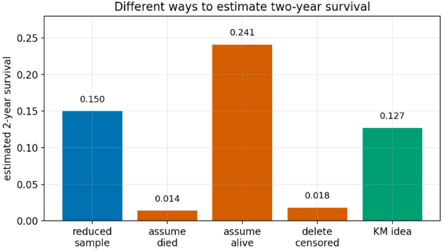
The estimator is called a **product-limit estimator** because the survival curve is built as a product of many local survival proportions.

A two-year survival example

A study has two cohorts.

- 100 patients recruited in year 0 and followed for 2 years.
- 1000 patients recruited in year 1 and followed for 1 year before the study ends.
- Deaths: 70 in year 1 from cohort 1, 15 in year 2 from cohort 1, and 750 in year 1 from cohort 2.

Naive approaches can give very different answers depending on how censored patients are handled.



Two-year example: product-limit calculation

The product-limit idea estimates

$$S(2) = \mathbb{P}(T > 2) = \mathbb{P}(T > 1)\mathbb{P}(T > 2 \mid T > 1).$$

The Kaplan–Meier-type calculation is

$$\widehat{\mathbb{P}}(T > 1) = \frac{30 + 250}{1100} = \frac{280}{1100},$$

where 30 from cohort 1 and 250 from cohort 2 survive year 1.

Then

$$\widehat{\mathbb{P}}(T > 2 \mid T > 1) = \frac{15}{30}.$$

Hence

$$\widehat{S}(2) = \frac{280}{1100} \times \frac{15}{30} = 0.127.$$

Definition: Kaplan–Meier estimator

Let $y_{(1)} < \cdots < y_{(k)}$ be the distinct uncensored event times. Let

$$N_j = \#\{i : Y_i \geq y_{(j)}\}, \qquad d_j = \#\{i : Y_i = y_{(j)}, \delta_i = 1\}.$$

The Kaplan–Meier estimator is

$$\boxed{\widehat{S}(t) = \prod_{j : y_{(j)} \leq t} \left(1 - \frac{d_j}{N_j}\right).}$$

This estimator was introduced by Kaplan and Meier (1958) and is also called the product-limit estimator.

What does each factor mean?

At event time $y_{(j)}$,

$$\frac{d_j}{N_j}$$

is the estimated conditional probability of failure at that time among those still at risk.

Therefore,

$$1 - \frac{d_j}{N_j}$$

is the estimated conditional probability of surviving through that event time.

The KM curve falls only at uncensored event times. It stays flat at censoring times, but censoring changes later risk sets.

Small worked example

Consider the data

$$3, \quad 2^+, \quad 0, \quad 1, \quad 5^+, \quad 3, \quad 5,$$

where $+$ denotes right censoring.

The uncensored event times are

$$0, \quad 1, \quad 3, \quad 5.$$

Event time $y_{(j)}$	N_j	d_j	factor $1 - d_j/N_j$
0	7	1	6/7
1	6	1	5/6
3	4	2	1/2
5	2	1	1/2

Small worked example: calculations

Using

$$\widehat{S}(t) = \prod_{y_{(j)} \le t} \left(1 - \frac{d_j}{N_j}\right),$$

we get

$$\widehat{S}(0) = 1 - \frac{1}{7} = \frac{6}{7} = 0.86,$$

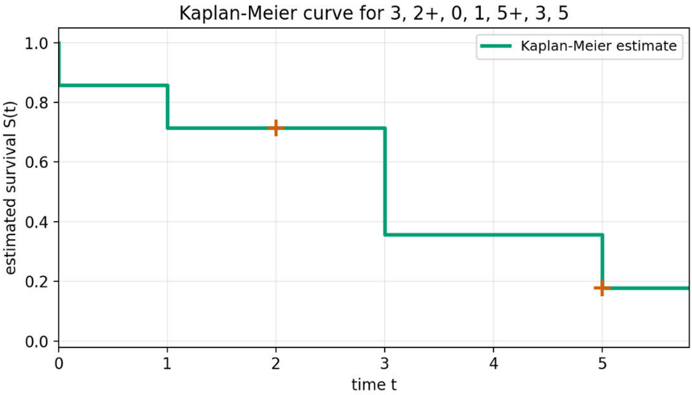
$$\widehat{S}(1) = \frac{6}{7}\left(1 - \frac{1}{6}\right) = \frac{5}{7} = 0.71,$$

$$\widehat{S}(3) = \frac{5}{7}\left(1 - \frac{2}{4}\right) = \frac{5}{14} = 0.36,$$

$$\widehat{S}(5) = \frac{5}{14}\left(1 - \frac{1}{2}\right) = \frac{5}{28} = 0.18.$$

At time 3 there are two failures, so the drop is larger: multiply by $1 - 2/4 = 1/2$.

Small worked example: the KM curve



- The curve is a step function.
- It drops at $0, 1, 3, 5$, the event times.
- The censoring marks at 2 and 5 do not themselves create a drop.

Handling ties

Suppose several events occur at the same observed time $y_{(j)}$.

- Count all failures at that time in d_j.
- Use one multiplicative factor:

$$1 - \frac{d_j}{N_j}.$$

- Censored observations at the same recorded time are usually treated as occurring just after failures at that time, unless a study-specific convention says otherwise.

In the small example

At time 3, two individuals fail and $N_j = 4$, so the factor is

$$1 - \frac{2}{4} = \frac{1}{2}.$$

KM as a nonparametric MLE

The right-censored likelihood, ignoring the censoring distribution under independent censoring, has the form

$$L \propto \prod_{i=1}^{n} f(y_i)^{\delta_i} S(y_i)^{1-\delta_i}.$$

Kaplan and Meier showed that, among distributions assigning probability mass to the observed uncensored event times, the product-limit estimator is the nonparametric maximum likelihood estimator (Kaplan and Meier, 1958; Wang, 2005).

So the KM estimator is not just a heuristic curve. It also arises from a likelihood principle.

A likelihood intuition for KM

At event time $y_{(j)}$, think of the discrete hazard

$$h_j = \mathbb{P}(T = y_{(j)} \mid T \geq y_{(j)}).$$

Among N_j people at risk, d_j fail at that time. The local likelihood contribution behaves like

$$h_j^{d_j}(1 - h_j)^{N_j - d_j}.$$

Maximizing this gives

$$\widehat{h}_j = \frac{d_j}{N_j}.$$

Therefore the estimated survival through that event time is

$$1 - \widehat{h}_j = 1 - \frac{d_j}{N_j}.$$

Multiplying these local survival factors gives the KM estimator.

Efron's redistribution-of-mass construction

The product-limit formula can also be understood through a constructive algorithm due to Efron (1967). A clear four-step version is discussed in Sun (n.d., Unit 5, pp. 5–6).

1. Arrange the observations in increasing order; if there are ties, put failures before censored observations.
2. Put mass $1/n$ at each observation.
3. Move from left to right. Whenever a censored observation is reached, redistribute its current mass equally among observations to its right.
4. Continue until all censoring masses have been redistributed. If the largest observation is censored, its mass remains beyond the largest observed time.

Censoring does not destroy probability mass. It says: "the failure happened somewhere later." Efron's algorithm literally moves that mass to the right.

Nelson–Aalen estimator

Recall the cumulative hazard function

$$H(t) = \int_0^t h(u)\,du = -\log S(t).$$

At event time v_j, the estimated discrete hazard is

$$\widehat{h}_j = \frac{d_j}{Y(v_j)}.$$

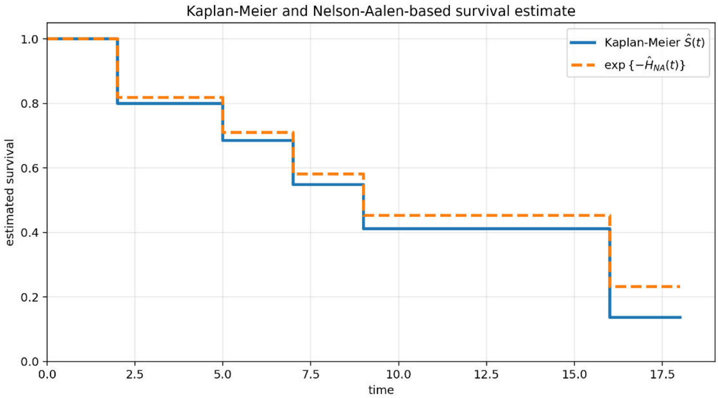
The Nelson–Aalen estimator of cumulative hazard is

$$\boxed{\widehat{H}_{NA}(t) = \sum_{j:v_j \le t} \frac{d_j}{Y(v_j)}.}$$

A survival estimate based on it is

$$\widehat{S}_{NA}(t) = \exp\{-\widehat{H}_{NA}(t)\}.$$

KM and Nelson–Aalen on the same data



- KM multiplies local survival factors: $\prod(1 - d_j/Y_j)$.
- Nelson–Aalen adds local hazard increments: $\sum d_j/Y_j$.
- The two are close when each d_j/Y_j is small.

Reading a KM survival curve

A Kaplan–Meier curve is interpreted as an estimated survival probability over time.

- $\widehat{S}(t) = 0.80$: estimated 80% surviving beyond time t.
- Vertical drops: observed event times.
- Larger drops: more failures or a smaller risk set.
- Censoring marks: individuals leaving observation without event.
- Flat segments: no event observed during that interval.

The right tail of a KM curve can be unstable when very few subjects remain at risk.

Median survival

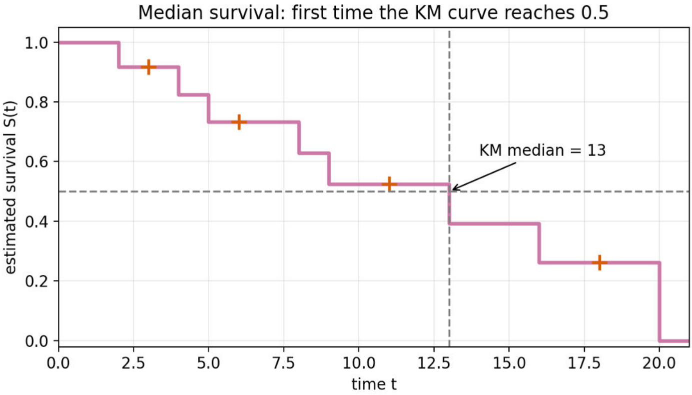
The population median survival time m is defined by

$$S(m) \leq 0.5 \quad \text{and} \quad S(m^{-}) > 0.5.$$

The Kaplan–Meier median is the first time at which the estimated survival curve reaches or goes below 0.5:

$$\widehat{m} = \inf\{t : \widehat{S}(t) \leq 0.5\}.$$

Median survival: graphical illustration



Using KM for parametric goodness-of-fit

The Kaplan–Meier estimator is nonparametric, so it can be used as a diagnostic reference for a proposed parametric model.

1. Fit or hypothesize a parametric family for T.
2. Find a transformation of $S(t)$, $F(t)$, or $H(t)$ that should look simple under that family.
3. Replace $S(t)$ by $\widehat{S}(t)$, usually the KM estimator.
4. Check the transformed plot visually.

> This is a graphical goodness-of-fit diagnostic. It can reveal serious mismatch, but it is not by itself a formal test.

The idea is discussed, for example, in Sun (n.d., Unit 5, pp. 9–10).

GOF idea 1: exponential survival

Suppose

$$T \sim \text{Exponential}(\lambda).$$

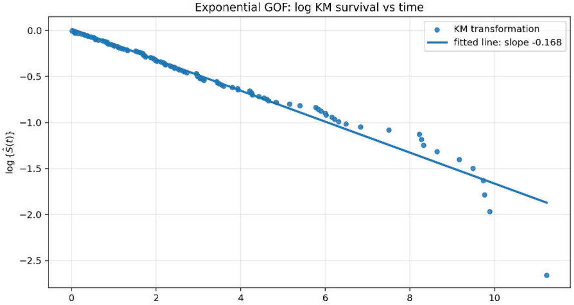
Then

$$S(t) = \exp(-\lambda t), \qquad \log S(t) = -\lambda t.$$

Therefore, under an exponential model,

$$\log\{\widehat{S}(t)\} \quad \text{plotted against} \quad t$$

should look approximately linear with negative slope.



GOF idea 2: Weibull survival

For a Weibull model, write

$$S(t) = \exp\{-(\lambda t)^p\}.$$

Then

$$-\log S(t) = (\lambda t)^p,$$

and hence

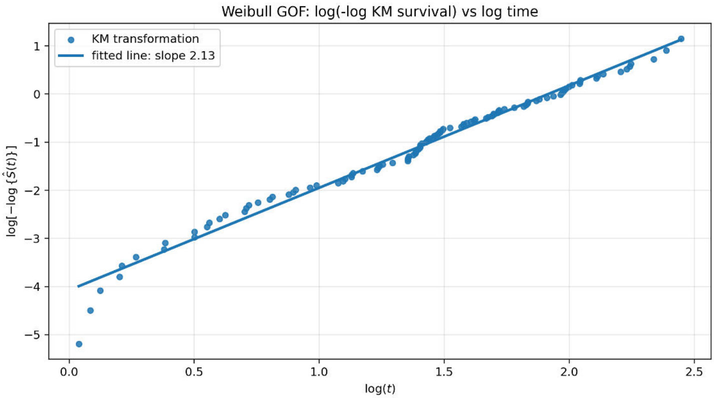
$$\log[-\log S(t)] = p \log \lambda + p \log t.$$

Therefore, under a Weibull model,

$$\log[-\log\{\widehat{S}(t)\}] \quad \text{plotted against} \quad \log t$$

should look approximately linear.

Weibull GOF plot



- If the plot is roughly straight, a Weibull assumption is plausible.
- The slope estimates the Weibull shape parameter p, at least heuristically.
- Strong curvature suggests the Weibull family may not be adequate.

A brief look at uncertainty: Greenwood's formula

A common large-sample variance estimate for the Kaplan–Meier estimator is Greenwood's formula:

$$\widehat{\mathbb{V}\mathrm{ar}}\{\widehat{S}(t)\} \approx \widehat{S}(t)^2 \sum_{j: y_{(j)} \le t} \frac{d_j}{N_j(N_j - d_j)}.$$

- The sum accumulates uncertainty over event times.
- The variance increases when risk sets become small.
- This motivates confidence intervals and confidence bands for survival curves.

Greenwood's formula and its heuristic derivation are discussed, for example, in Wang (2005, pp. 42–44).

Why risk sets control uncertainty

At a given event time $y_{(j)}$, the local failure count d_j is like a binomial count among N_j individuals at risk.

$$d_j \approx \mathrm{Binomial}(N_j, h_j).$$

- Large N_j: stable estimate of the local failure probability.
- Small N_j: unstable local estimate.
- Late event times often have small risk sets, so tail estimates are more variable.

The KM curve is usually most reliable where a reasonable number of subjects remain at risk.

Common practical decisions

When reporting a Kaplan–Meier analysis, mention:

- number of individuals at risk at selected times;
- number of events;
- number censored;
- median survival, if estimable;
- confidence intervals for survival probabilities or median survival;
- whether censoring is plausibly independent or non-informative.

> The Kaplan–Meier estimator assumes that censoring does not carry extra information about the event time after conditioning on the design/covariates being considered.

Key takeaway messages

1. The Kaplan–Meier estimator estimates $S(t) = \mathbb{P}(T > t)$ without specifying a parametric distribution.
2. The estimator is built from event times, risk sets, and event counts.
3. At each event time, the local survival factor is $1 - d_j/N_j$.
4. Censored observations do not create jumps but affect later risk sets.
5. The KM curve is a step function and the median survival is the first time it reaches or crosses 0.5.
6. The far right tail should be interpreted carefully, especially when few individuals remain at risk.

What comes next

Today we estimated one survival curve nonparametrically.

Question	Next tool
How do we compare two survival curves?	log-rank test
How do covariates affect survival?	Cox proportional hazards model
How do we model hazard ratios?	partial likelihood
How do we check proportional hazards?	residuals and diagnostics

The risk-set idea from today will reappear in the Cox model.

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Advanced Statistical Methods II

Lecture 11: From Kaplan–Meier to Regression for Survival Data

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First Semester, AY 2026–27

Outline of Today's Lecture

From one survival curve to many covariates

Choosing what to model

The proportional hazards idea

The Cox proportional hazards model

Hazard ratio interpretation

Risk sets and the intuition behind Cox estimation

Examples and model interpretation

When the Cox model is reasonable

Summary and next steps

Where we are

In the previous lectures, we learned how to estimate survival curves nonparametrically.

- Observed survival data: (Y_i, δ_i), where

$$Y_i = \min(T_i, C_i), \qquad \delta_i = \mathbb{I}(T_i \le C_i).$$

- Risk set at event time t_j: individuals still under observation and event-free just before t_j.
- Kaplan–Meier estimator:

$$\widehat{S}(t) = \prod_{t_j \le t} \left(1 - \frac{d_j}{n_j}\right).$$

Today: *how should survival depend on covariates* such as treatment, age, sex, disease stage, biomarker value, or machine temperature?

A motivating medical example

Suppose a cancer study records time to relapse.

Patient	Time	Event?	Treatment	Age	Stage
1	5.2	1	New	42	II
2	8.1	0	Control	67	III
3	3.4	1	New	71	III
4	10.7	0	Control	50	II
⋮	⋮	⋮	⋮	⋮	⋮

If we draw Kaplan–Meier curves for treatment groups, are we answering the whole question?

Not necessarily. Treatment groups may differ in age, stage, biomarker levels, or baseline prognosis.

Why Kaplan–Meier is not enough

Kaplan–Meier is excellent for estimating **one survival curve** or comparing a few groups.

But it becomes limited when many covariates are present.

- It handles groups, but not continuous covariates naturally.
- Splitting age into age groups loses information and creates arbitrary cut-points.
- With several covariates, stratified KM curves create many small cells.
- It does not give a compact covariate effect such as "per 10-year increase in age."

The curse of many Kaplan–Meier curves

Suppose we stratify by three covariates:

Covariate	Categories	Number of strata
Treatment	New / Control	2
Age group	Young / Middle / Old	3
Disease stage	I / II / III / IV	4
Total		$2 \times 3 \times 4 = 24$

Groups versus regression

Group comparison	Regression modelling
"Do the survival curves differ between groups?"	"How does survival change with covariates?"
Usually categorical groups	Categorical and continuous covariates
KM curves, log-rank test	Cox model, parametric regression models
Mostly unadjusted	Can adjust for multiple covariates
Visual comparison	Estimates, standard errors, confidence intervals

Regression does not replace Kaplan Meier. It answers a different, more covariate-focused question.

A familiar analogy

For ordinary continuous responses:

Group means $\bar{y}_1, \bar{y}_2$	$\rightarrow$	Linear regression $E(Y \mid X) = x^\top \beta$

For survival responses:

KM curves by group	$\rightarrow$	Survival regression $h(t \mid x)$

What is the survival-analysis analogue of the regression function?

To model the **hazard function**.

Three possible targets

For survival data, a covariate may affect several related quantities.

Target	Symbol	Interpretation
Survival function	$S(t \mid x)$	Probability of surviving beyond t
Density	$f(t \mid x)$	Distribution of event times
Hazard function	$h(t \mid x)$	Instantaneous event rate at t

The hazard is

$$h(t \mid x) = \lim_{\Delta t \downarrow 0} \frac{\mathbb{P}(t \leq T < t + \Delta t \mid T \geq t, x)}{\Delta t}.$$

It is a natural target because it describes the event risk among individuals who are still event-free.

Why model the hazard?

The hazard is local in time.

- At time t, it focuses on individuals who have survived up to t.
- It allows covariates to act on the instantaneous event rate.
- It provides a useful effect measure: the **hazard ratio**.

The Cox model keeps the risk-set idea and adds a regression structure for covariates.

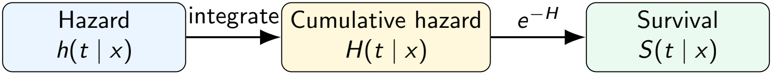
Survival and hazard are connected

If $H(t \mid x)$ is the cumulative hazard,

$$H(t \mid x) = \int_0^t h(u \mid x)\, du,$$

then

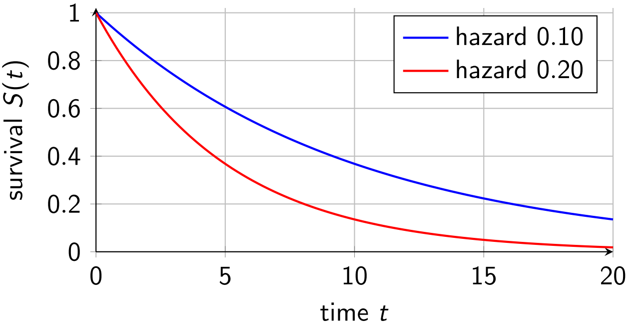
$$S(t \mid x) = \exp\{-H(t \mid x)\}.$$



So a model for $h(t \mid x)$ automatically implies a model for $S(t \mid x)$.

A small visual intuition

Suppose a baseline group has constant hazard $h_0(t) = 0.10$. A higher-risk group has hazard 0.20.



A larger hazard means a faster decline in survival, but the survival curves themselves are not separated by a constant vertical distance.

The proportional hazards idea

For two covariate values x_a and x_b,

$$\frac{h(t \mid x_a)}{h(t \mid x_b)} \quad \text{does not depend on } t.$$

The shapes over time are controlled by a common time pattern; covariates multiply that pattern.

Multiplicative rather than additive effects

A proportional hazards model is multiplicative:

$$h(t \mid x) = h_0(t) \times \text{relative risk depending on } x.$$

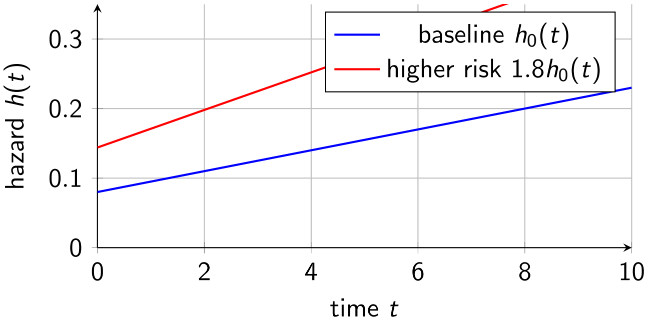
For example, if treatment reduces hazard by 30%, then

$$h(t \mid \text{new treatment}) = 0.70\, h(t \mid \text{control})$$

for all t, under proportional hazards.

This does **not** mean survival probabilities are 30% larger at all times.
The hazard ratio is a ratio of instantaneous event rates.

Visualizing proportional hazards



The two hazard curves can change over time, but their ratio is always 1.8.

Why this idea is useful

Proportional hazards is useful because it separates two questions.

Question	Model component
How does baseline risk evolve over time?	$h_0(t)$
How do covariates change risk?	$\exp(x^\top \beta)$

We do not have to fully parametrize the time pattern $h_0(t)$ in the Cox model.

This is why the Cox model is often called **semiparametric**.

The Cox model

The Cox proportional hazards model is

$$\boxed{h(t \mid x) = h_0(t) \exp(x^\top \beta)}$$

where

- $h_0(t)$ is an unspecified baseline hazard;
- $x = (x_1, \ldots, x_p)^\top$ is a vector of covariates;
- $\beta = (\beta_1, \ldots, \beta_p)^\top$ is the regression coefficient vector.

Cox's key idea was to estimate covariate effects without specifying a parametric form for $h_0(t)$ (Cox, 1972).

What does baseline hazard mean?

If $x = 0$, then

$$h(t \mid x = 0) = h_0(t).$$

So $h_0(t)$ is the hazard for the baseline covariate profile.

Interpretation

If x contains treatment $(1 = \text{new}, 0 = \text{control})$ and age centered at 60, then $h_0(t)$ is the hazard for a control patient aged 60.

If covariates are not centered, $x = 0$ may not be scientifically meaningful. The hazard ratios are usually more important than the literal baseline profile.

Why the exponential form?

The model uses

$$\exp(x^\top \beta)$$

for the relative risk.

This has several advantages:

- It is always positive, so hazards remain nonnegative.
- Regression effects become additive on the log-hazard scale:

$$\log h(t \mid x) = \log h_0(t) + x^\top \beta.$$

- Coefficients have hazard-ratio interpretations.

Linear predictors are convenient; exponentiation converts them into multiplicative hazard effects.

A model with two covariates

Suppose

$$x_1 = \mathbb{I}(\text{new treatment}), \qquad x_2 = \text{age in decades above 60}.$$

Then

$$h(t \mid x_1, x_2) = h_0(t) \exp(\beta_1 x_1 + \beta_2 x_2).$$

Coefficient	Meaning	Hazard ratio
β_1	New treatment versus control	e^{β_1}
β_2	10-year increase in age	e^{β_2}

Example

If $\beta_1 = -0.357$, then $e^{\beta_1} = 0.70$: the new treatment has 30% lower hazard than control, adjusting for age.

Cox model implies proportional hazards

For two individuals with covariates x_a and x_b,

$$\frac{h(t \mid x_a)}{h(t \mid x_b)} = \frac{h_0(t)\exp(x_a^\top \beta)}{h_0(t)\exp(x_b^\top \beta)}.$$

Therefore,

$$\boxed{\frac{h(t \mid x_a)}{h(t \mid x_b)} = \exp\{(x_a - x_b)^\top \beta\}}$$

The baseline hazard cancels. The hazard ratio does not depend on time.

Connection to survival curves

Let

$$H_0(t) = \int_0^t h_0(u)\,du, \qquad S_0(t) = \exp\{-H_0(t)\}.$$

Under the Cox model,

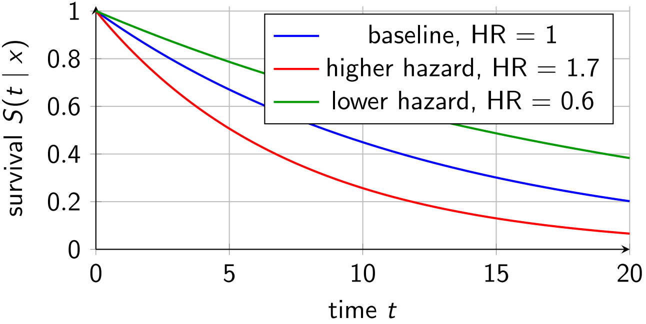
$$H(t \mid x) = H_0(t)\exp(x^\top \beta),$$

so

$$\boxed{S(t \mid x) = S_0(t)^{\exp(x^\top \beta)}.}$$

The model creates a family of survival curves by transforming one baseline survival curve.

Here is a stylized example with $S_0(t) = \exp(-0.08t)$.



The survival curves are not parallel; the proportionality is on the hazard scale.

The hazard ratio

For a one-unit increase in covariate x_k, holding other covariates fixed,

$$\boxed{\mathrm{HR}_k = e^{\beta_k}.}$$

Hazard ratio	Interpretation
$\mathrm{HR} = 1$	no hazard difference
$\mathrm{HR} > 1$	higher instantaneous event rate
$\mathrm{HR} < 1$	lower instantaneous event rate

A hazard ratio is not a probability ratio, not an odds ratio, and not directly a ratio of median survival times.

Example: treatment effect

Suppose the Cox model is

$$h(t \mid x) = h_0(t)\exp(\beta x),$$

where $x = 1$ for new treatment and $x = 0$ for control.

If $\hat{\beta} = -0.5108$, then

$$e^{\hat{\beta}} = e^{-0.5108} \approx 0.60.$$

Interpretation

At any time t, among patients still event-free just before t, the new-treatment group has about 40% lower instantaneous relapse rate than the control group.

This interpretation assumes proportional hazards.

Example: age effect

Suppose age is measured in years and

$$\hat{\beta}_{\text{age}} = 0.035.$$

Then the one-year hazard ratio is

$$\exp(0.035) = 1.036.$$

A one-year interpretation is often too small to communicate clearly. For a 10-year increase,

$$\exp(10 \times 0.035) = \exp(0.35) = 1.42.$$

A 10-year older patient has about 42% higher instantaneous event rate, holding other covariates fixed.

Example: continuous biomarker

Let x be a biomarker standardized to mean 0 and variance 1.

If

$$\hat{\beta} = 0.50,$$

then

$$\exp(0.50) = 1.65.$$

Interpretation

A one standard deviation increase in the biomarker is associated with a 65% higher hazard, adjusting for other covariates.

For continuous covariates, always say what a "one-unit" increase means.

Comparing two covariate profiles

For two patients a and b,

$$\mathrm{HR}(a : b) = \exp\{(x_a - x_b)^\top \beta\}.$$

Suppose

$$\beta_{\mathrm{treat}} = -0.36, \qquad \beta_{\mathrm{age10}} = 0.30.$$

Compare:

- Patient A: new treatment, age 70;
- Patient B: control, age 60.

Then

$$\log \mathrm{HR}(A : B) = (-0.36)(1) + (0.30)(1) = -0.06,$$

so

$$\mathrm{HR}(A : B) = e^{-0.06} = 0.94.$$

Treatment benefit and age disadvantage nearly cancel.

A common misinterpretation

Suppose $\mathrm{HR} = 0.60$ for treatment.

Wrong statement: "The treated group has 40% higher probability of survival at every time."

Better statement:

At each time t, among individuals who are still at risk just before t, the treated group has 40% lower instantaneous event rate than the control group, under the proportional hazards model.

The survival-probability difference depends on time and on the baseline hazard.

Risk sets return

The Kaplan–Meier estimator used the risk set $\mathcal{R}(t_j)$ at each event time.

The Cox model also uses risk sets.

At an observed event time t_j, the risk set is

$$\mathcal{R}(t_j) = \{i : Y_i \geq t_j\}.$$

It contains all individuals who are still under observation and event-free just before t_j.

The Cox model asks: among the individuals in the risk set, whose covariates make them more likely to be the one who fails now?

A small risk-set example

At time $t = 6$ months, suppose patient 2 fails. The risk set is:

Patient	In risk set?	Treatment	Linear score $\eta_i = x_i^\top \beta$
1	Yes	Control	0.00
2	Yes	New	-0.40
3	Yes	Control	0.30
4	No, censored earlier	New	–
5	Yes	New	-0.10

The Cox model assigns relative risk score

$$\exp(\eta_i).$$

The failed subject is compared only with subjects in the same risk set.

Why the baseline hazard cancels

At an event time t_j, suppose exactly one individual fails.

Under the Cox model,

$$h_i(t_j) = h_0(t_j)\exp(x_i^\top \beta).$$

Given that someone in $\mathcal{R}(t_j)$ fails at t_j, the approximate probability that subject i is the one who fails is

$$\frac{h_0(t_j)\exp(x_i^\top \beta)}{\sum_{\ell \in \mathcal{R}(t_j)} h_0(t_j)\exp(x_\ell^\top \beta)}.$$

Thus

$$\boxed{\frac{\exp(x_i^\top \beta)}{\sum_{\ell \in \mathcal{R}(t_j)} \exp(x_\ell^\top \beta)}}$$

The unknown $h_0(t_j)$ cancels.

Example 1: clinical trial

Outcome: time to disease progression.

Covariate	Possible coding
Treatment	1 = new drug, 0 = standard care
Age	years, or decades above 60
Stage	indicator variables for stages III and IV
Biomarker	standardized value

A possible model:

$$h(t \mid x) = h_0(t) \exp(\beta_1 \mathrm{Trt} + \beta_2 \mathrm{Age10} + \beta_3 \mathrm{StageIII} + \beta_4 \mathrm{StageIV}).$$

The treatment hazard ratio is adjusted for age and disease stage.

Example 2: reliability engineering

Outcome: time until machine failure.

Possible covariates:

- operating temperature;
- vibration level;
- load level;
- manufacturer;
- maintenance schedule.

A Cox model might be

$$h(t \mid x) = h_0(t)\exp(\beta_1 \text{Temp} + \beta_2 \text{Vibration} + \beta_3 \text{Load}).$$

Interpretation

If $e^{\beta_1} = 1.25$ per 10 degrees, then each 10-degree increase in temperature is associated with a 25% higher failure hazard, holding vibration and load fixed.

Example 3: customer churn

Outcome: time until a customer leaves a service.

Possible covariates:

- monthly price;
- number of support tickets;
- usage frequency;
- plan type;
- region.

> Survival methods are not only for medicine. They apply whenever the response is a time until an event, with censoring.

Here, censoring may mean the customer was still active when the dataset was closed.

Mini interpretation exercise

A fitted Cox model gives:

Covariate	Estimate	Hazard ratio	Interpretation?
Treatment	-0.45	0.64	?
Age10	0.28	1.32	?
BiomarkerSD	0.52	1.68	?

Write one sentence interpreting each hazard ratio. Be precise about "holding other covariates fixed."

Solution to the mini exercise

- Treatment: treated individuals have about 36% lower hazard than controls, holding age and biomarker fixed.
- Age10: each 10-year increase in age is associated with about 32% higher hazard, holding treatment and biomarker fixed.
- BiomarkerSD: each one-standard-deviation increase in the biomarker is associated with about 68% higher hazard, holding treatment and age fixed.

Do not say "36% lower probability of death at every time." The model is about hazard, not direct survival probability.

What the Cox model assumes

The Cox model is powerful, but it is not assumption-free.

Assumption	Meaning
Proportional hazards	Covariate hazard ratios are constant over time
Log-linear covariate effect	$x^\top\beta$ is appropriate on the log-hazard scale
Independent censoring	Censoring is not informative after conditioning on covariates

The Cox model avoids specifying $h_0(t)$, but it still imposes structure on covariate effects.

What proportional hazards does not mean

Proportional hazards does not mean:

- survival curves are parallel;
- survival differences are constant over time;
- median survival times have a fixed ratio;
- the treatment effect is causal without a suitable study design or adjustment strategy.

It means that for two fixed covariate profiles,

$$\frac{h(t \mid x_a)}{h(t \mid x_b)}$$

is constant in t.

When proportional hazards may fail

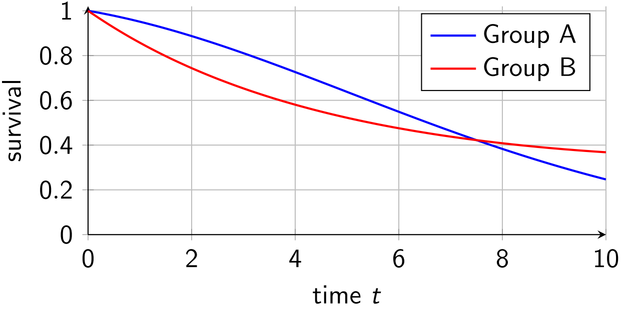
Examples where proportional hazards may be questionable:

- A surgery has high early risk but long-term benefit.
- A drug works strongly at first but its effect weakens later.
- Two KM curves cross substantially.
- A machine component has burn-in failures and then stabilizes.

> If the hazard ratio changes over time, a single Cox hazard ratio may be an oversimplification.

Possible remedies include time-varying coefficients, stratified Cox models, or alternative survival models.

A visual warning: crossing survival curves



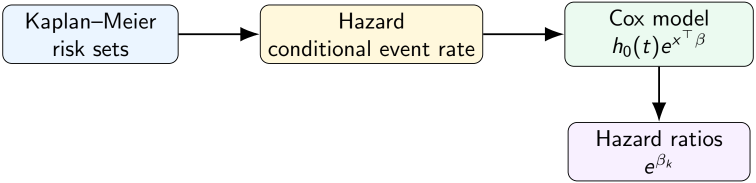
Crossing or converging curves suggest that the relative hazard may not be constant over time.

Kaplan–Meier remains useful even when regression is needed.

- Initial exploratory plots by major groups.
- Visual check for crossing curves.
- Communication of unadjusted survival experience.
- Comparison with model-based adjusted survival curves.

A good survival analysis often uses both KM curves and regression models.

Concept map



The Cox model is best understood as risk-set-based regression for event times.

Key Takeaway Messages

1. KM curves estimate survival nonparametrically, but they do not easily adjust for multiple covariates.
2. Survival regression models the event-time distribution conditional on covariates.
3. The Cox model specifies

$$h(t \mid x) = h_0(t) \exp(x^\top \beta),$$

with unspecified baseline hazard.

4. Hazard ratios compare instantaneous event rates among those still at risk.
5. The proportional hazards assumption means hazard ratios are constant over time.
6. Risk sets provide the bridge from Kaplan–Meier estimation to Cox partial likelihood.

What comes next

Topic	Why it matters
Partial likelihood	Estimating β without specifying $h_0(t)$
Ties	Multiple events at the same observed time
Wald / score / likelihood ratio tests	Inference for covariate effects
Schoenfeld residuals	Checking proportional hazards
Adjusted survival curves	Communicating fitted models

Today: model intuition. Next: estimation and diagnostics.

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